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Practical 01

Aim: Write a program for implementing Client Server communication model using TCP.

Server-side Code

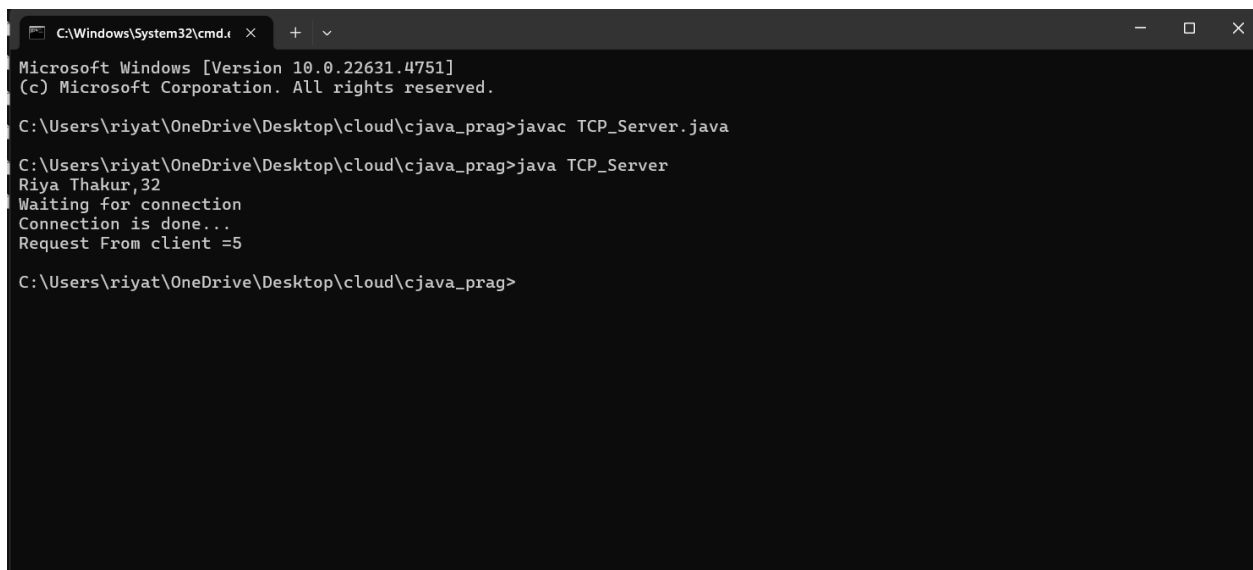
```
import java.io.*;
import java.net.*;

public class TCP_Server {
    public static void main(String[] args) {
        try {
            System.out.println("Riya Thakur,32");
            int port =5555 ;
            ServerSocket ss=new ServerSocket (port);
            System.out.println("Waiting for connection");
            Socket s= ss.accept();
            System.out.println("Connection is done...");

            BufferedReader br = new BufferedReader(new InputStreamReader(s.getInputStream()));
            int num = Integer.parseInt(br.readLine());
            System.out.println("Request From client =" + num);
            int f = 1;
            for(int i=1;i<=num;i++)
            {
                f= f * i;
            }
            PrintWriter pw = new PrintWriter(new OutputStreamWriter(s.getOutputStream()));
            pw.println(f);
        }
    }
}
```

```
pw.flush();  
}  
catch (Exception e)  
{  
    System.out.println("Error : " + e);  
}  
}}
```

Server output:-



```
C:\Windows\System32\cmd.exe  X  +  -  
Microsoft Windows [Version 10.0.22631.4751]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>javac TCP_Server.java  
  
C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>java TCP_Server  
Riya Thakur,32  
Waiting for connection  
Connection is done...  
Request From client =5  
  
C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>
```

Client-Side Code

```
import java.io.*;  
import java.net.*;  
  
public class TCP_Client {  
    public static void main(String[] args) {  
        System.out.println("Riya Thakur ,32c");  
        try {  
            int port = 5555;  
            InetAddress addr = InetAddress.getLocalHost();
```

```
Socket s = new Socket(addr,port);

System.out.println("Connection is done ...");

BufferedReader br1 = new BufferedReader(new InputStreamReader(System.in));

System.out.println("Enter The Number :");

int num = Integer.parseInt(br1.readLine());

PrintWriter pw = new PrintWriter(new OutputStreamWriter(s.getOutputStream()));

pw.println(num);

pw.flush();

BufferedReader br2 = new BufferedReader(new InputStreamReader(s.getInputStream()));

int ans = Integer.parseInt(br2.readLine());

System.out.println("Response From Server =" + ans);

}

catch (Exception e)

{

System.out.println("Error : " + e);

}

}

}
```

Client output:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.4751]
(c) Microsoft Corporation. All rights reserved.

C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>javac TCP_Client.java

C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>java TCP_Client
Riya Thakur ,32
Connection is done ...
Enter The Number :
5
Response From Server =120

C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>|
```

Practical 02

Aim: A client server based program using UDP to find the factorial of the entered number(also implement iterative server).

Server Side Code:

```
import java.io.*;
import java.net.*;

public class UDP_Factorial_server {

    public static void main(String[] args) {
        try
        {
            int port = 1234;

            DatagramSocket ds = new DatagramSocket(port);

            System.out.println("Waiting for Client ...");

            while(true)
            {
                byte b[] = new byte[1024];

                DatagramPacket dp1 = new DatagramPacket(b, b.length);

                ds.receive(dp1);

                String str = new String(dp1.getData(),0,dp1.getLength());

                System.out.println("Request From Client="+str);

                if(str.equals("end"))
                {
                    System.exit(0);
                }
            }
        }
    }
}
```

```

int num = Integer.parseInt(str);

int f = 1;

for(int i=1;i<=num;i++)
{
    f = f * i;
}

String res = "Factorial="+f;

byte b1[] = new byte[1024];

b1 = res.getBytes();

InetAddress addr1 = InetAddress.getByName(null);

DatagramPacket dp2 = new DatagramPacket(b1, b1.length,addr1,5678);

ds.send(dp2); }}

catch (Exception e)

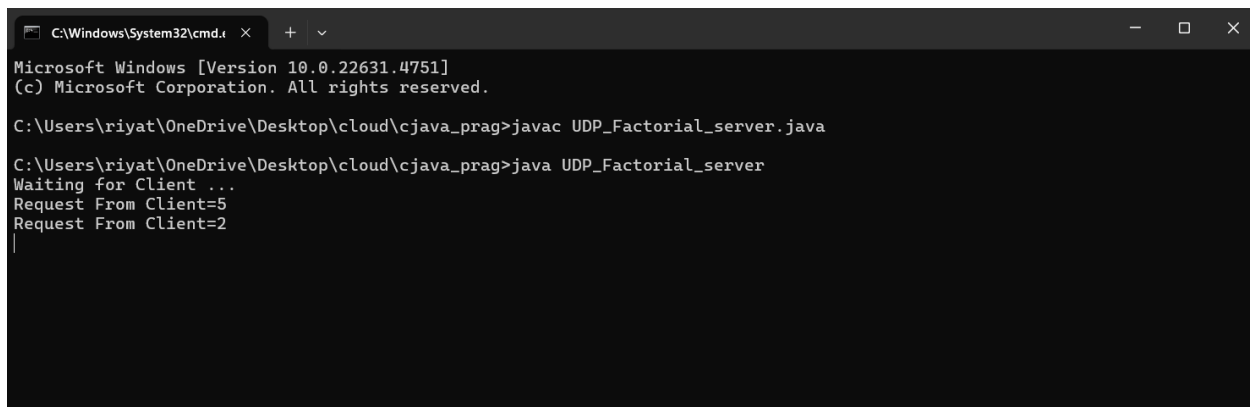
{

    System.out.println("Error=" + e);

}}

```

Server Side Output:



```

C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.4751]
(c) Microsoft Corporation. All rights reserved.

C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>javac UDP_Factorial_server.java

C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>java UDP_Factorial_server
Waiting for Client ...
Request From Client=5
Request From Client=2

```

Client Side Code:

```

import java.io.*;

```

```
import java.net.*;

public class UDP_Factorial_Client {

public static void main(String[] args) {

try

{

int port = 5678;

DatagramSocket ds = new DatagramSocket(port);

while(true)

{

System.out.println("Enter The Number :");

BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

String str = br.readLine();

byte b[] = new byte[1024];

b = str.getBytes();

InetAddress addr = InetAddress.getByName(null);

DatagramPacket dp1 = new DatagramPacket(b, b.length,addr,1234);

ds.send(dp1);

if(str.equals("end"))

{

System.exit(0);

}

byte b1[] = new byte[1024];

DatagramPacket dp2 = new DatagramPacket(b1, b1.length);

ds.receive(dp2);

String res = new String(dp2.getData(),0,dp2.getLength());

System.out.println("Response From Server : " +res);

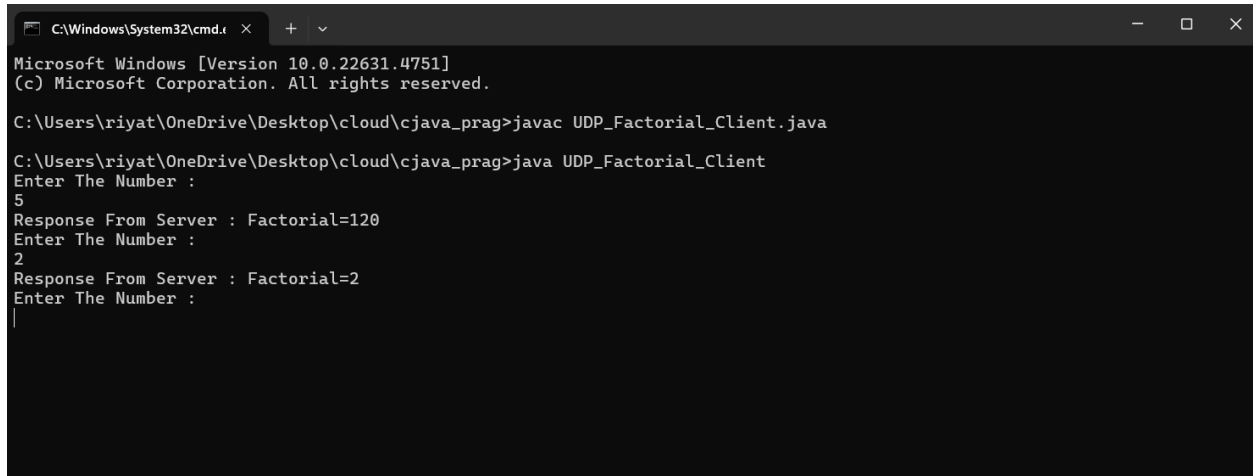
}}

catch (Exception e)
```



```
{  
System.out.println("Error=" + e);  
}}}
```

Client Side Output:



```
C:\Windows\System32\cmd.exe  
Microsoft Windows [Version 10.0.22631.4751]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>javac UDP_Factorial_Client.java  
  
C:\Users\riyat\OneDrive\Desktop\cloud\cjava_prag>java UDP_Factorial_Client  
Enter The Number :  
5  
Response From Server : Factorial=120  
Enter The Number :  
2  
Response From Server : Factorial=2  
Enter The Number :  
|
```

Practical 03

Aim: A program that finds the addition, subtraction, multiplication and division of the entered two numbers using RPC.

Server Side Code:

```
import java.io.*;
import java.net.*;

public class TCP_server {
    public static void main(String[] args) {
        System.out.println("RiyaThakur,Seat no:31011022032");
        try
        {
            int port = 8888;
            ServerSocket ss = new ServerSocket(port);
            System.out.println("Waiting for Client ...");
            Socket s = ss.accept();
            System.out.println("Connection is done ...");
            while(true)
            {
                BufferedReader br = new BufferedReader(new
                InputStreamReader(s.getInputStream()));
                int n1 = Integer.parseInt(br.readLine());
                int n2 = Integer.parseInt(br.readLine());
```

```
String sign = br.readLine();
System.out.println("Data From Client :");
System.out.println("n1="+n1);
System.out.println("n2="+n2);
System.out.println("sign="+sign);
int ans = 0;
if(sign.equals("+"))
{
    ans = n1 + n2;
}
else if(sign.equals("-"))
{
    ans = n1 - n2;
}
else if(sign.equals("*"))
{
    ans = n1 * n2;
}
else if(sign.equals("/"))
{
    ans = n1 / n2;
}

else
{

```

```
ans = 0;
}
String res = "Answer =" + ans;
PrintWriter pw = new PrintWriter(new
OutputStreamWriter(s.getOutputStream()));
pw.println(res);
pw.flush();
}
}
catch (Exception ex)
{
System.out.println("Error :" + ex);
}}}
```

Client Side Code:

```
import java.io.*;
import java.net.*;
public class TCP_client {
public static void main(String[] args)
{
System.out.println("RiyaThakur,Seat no:31011022032");
try
{
int port = 8888;
```

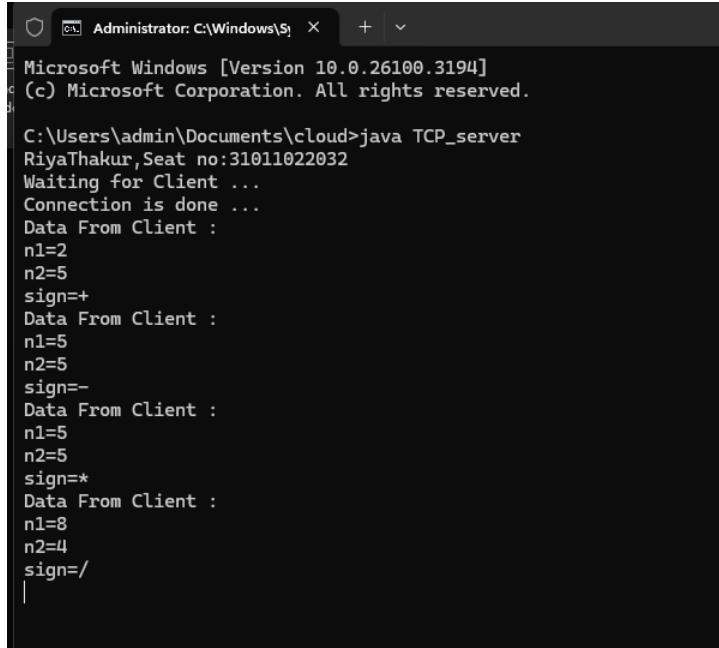
```
InetAddress addr = InetAddress.getByName(null);
System.out.println("IP-Address :" + addr);
Socket s = new Socket(addr, port);
System.out.println("Connection is done ...");
while(true)
{

    BufferedReader br = new BufferedReader(new
    InputStreamReader(System.in ));
    System.out.println("Enter The TWO Numbers :");
    String str = br.readLine();
    if(str.equals("end"))
    {
        System.exit(0);
    }
    int n1 = Integer.parseInt(str);
    int n2 = Integer.parseInt(br.readLine());
    System.out.println("Enter The Operation Sign(+,-,*,/)");
    String sign = br.readLine();

    PrintWriter pw = new PrintWriter(new
    OutputStreamWriter(s.getOutputStream()));
    pw.println(n1);
    pw.println(n2);
    pw.println(sign);
    pw.flush();
```

```
BufferedReader br1 = new BufferedReader(new
InputStreamReader(s.getInputStream()));
String res = br1.readLine();
System.out.println("REsponse From Server :"+res);
}}
catch(Exception ex)
{
System.out.println("Error :"+ex);
}}}
```

Server side output:



```
Administrator: C:\Windows\Sy X + v
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin\Documents\cloud>java TCP_server
RiyaThakur,Seat no:31011022032
Waiting for Client ...
Connection is done ...
Data From Client :
n1=2
n2=5
sign=+
Data From Client :
n1=5
n2=5
sign=-
Data From Client :
n1=5
n2=5
sign=*
Data From Client :
n1=8
n2=4
sign=/
|
```

Client Side Output:

```
Administrator: C:\Windows\Sy X + v
Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

C:\Users\admin\Documents\cloud>java TCP_client
RiyaThakur,Seat no:31011022032
IP-Address :localhost/127.0.0.1
Connection is done ...
Enter The TWO Numbers :
2
5
Enter The Operation Sign(+,-,*,/)
+
REsponse From Server :Answer =7
Enter The TWO Numbers :
5
5
Enter The Operation Sign(+,-,*,/)
-
REsponse From Server :Answer =0
Enter The TWO Numbers :
5
5
Enter The Operation Sign(+,-,*,/)
*
REsponse From Server :Answer =25
Enter The TWO Numbers :
8
4
Enter The Operation Sign(+,-,*,/)
/
REsponse From Server :Answer =2
Enter The TWO Numbers :
|
```

Practical 04

Aim: Implement UDP for sending broad cast sender and receiver using a multicast Socket.

Server Side Code:

```
import java.net.*;
import java.io.*;

public class Server1 {
    public static final int PORT = 1234;

    public static void main(String args[])throws Exception {
        MulticastSocket socket;
        DatagramPacket packet;
        InetAddress address;

        address = InetAddress.getByName("239.1.2.3");
        socket = new MulticastSocket();
        socket.joinGroup(address);
        byte[] data = null;
        for(;;)
        {
            Thread.sleep(10000); System.out.println("Sending ");
            String str = ("This is Riya Calling ");
            data= str.getBytes();
            address = InetAddress.getByName(null);
```



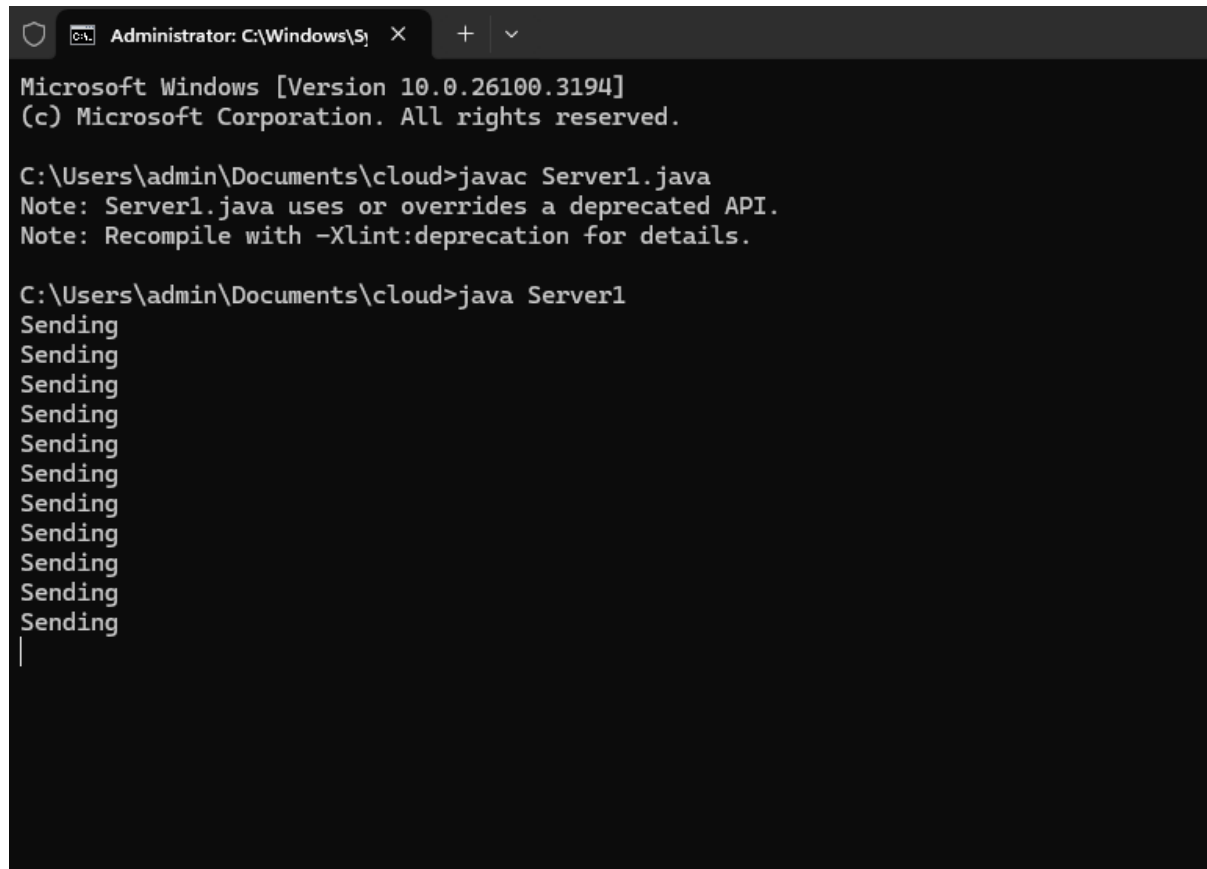
```
packet = new DatagramPacket(data, str.length(),address,PORT);
socket.send(packet);
}
}}
```

Client side code :

```
import java.net.*;
import java.io.*;
public class Client1 {
public static final int PORT = 1234;
public static void main(String args[])throws Exception {
MulticastSocket socket;
DatagramPacket packet;
InetAddress address;
address = InetAddress.getByName("239.1.2.3");
socket = new MulticastSocket(PORT);
socket.joinGroup(address);
byte[] data = new byte[100];
packet = new DatagramPacket(data,data.length);
for(;;)
{
socket.receive(packet);
String str = new String(packet.getData());
System.out.println("Message received from "+ packet.getAddress() +
"Message is : "+str);
}
```

```
}  
}}
```

Server Side output :



```
Administrator: C:\Windows\Sy X + v  
Microsoft Windows [Version 10.0.26100.3194]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\admin\Documents\cloud>javac Server1.java  
Note: Server1.java uses or overrides a deprecated API.  
Note: Recompile with -Xlint:deprecation for details.  
  
C:\Users\admin\Documents\cloud>java Server1  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
Sending  
|
```

Client side output :

Microsoft Windows [Version 10.0.26100.3194]
(c) Microsoft Corporation. All rights reserved.

```
C:\Users\admin\Documents\cloud>javac Client.java
Note: Client.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
```

[illegible]

Practical 05

Aim:

- a) Show the implementation of web services.
- b) Implementing “Calculator” Web Service.

Step 1 : Go to file → new project → java web → Web application.

Step 2 : Click on project name → new → Web Service → allot a name → Give package a name and click finish.

Step 3: Write Code then clean and build and deploy

Cal_s

```
package org.me;
```

```
import javax.jws.WebService;
```

```
import javax.jws.WebMethod;
```

```
import javax.jws.WebParam;
```

```
/**
```

```
*
```

```
* @author riyat
```

```
*/
```

```
@WebService(serviceName = "Cal_s")
```

```
public class Cal_s {
```

```
@WebMethod(operationName = "Addition")

public int add(@WebParam( name = "number1") int n1,@WebParam(name = "number2") int n2 ){

return n1+n2;

}

@WebMethod (operationName = "subtraction")

public int sub(@WebParam(name = "number1")int n1,@WebParam(name = "number2") int n2){

return n1-n2;

}

@WebMethod (operationName = "division")

public int div(@WebParam(name = "number1")int n1,@WebParam(name = "number2") int n2){

return n1/n2;

}

@WebMethod (operationName = "multiplication")

public int multi(@WebParam(name = "number1")int n1,@WebParam(name = "number2") int n2){

return n1*n2;

}

}
```

Step 4: Go to web service and right click on your service and test web service.

Output:

Somaiya Vidyavihar

Cal_s Web Service Tester

localhost:9999/Cal_s/Cal_s?Tester

Cal_s Web Service Tester

This form will allow you to test your web service implementation ([WSDL File](#))

To invoke an operation, fill the method parameter(s) input boxes and click on the button labeled with the method name.

Methods :

public abstract int org.me.CalS.addition(int,int)

addition (2) 2

public abstract int org.me.CalS.subtraction(int,int)

subtraction (2) 2

public abstract int org.me.CalS.multiplication(int,int)

multiplication (2) 2

public abstract int org.me.CalS.division(int,int)

division (2) 2

Somaiya Vidyavihar

Method invocation trace

localhost:9999/Cal_s/Cal_s?Tester

addition Method invocation

Method parameter(s)

Type	Value
int	2
int	2

Method returned

int : "4"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body xmlns:ns2="http://me.org/">
    <ns2:Addition>
      <number1>2</number1>
      <number2>2</number2>
    </ns2:Addition>
  </S:Body>
</S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body xmlns:ns2="http://me.org/">
    <ns2:AdditionResponse>
      <return>4</return>
    </ns2:AdditionResponse>
  </S:Body>
</S:Envelope>
```

Somaiya Vidyavihar

Method invocation trace

localhost:9999/Cal_s/Cal_s?Tester

subtraction Method invocation

Method parameter(s)

Type	Value
int	2
int	2

Method returned

int : "0"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body xmlns:ns2="http://me.org/">
    <ns2:subtraction>
      <number1>2</number1>
      <number2>2</number2>
    </ns2:subtraction>
  </S:Body>
</S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body xmlns:ns2="http://me.org/">
    <ns2:subtractionResponse>
      <return>0</return>
    </ns2:subtractionResponse>
  </S:Body>
</S:Envelope>
```

Somaiya Vidyavihar

Method invocation trace

localhost:9999/Cal_s/Cal_s?Tester

multiplication Method invocation

Method parameter(s)

Type	Value
int	2
int	2

Method returned

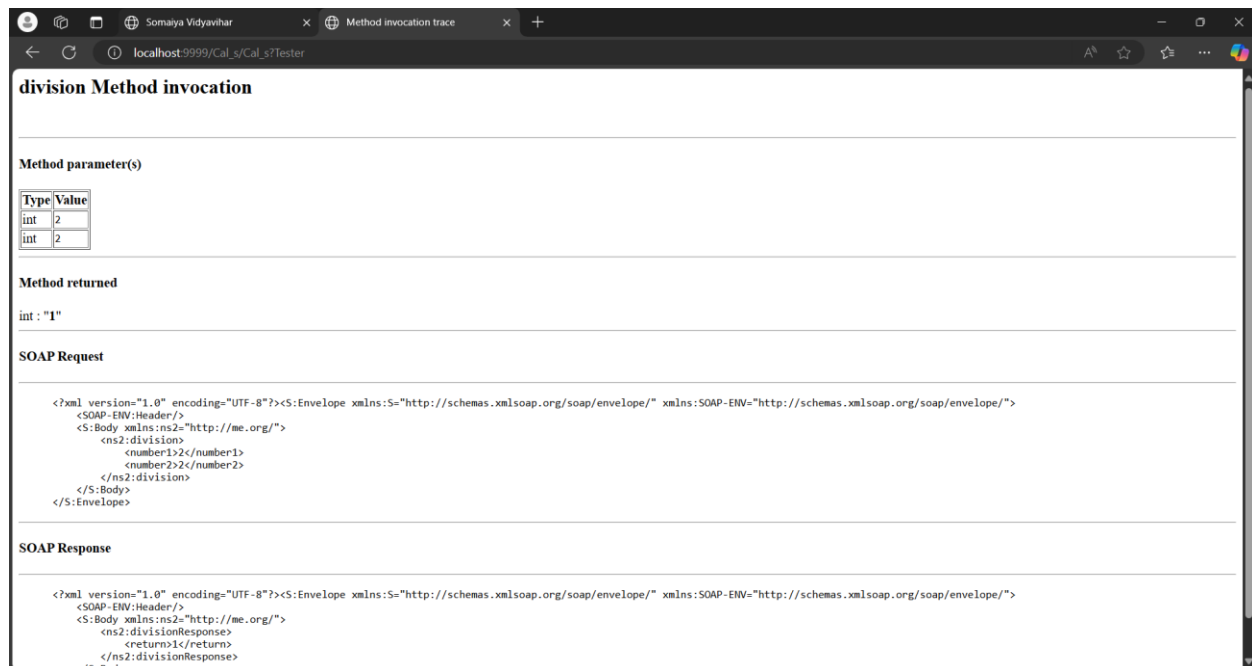
int : "4"

SOAP Request

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body xmlns:ns2="http://me.org/">
    <ns2:multiplication>
      <number1>2</number1>
      <number2>2</number2>
    </ns2:multiplication>
  </S:Body>
</S:Envelope>
```

SOAP Response

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-ENV="http://schemas.xmlsoap.org/soap/envelope/">
  <SOAP-ENV:Header/>
  <S:Body xmlns:ns2="http://me.org/">
    <ns2:multiplicationResponse>
      <return>4</return>
    </ns2:multiplicationResponse>
  </S:Body>
</S:Envelope>
```



To create Client:

Step 1: Go to files → new project → Java → Java Application → Give project name.

Step 2: Right click on your Project → new → Web Service Client → In project, browse your Web service → give a package name

Step 3: Write the code in the java file → Write the code → Clean and build → Click on the project and run.

Client side Code:

Calculator_Client

```
package calculator_client;

import java.util.Scanner;
```



```
public class Calculator_Client {

    public static void main(String[] args) {

        Scanner sc = new Scanner (System.in);

        System.out.println("Enter num1 :");

        int num1 = sc.nextInt();

        System.out.println("Enter num2 :");


        int num2 = sc.nextInt();

        org.me.CalS_Service service = new org.me.CalS_Service();

        org.me.CalS port = service.getCalSPort();

        System.out.println("Output in:" + port.addition(num1,num2));

        System.out.println("Output in:" + port.subtraction(num1,num2));

        System.out.println("Output in:" + port.division(num1,num2));

        System.out.println("Output in:" + port.multiplication(num1,num2));

    }

}
```

Output:



The screenshot shows an IDE output window with the following content:

```
Output
Java DB Database Process x GlassFish Server x Retriever Output x Calculator_Client (run) x
wimport-client-generator
compile:
run:
Enter num1 :
5
Enter num2 :
5
Output in:10
Output in:0
Output in:1
Output in:25
BUILD SUCCESSFUL (total time: 21 seconds)
```

Practical 06

Aim: Implement Xen Server virtualization and manage with Xen Center

Steps:

Step 1: Go to Xenserver.com and download Xenserver8 -ISO

Downloads

XenServer 8 - Installation ISO

December 10, 2024

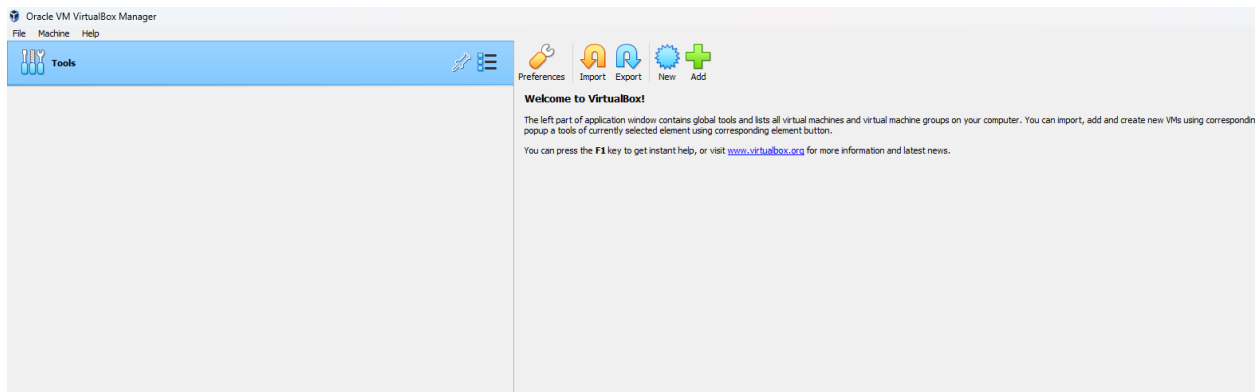
Download the latest XenServer 8 installation ISO. This includes all updates in the Normal channel to December 9, 2024

DOWNLOAD XENSERVR 8 - ISO

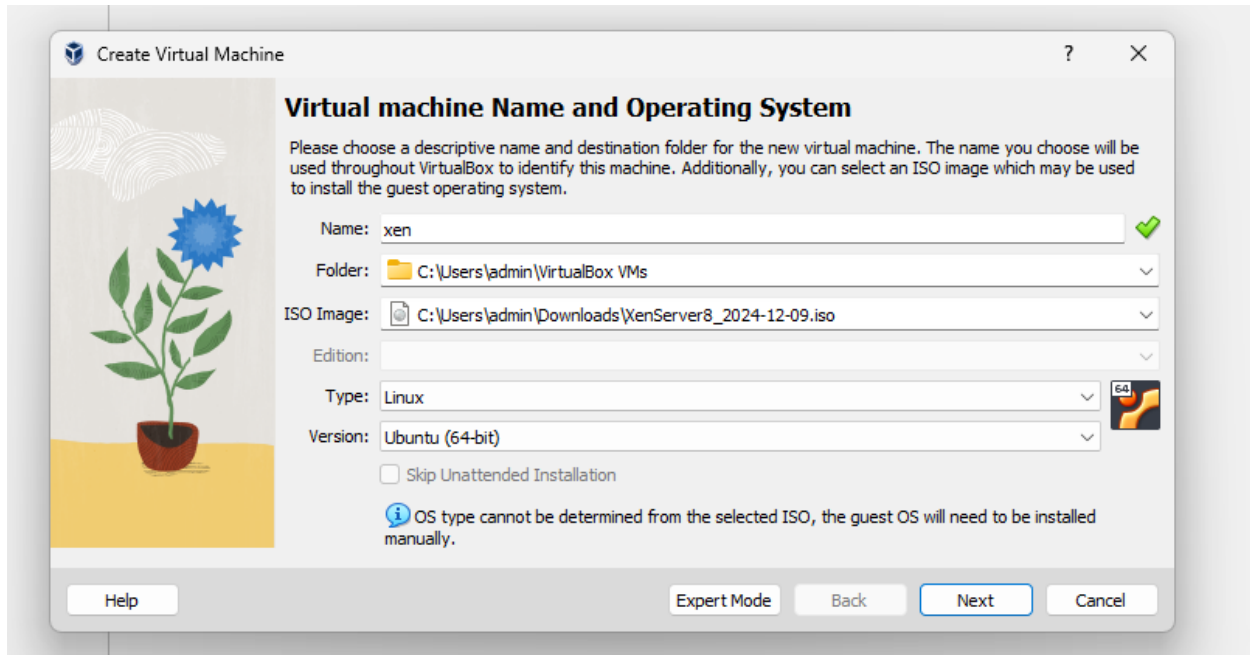
Checksum

SHA-256: 8364422457efa9a792a77dfde17b29d00e347a745a28cefe6106c1c6e67e0ac8

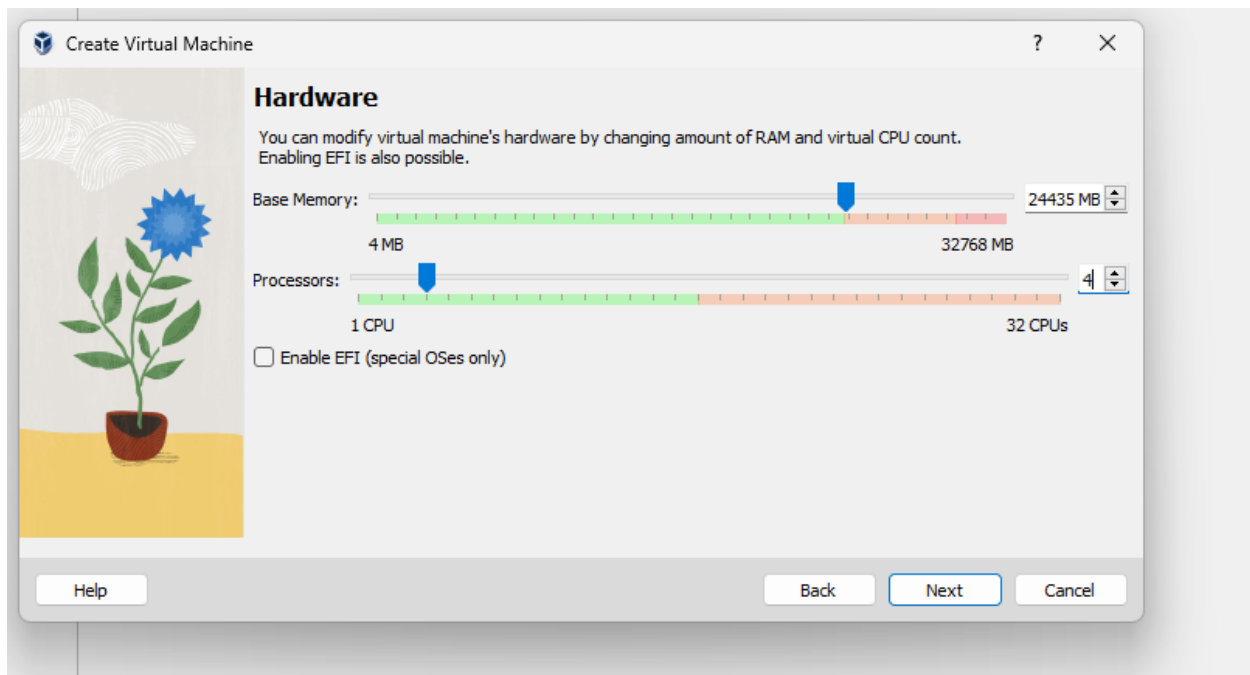
Step 2 : Now go to Oracle VM VirtualBox Manager and click New.



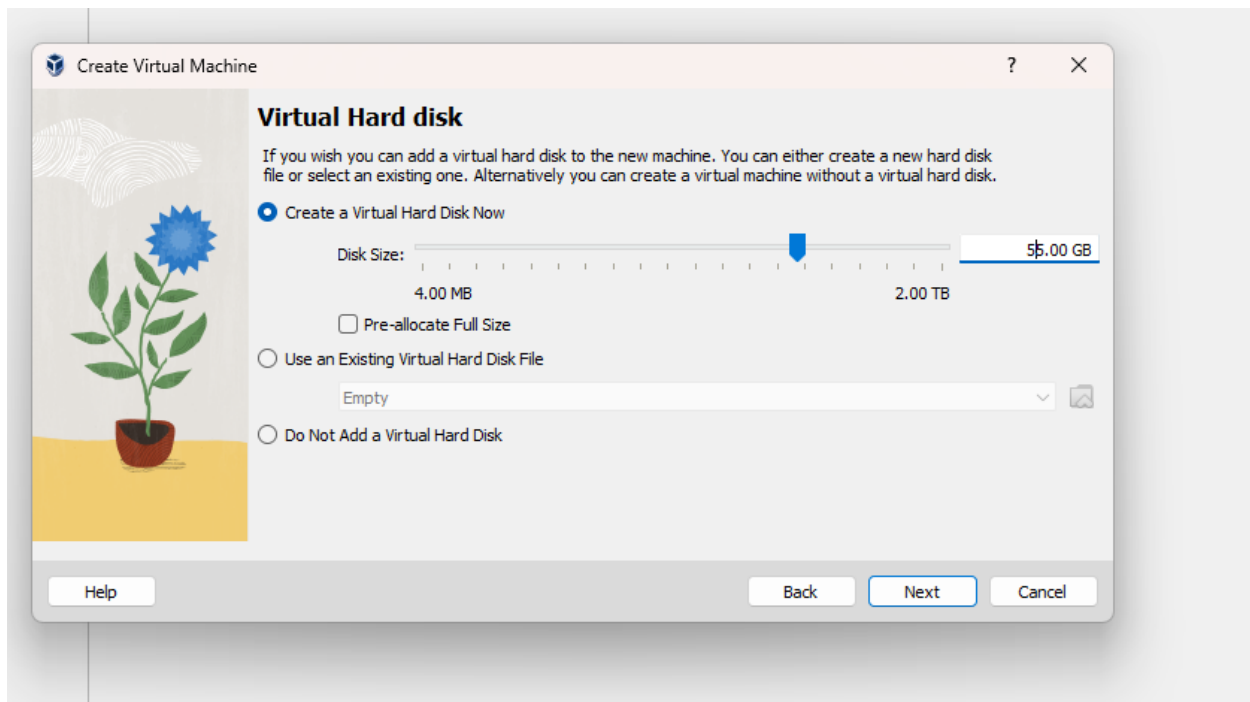
Step 3 : Fill name as Xen, Type as Linux and Version as Ubuntu(64 bit) and click Next.



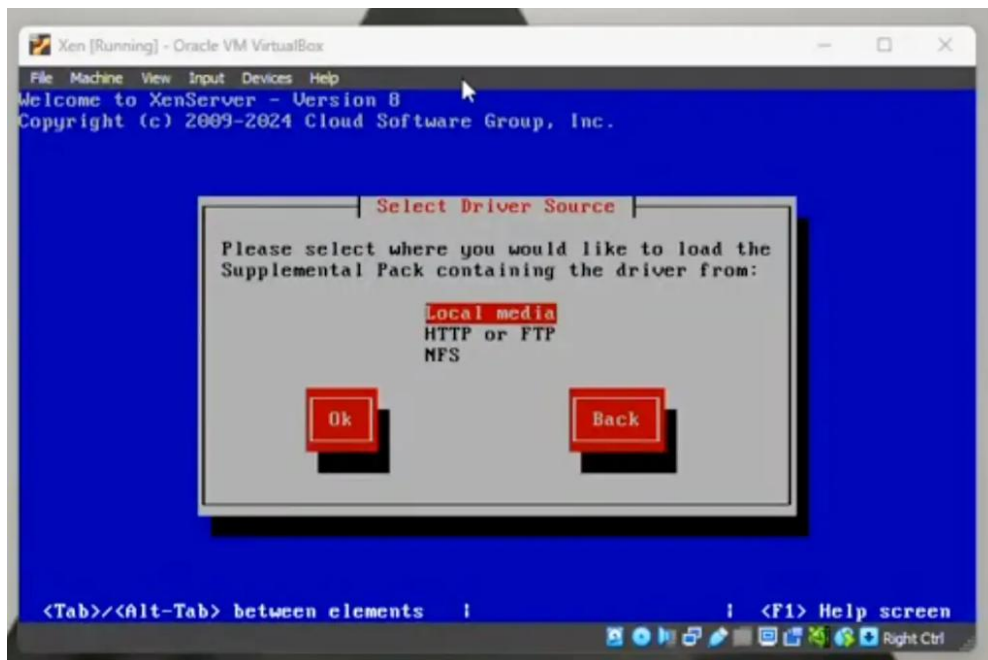
Step 4 : Adjust your base memory and processor.



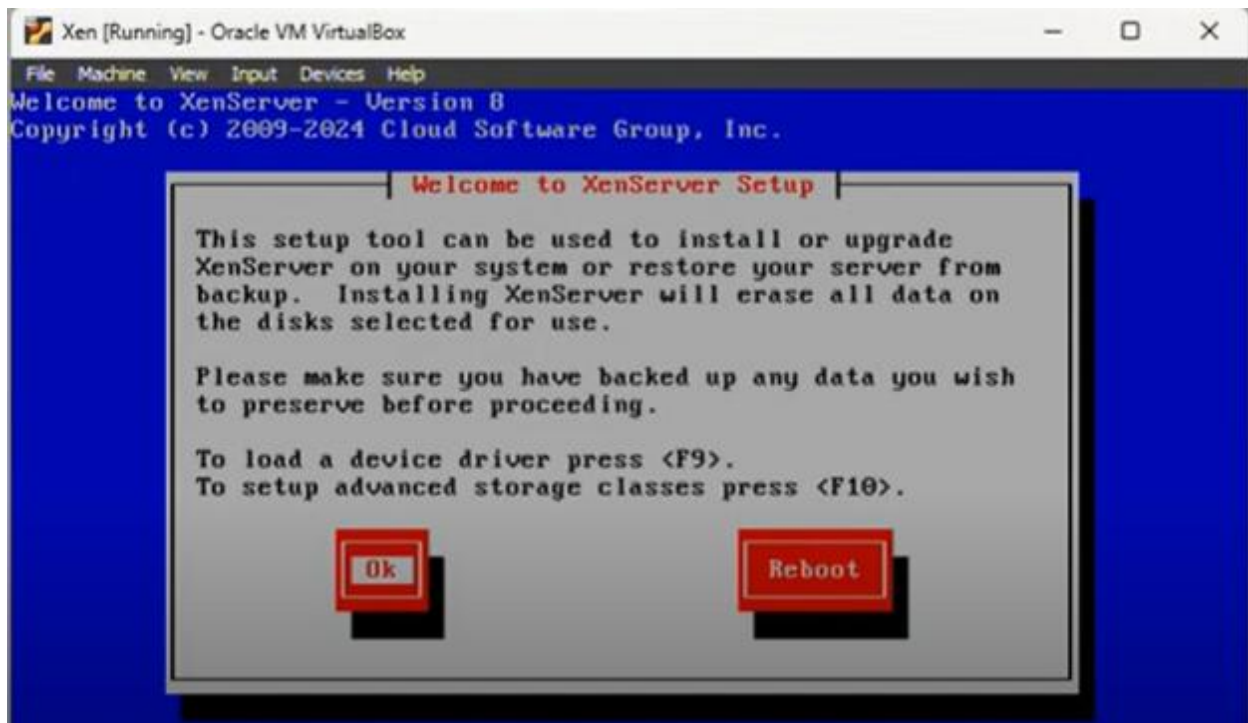
Step 5 : Adjust Virtual hard disk space and click next.



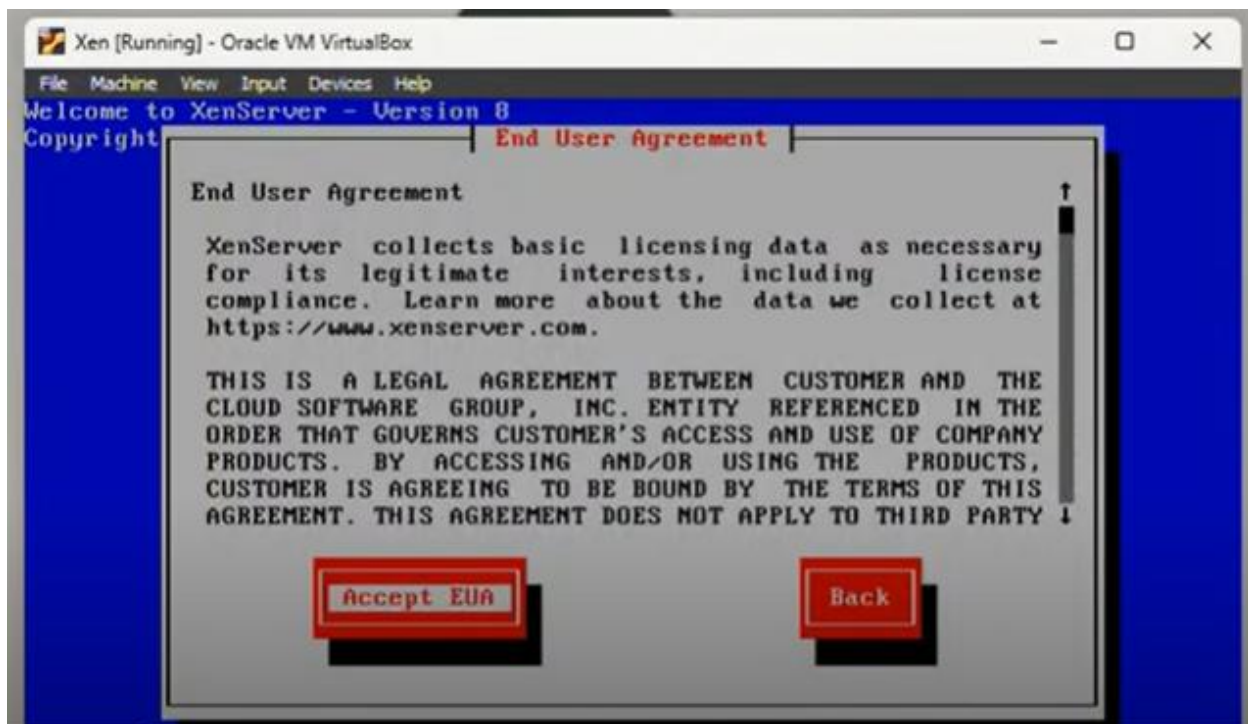
Step 6 : Select Back



Step 7 : Click ok.



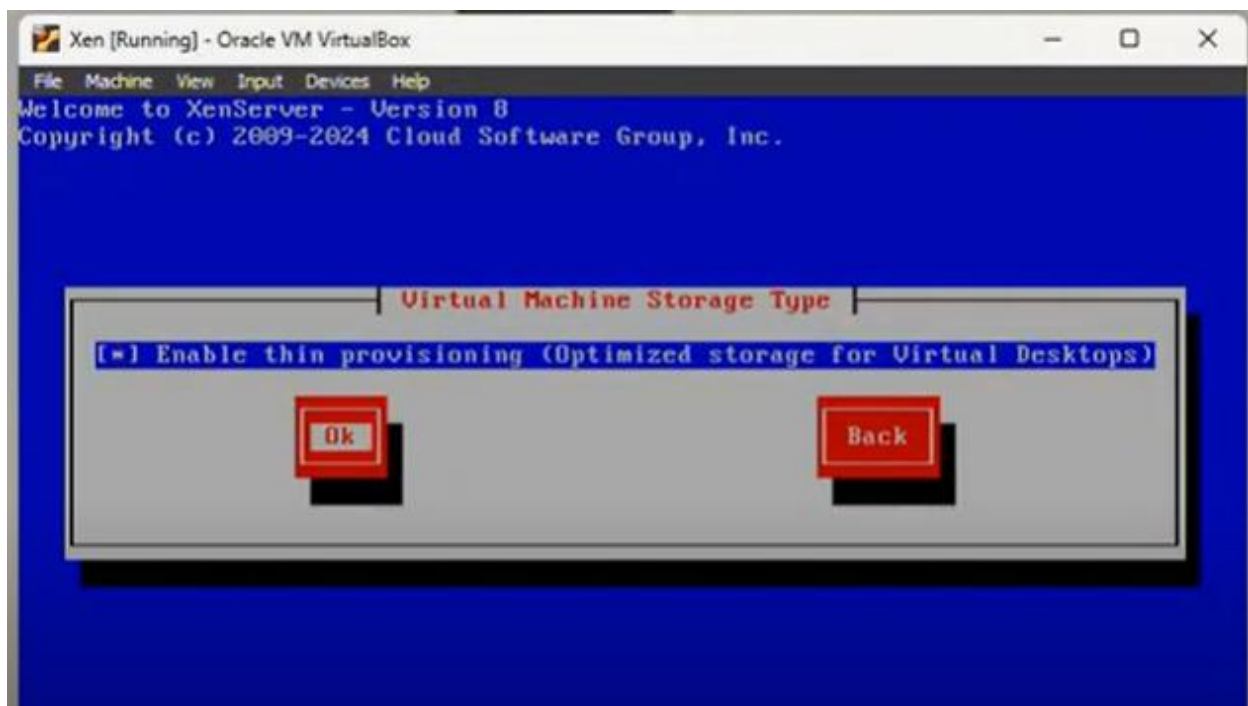
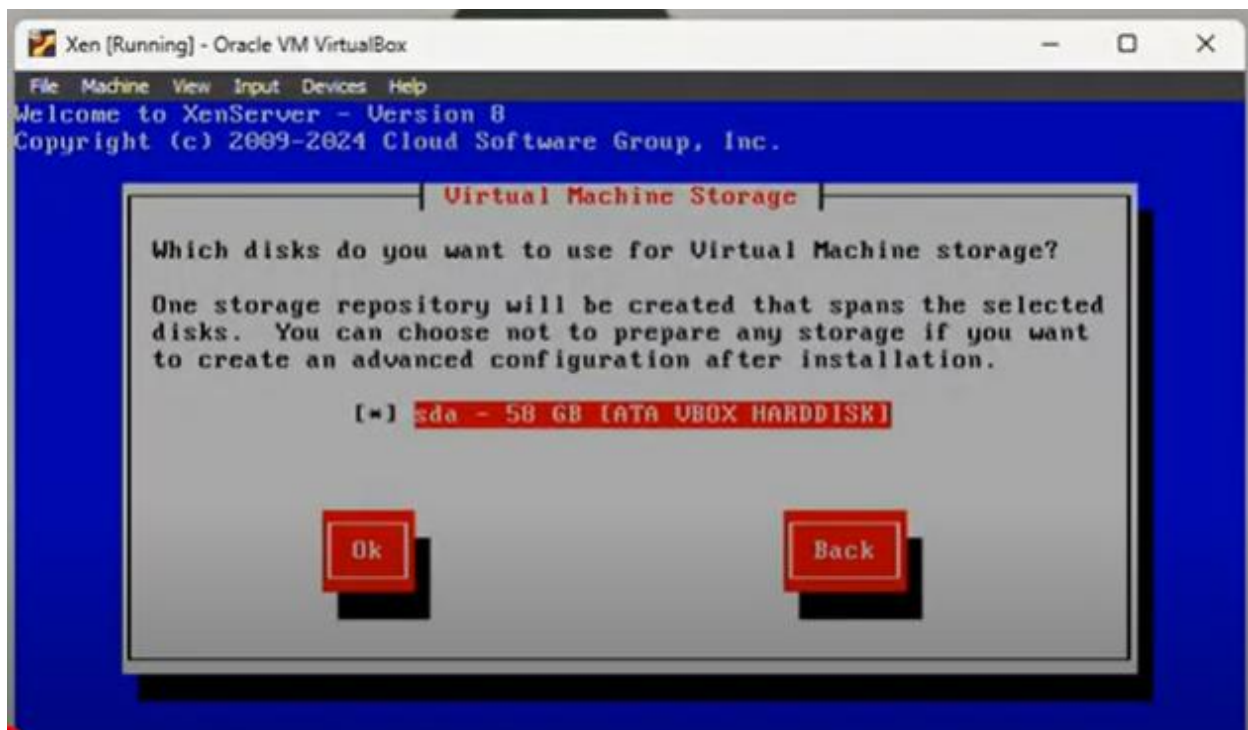
Step 8 : Click Accept



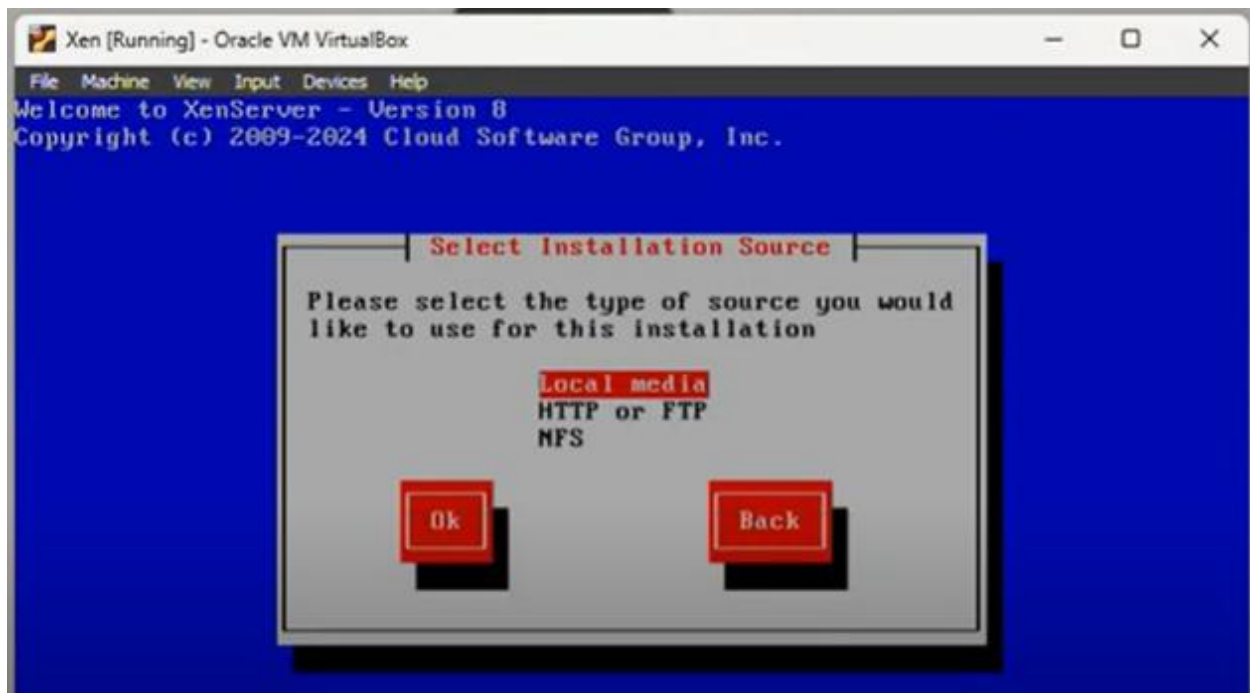
Step 9 : Click ok



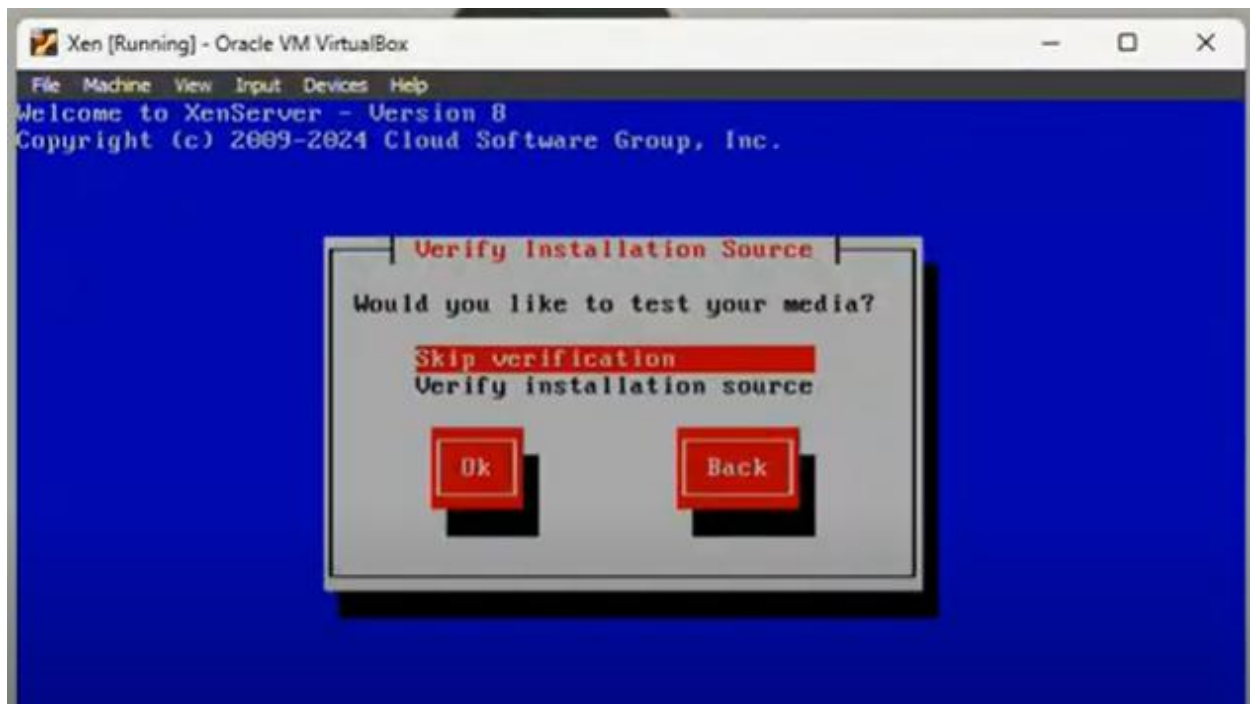
Step 10 : Click on space bar and then click ok



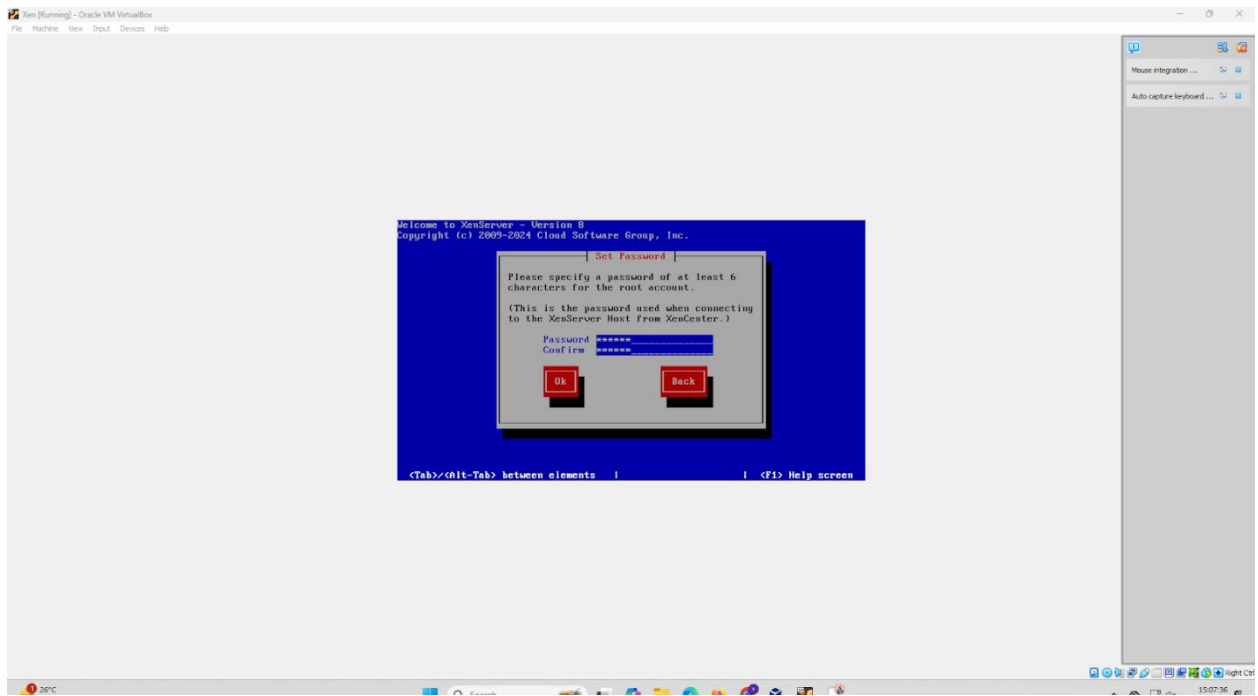
Step 11 : Select local media .



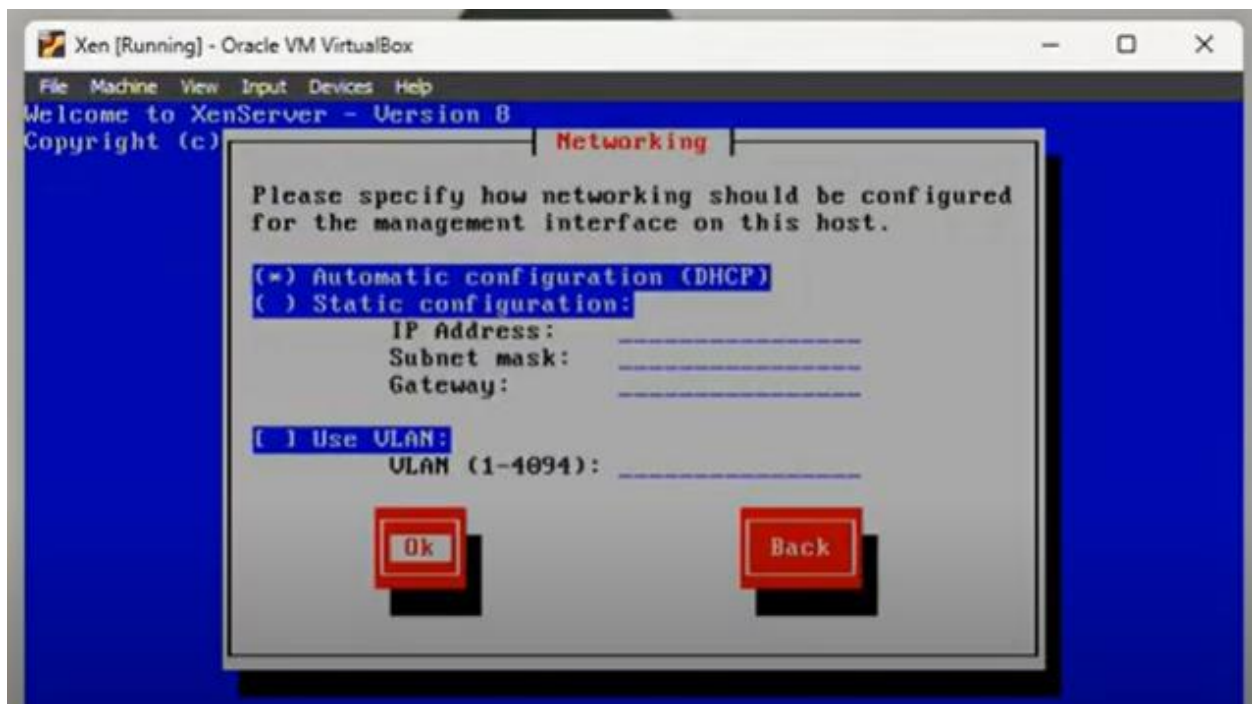
Step 12: Select skip verification

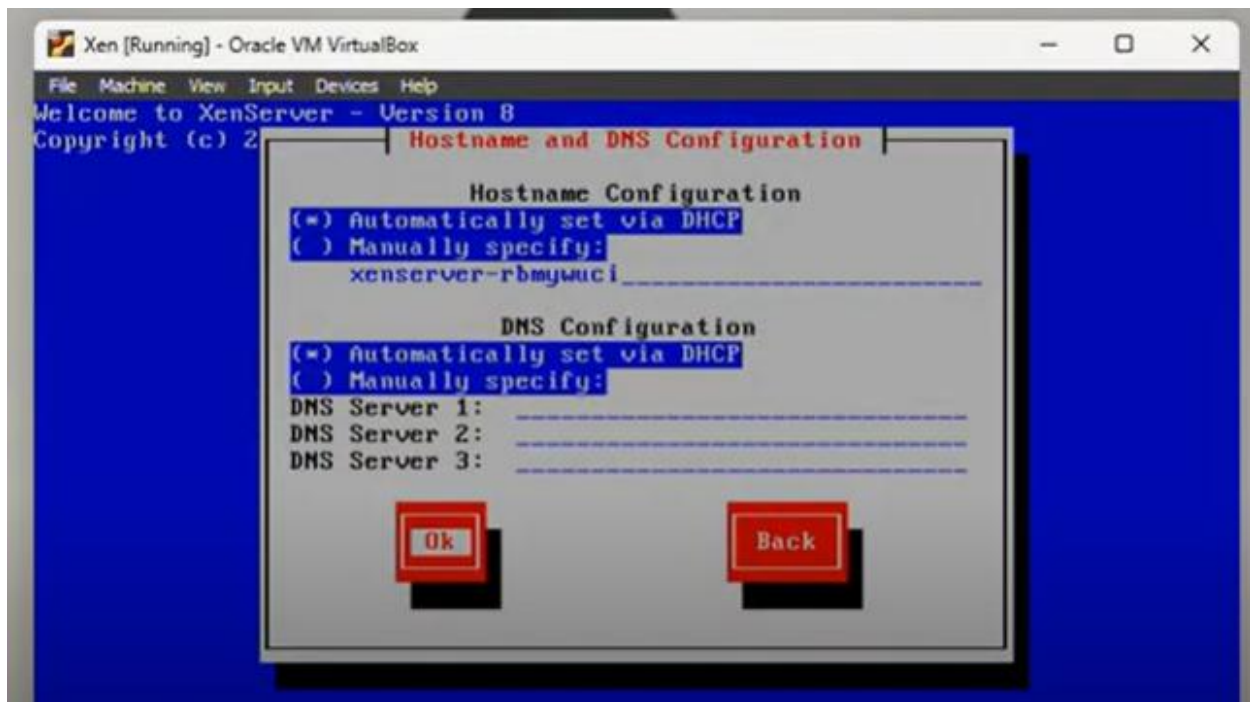


Step 13: Enter password and click ok.

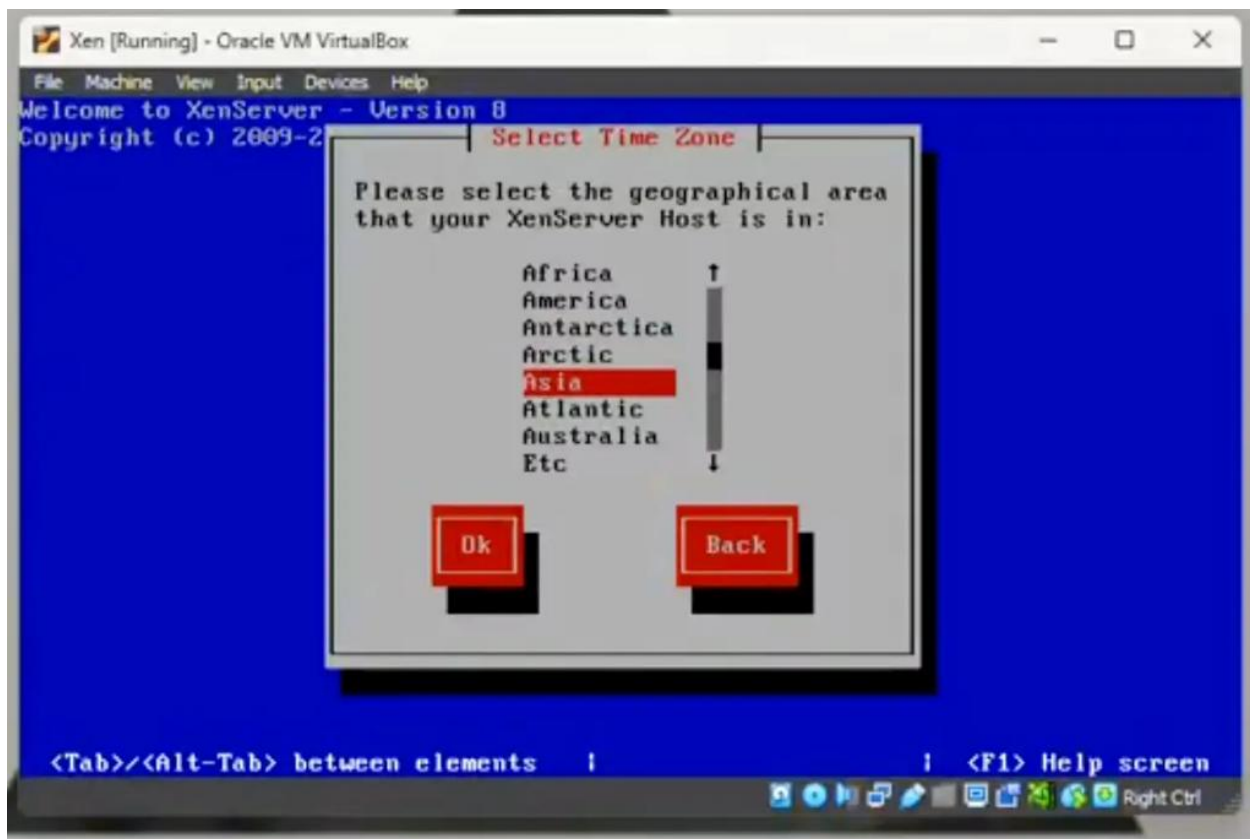


Step 14: Select automatic configuration and click ok





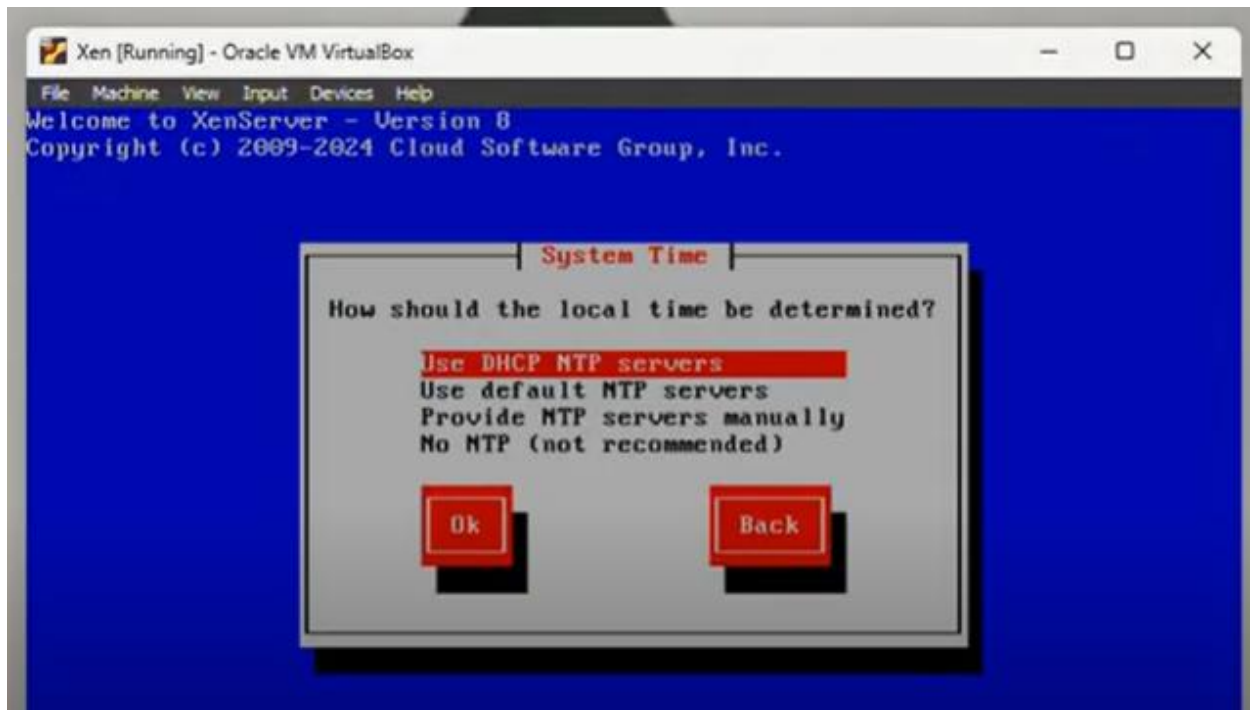
Step 15: Select Asia



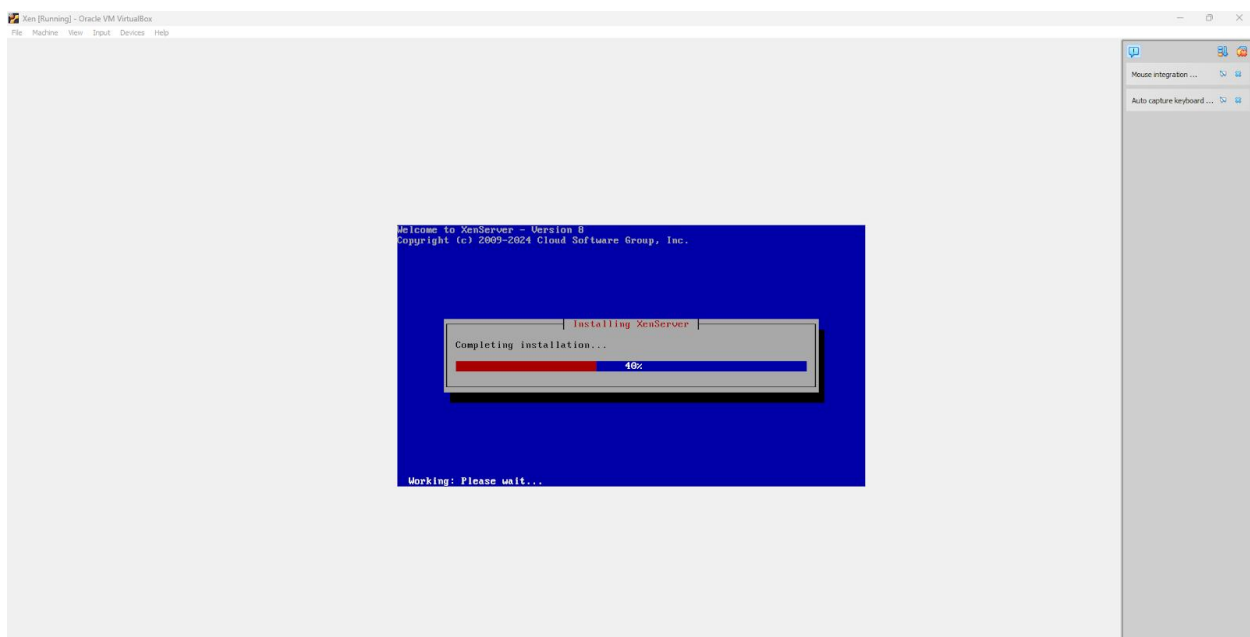
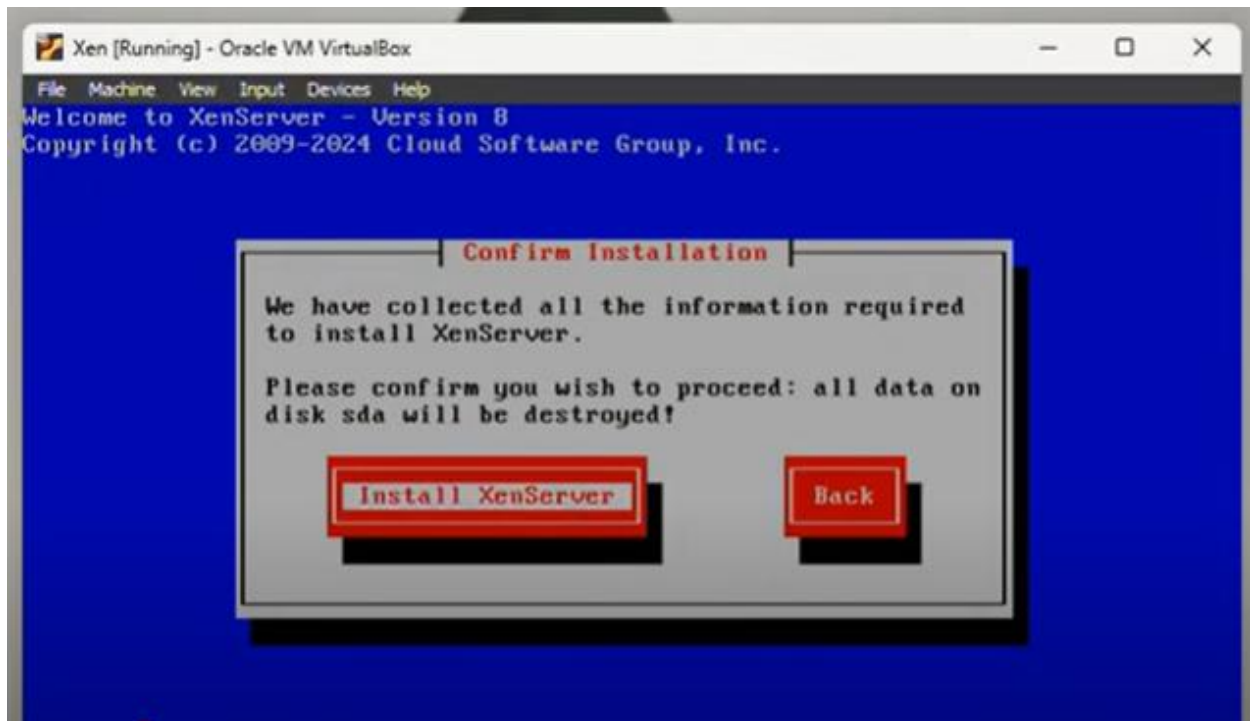
Step 16 : Select Kolkata

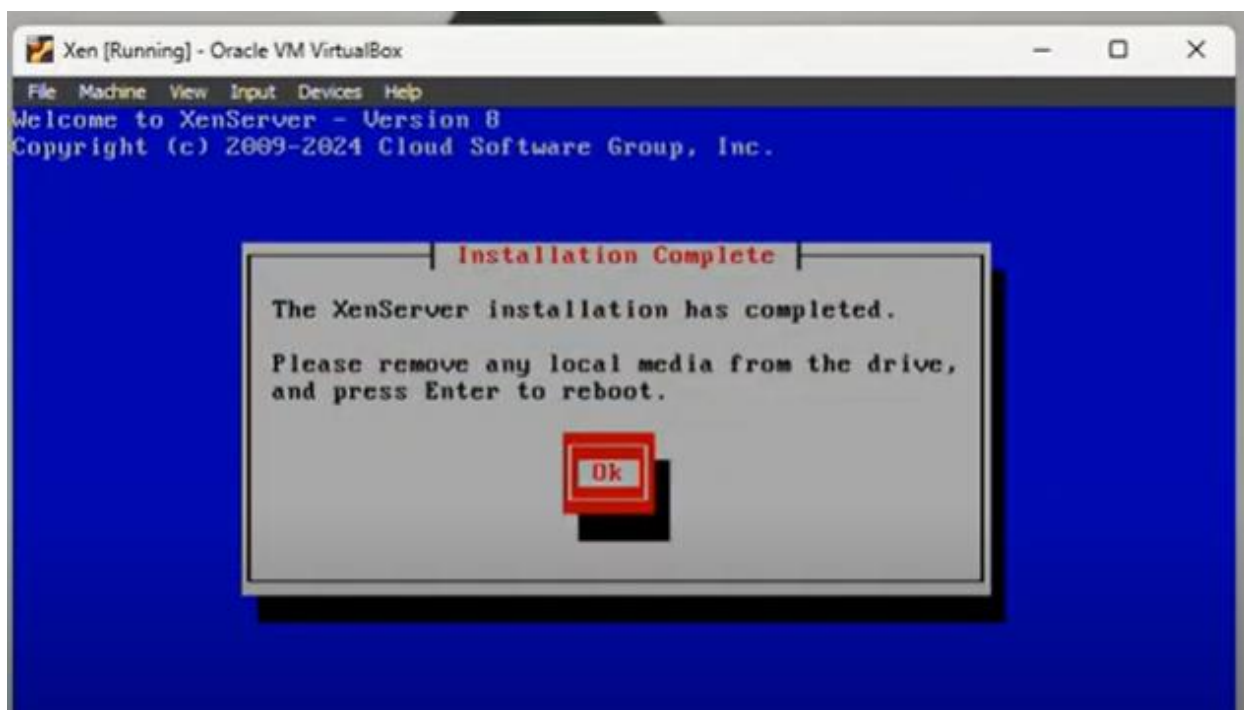
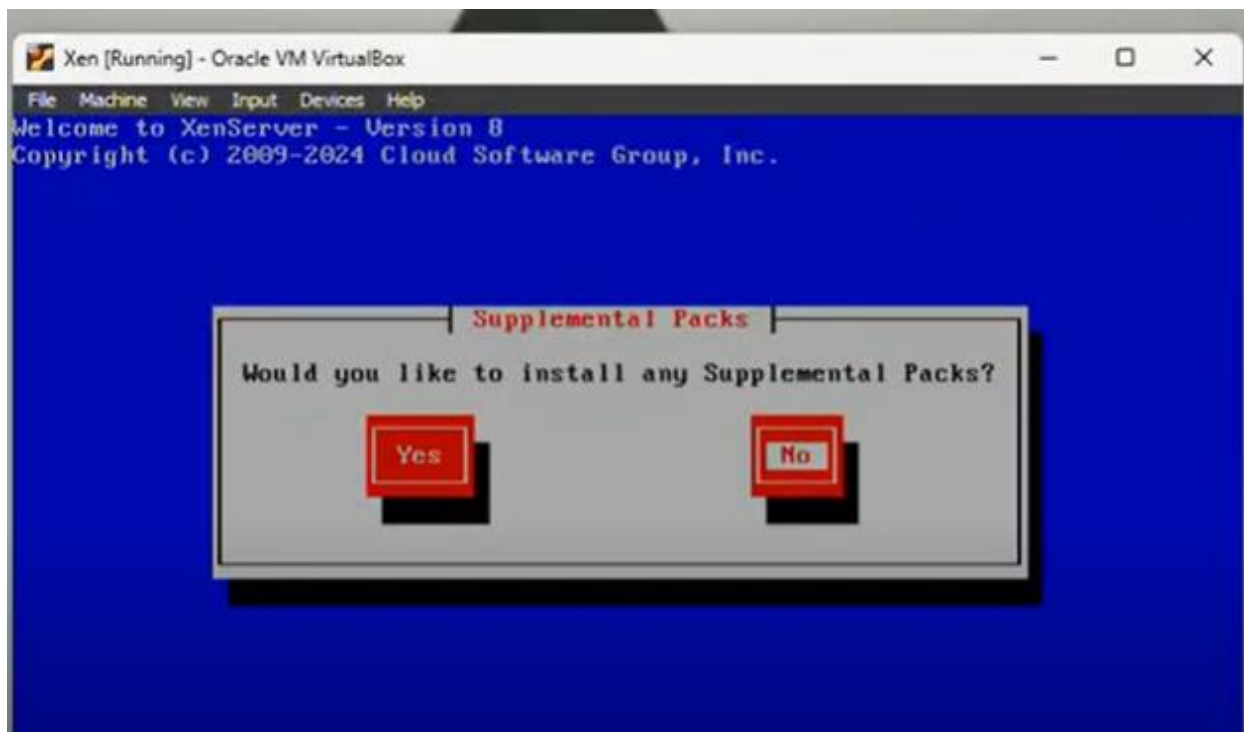


Step 17 : Select DHCP NTP



Step 18 : Click to install XenServer





```
File /usr/lib/modules/4.18.0-345.el8.x86_64/modules.order: No such file or directory
[FAILED] Failed to start Load Kernel Modules.
See 'systemctl status systemd-modules-load.service' for details.
[ OK ] Started Journal Service.
    Starting Apply Kernel Variables...
    Starting Create Static Device Nodes in /dev...
[ OK ] Started Apply Kernel Variables.
[ OK ] Started Create Static Device Nodes in /dev.
[ OK ] Started Setup Virtual Console.
[ OK ] Started dracut cmdline hook.
    Starting dracut pre-udev hook...
[ OK ] Started dracut pre-udev hook.
    Starting udev Kernel Device Manager...
[ OK ] Started udev Kernel Device Manager.
    Starting udev Coldplug all Devices...
    Mounting Configuration File System...
[ OK ] Mounted Configuration File System.
[ OK ] Started udev Coldplug all Devices.
    Starting Show Plymouth Boot Screen...
    Starting dracut initqueue hook...
[ OK ] Reached target System Initialization.
```

XenServer 8

15:18:24

localhost.localdomain

Configuration

Customize System

Status Display

Network and Management Interface
Authentication
Virtual Machines
Disks and Storage Repositories
Resource Pool Configuration
Hardware and BIOS Information
Keyboard and Timezone
Remote Service Configuration
Backup, Restore and Update
Technical Support
Reboot or Shutdown
Local Command Shell

innotek GmbH
VirtualBox

XenServer 8

Management Network Parameters

Device	eth0
IP address	10.0.2.15
Netmask	255.255.255.0
Gateway	10.0.2.2

Press <Enter> to display the SSL key
fingerprints for this host

<Enter> OK <Up/Down> Select

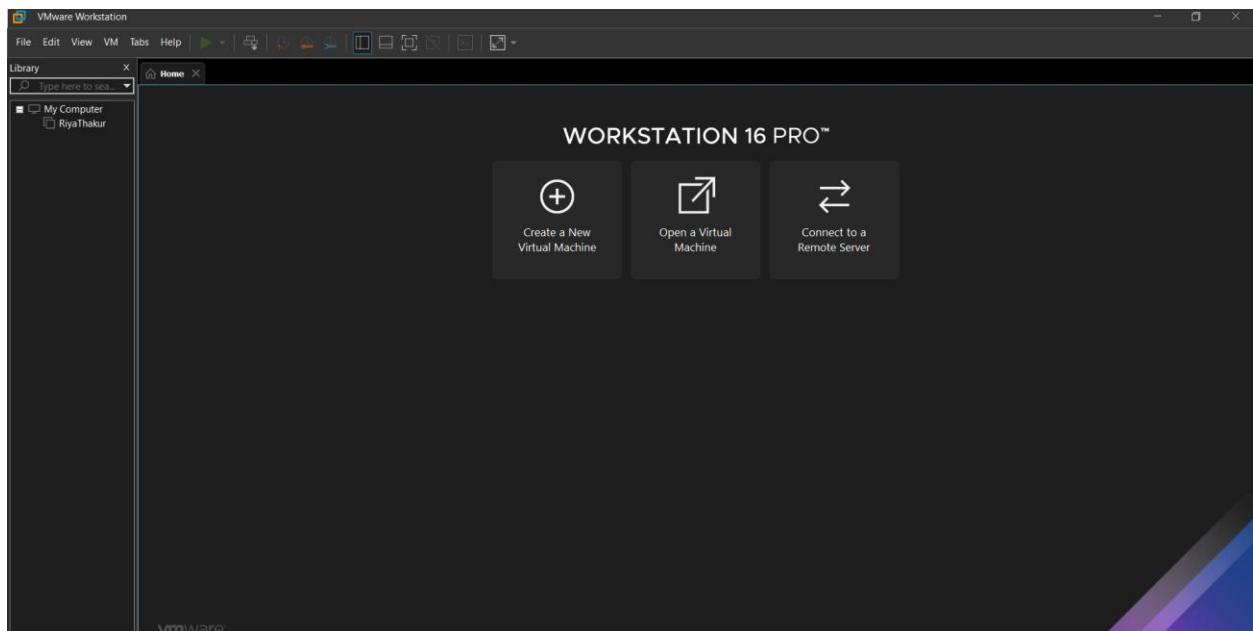
<Enter> Fingerprints <F5> Refresh

Practical 07

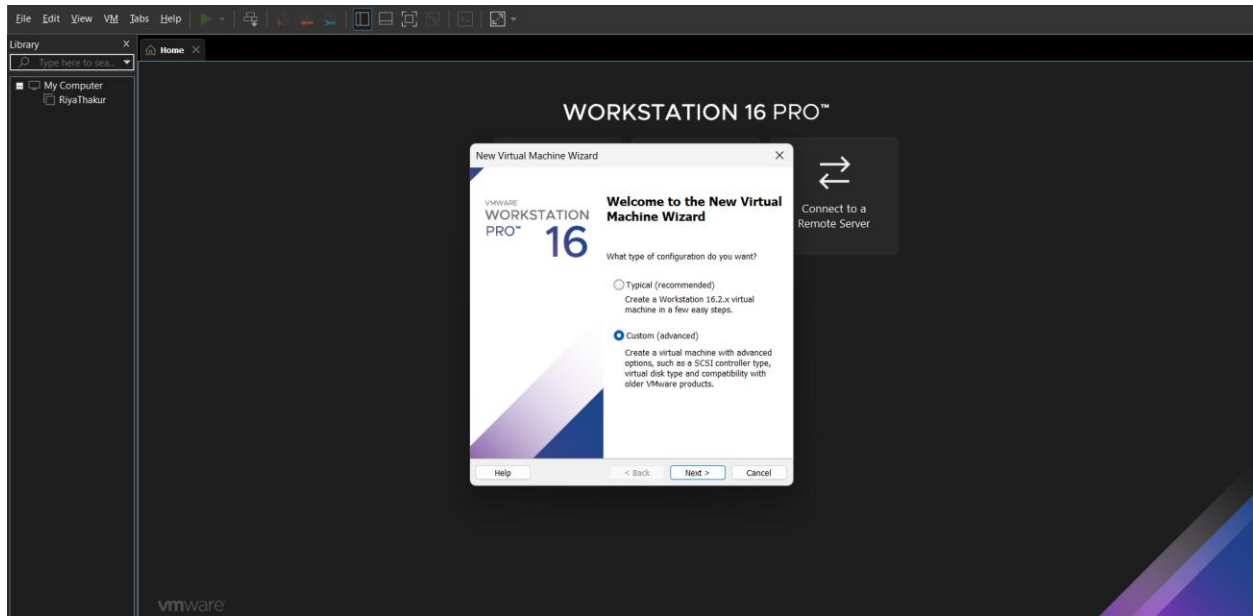
Aim: Implement virtualization using VMWare ESXi Server and managing with vCenter.

Steps:

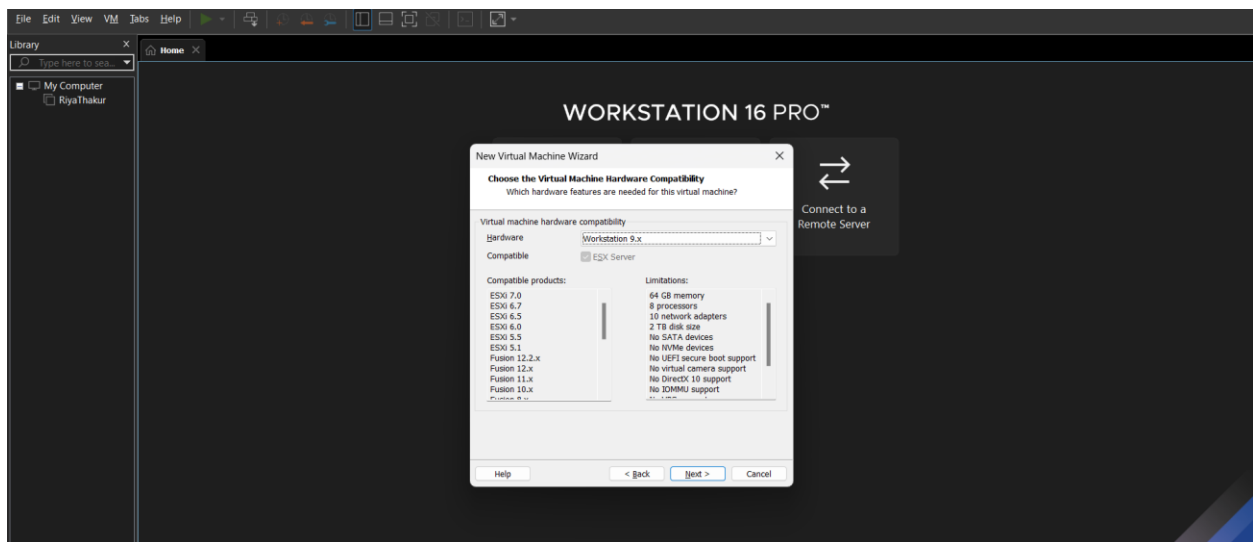
Step 1: Open VM Workstation 16 Pro



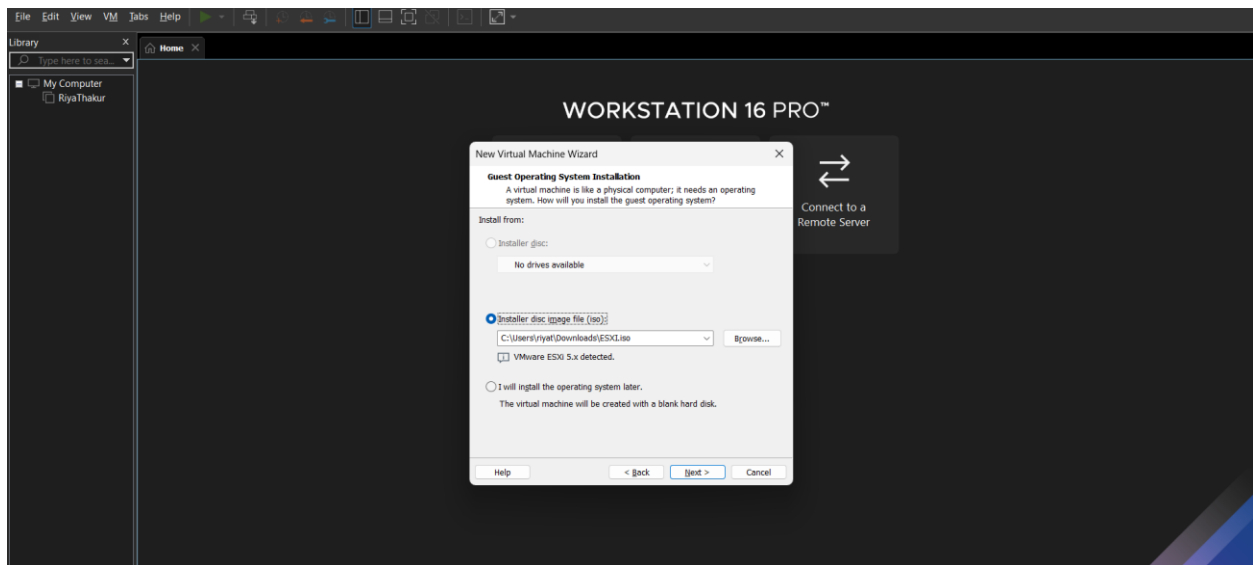
Step 2: Click Create new Virtual Machine and select custom and click next.



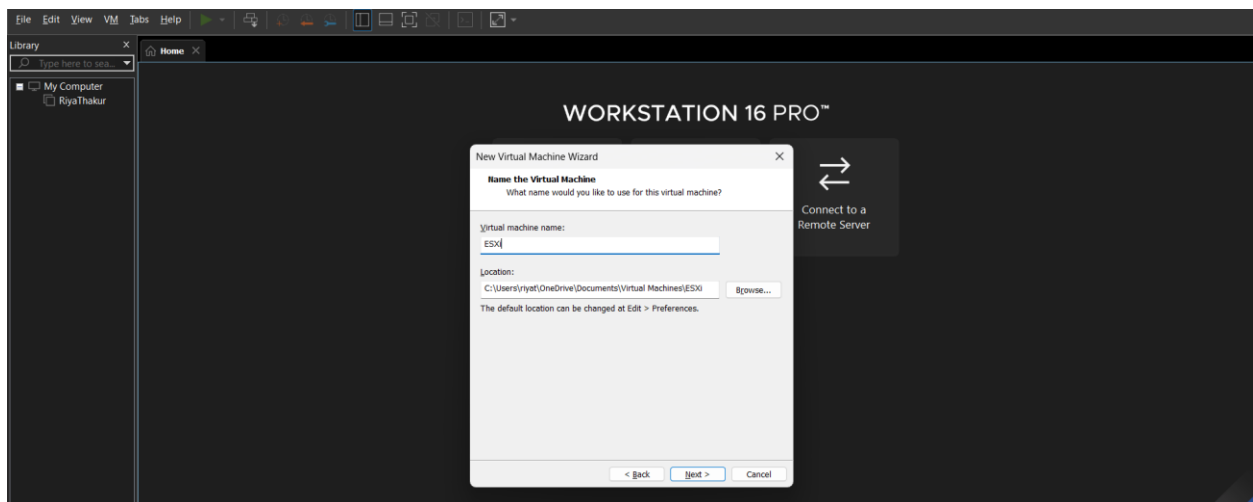
Step3: Select Workstation9.x and click next



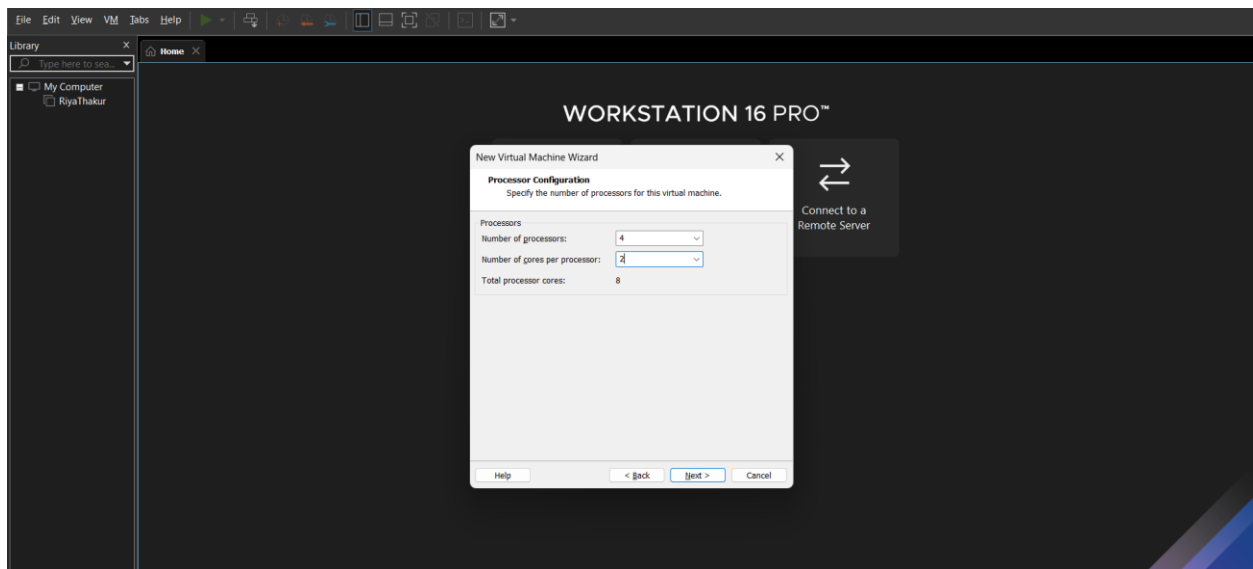
Step 4 : Browse ESXI.iso downloaded file , select it and click next



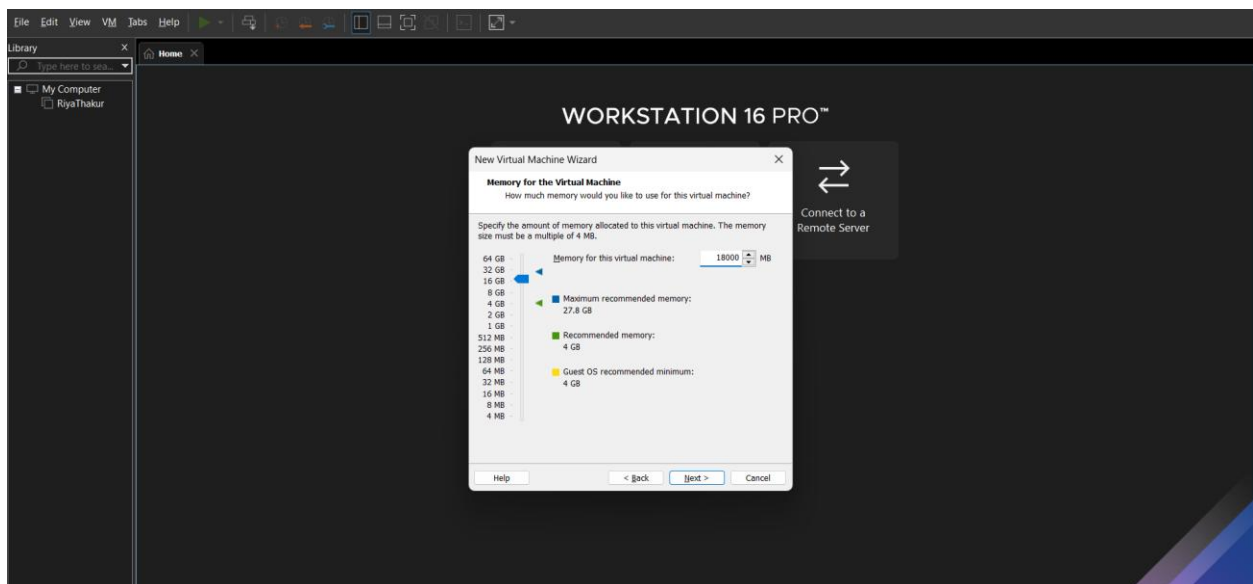
Step 5: Give your virtual machine a name and click next.



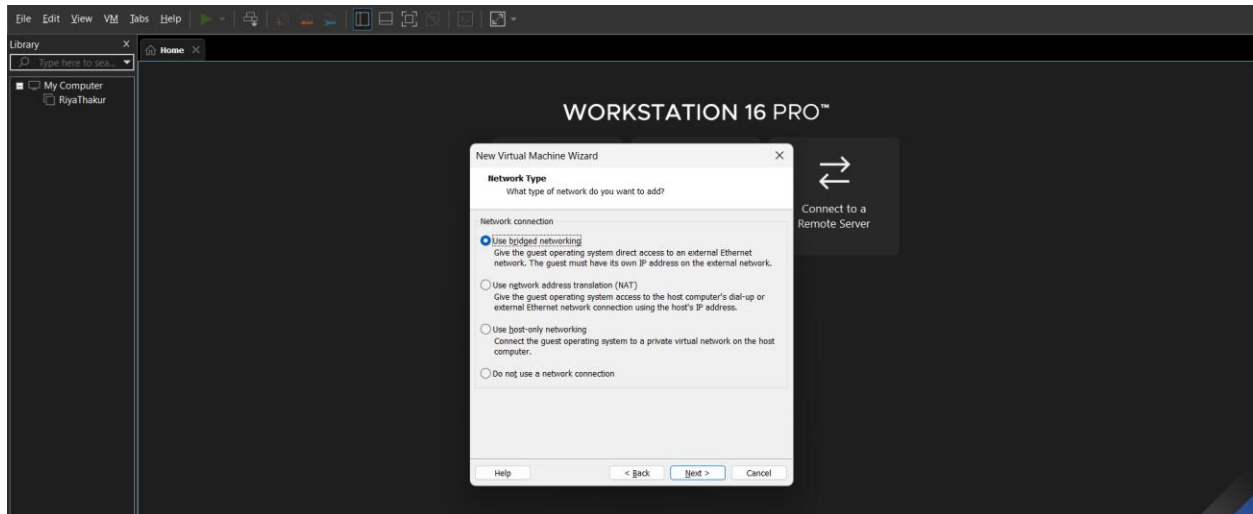
Step 6: Change Number of processor to 4 and Number of cores per processor to 2



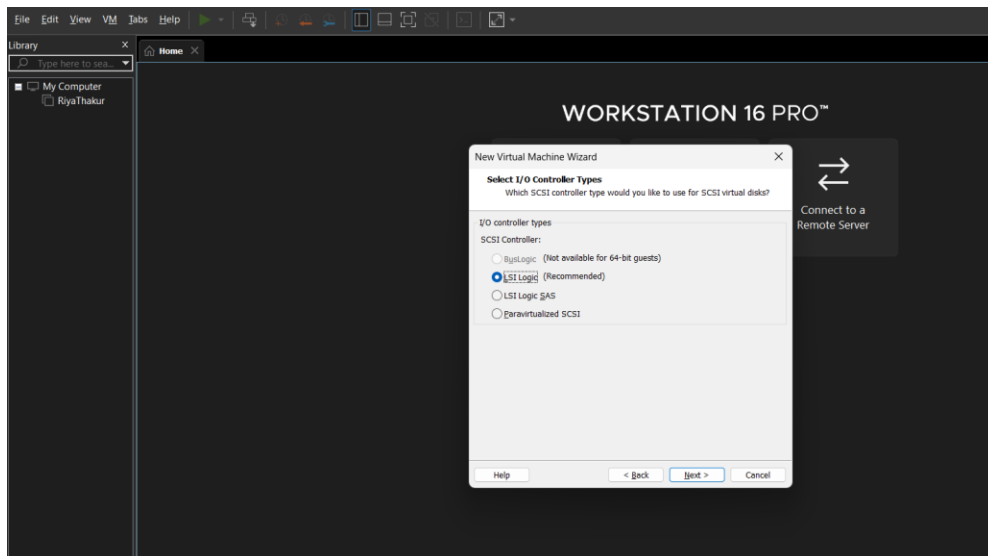
Step 7: Set memory for virtual machine as 18,000 Mb and click next.



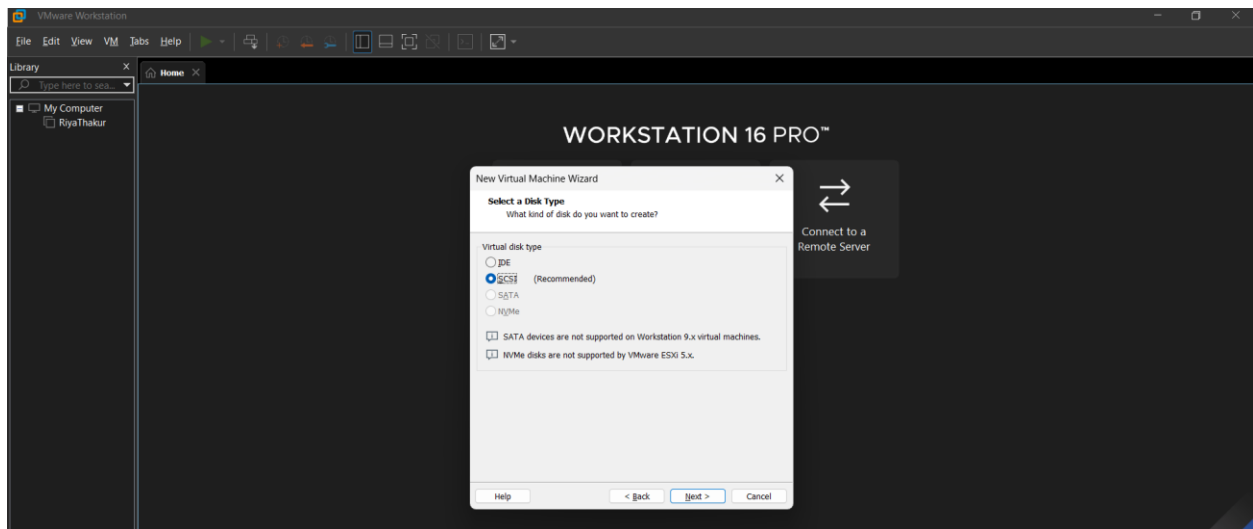
Step 8: Select Use bridged networking and click next



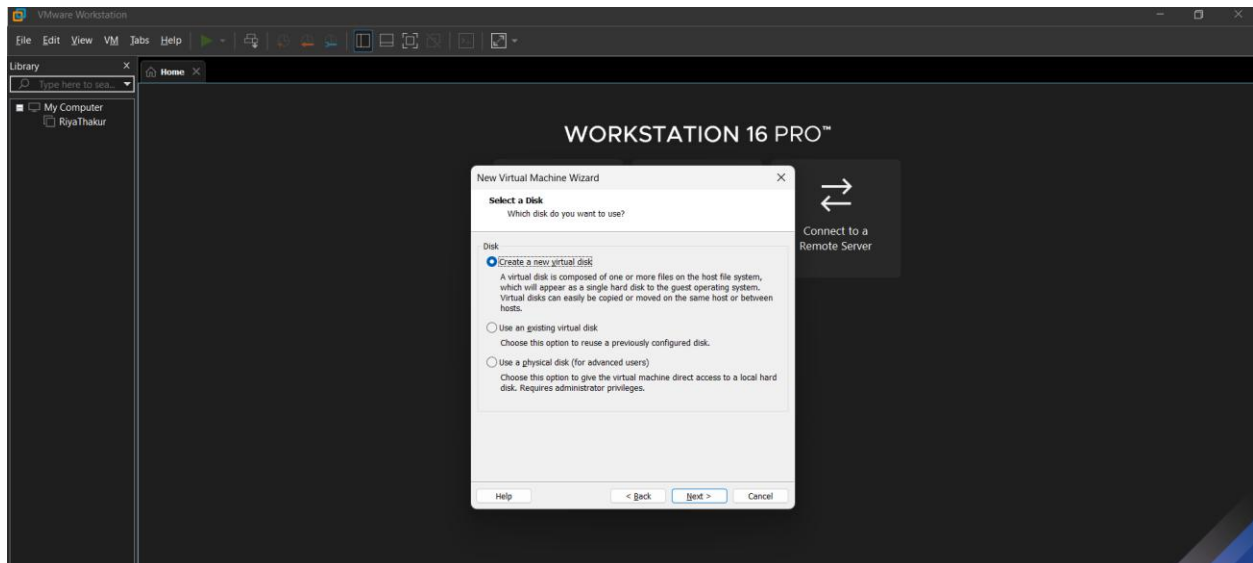
Step 9 : Select LSI Logic and click next.



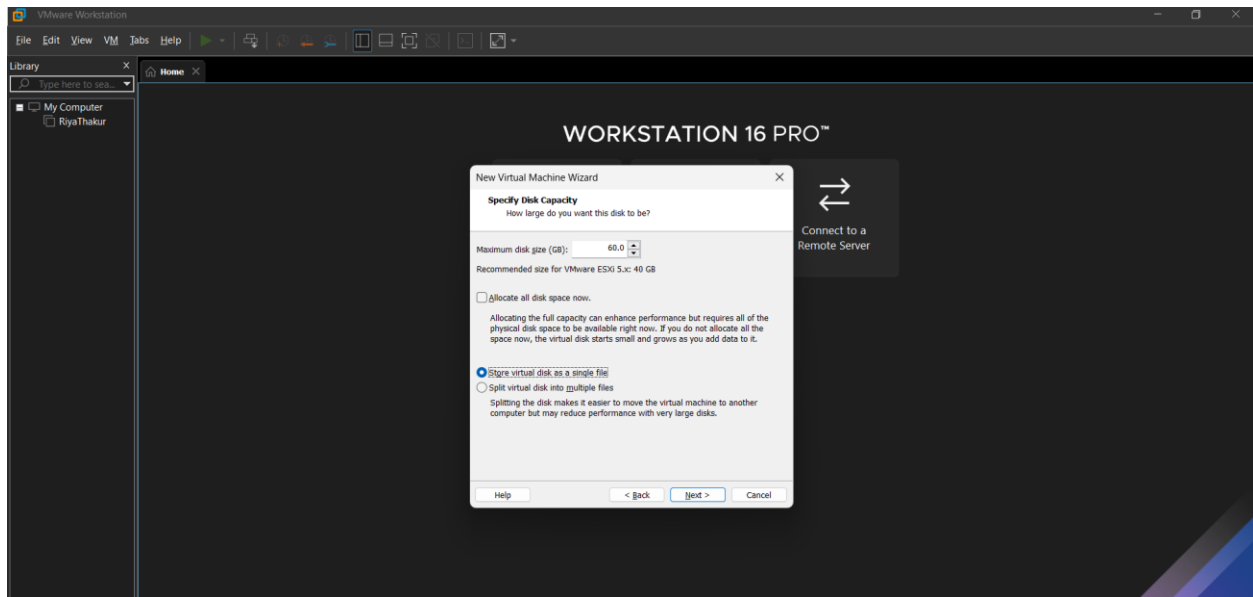
Step 10 : Select SCSI and click next



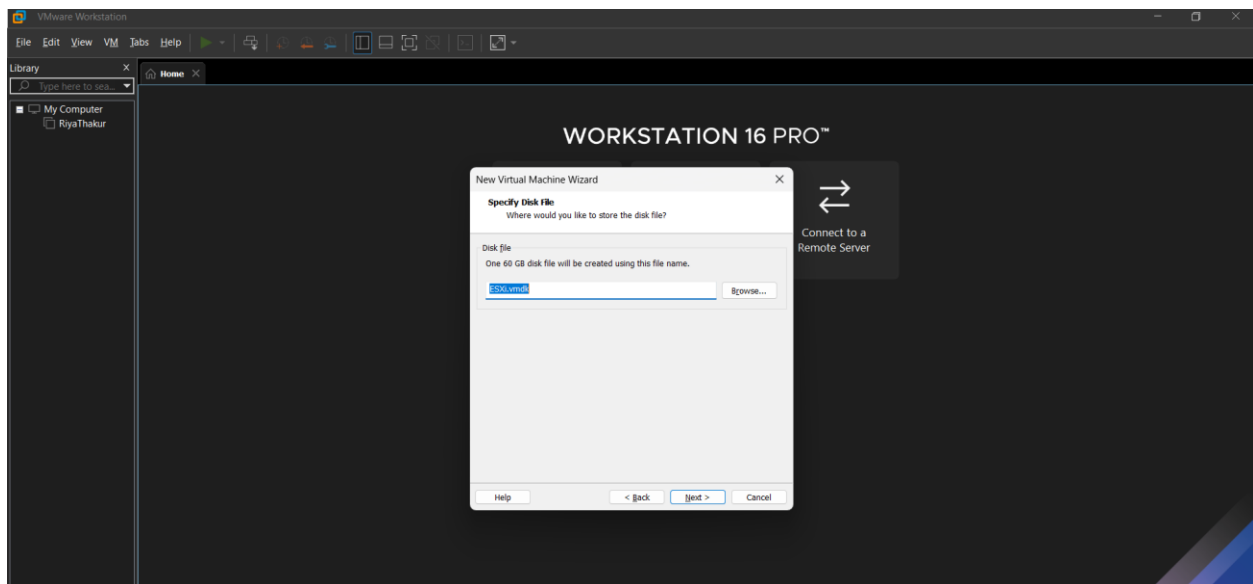
Step 11: Select create a new virtual disk and click next.



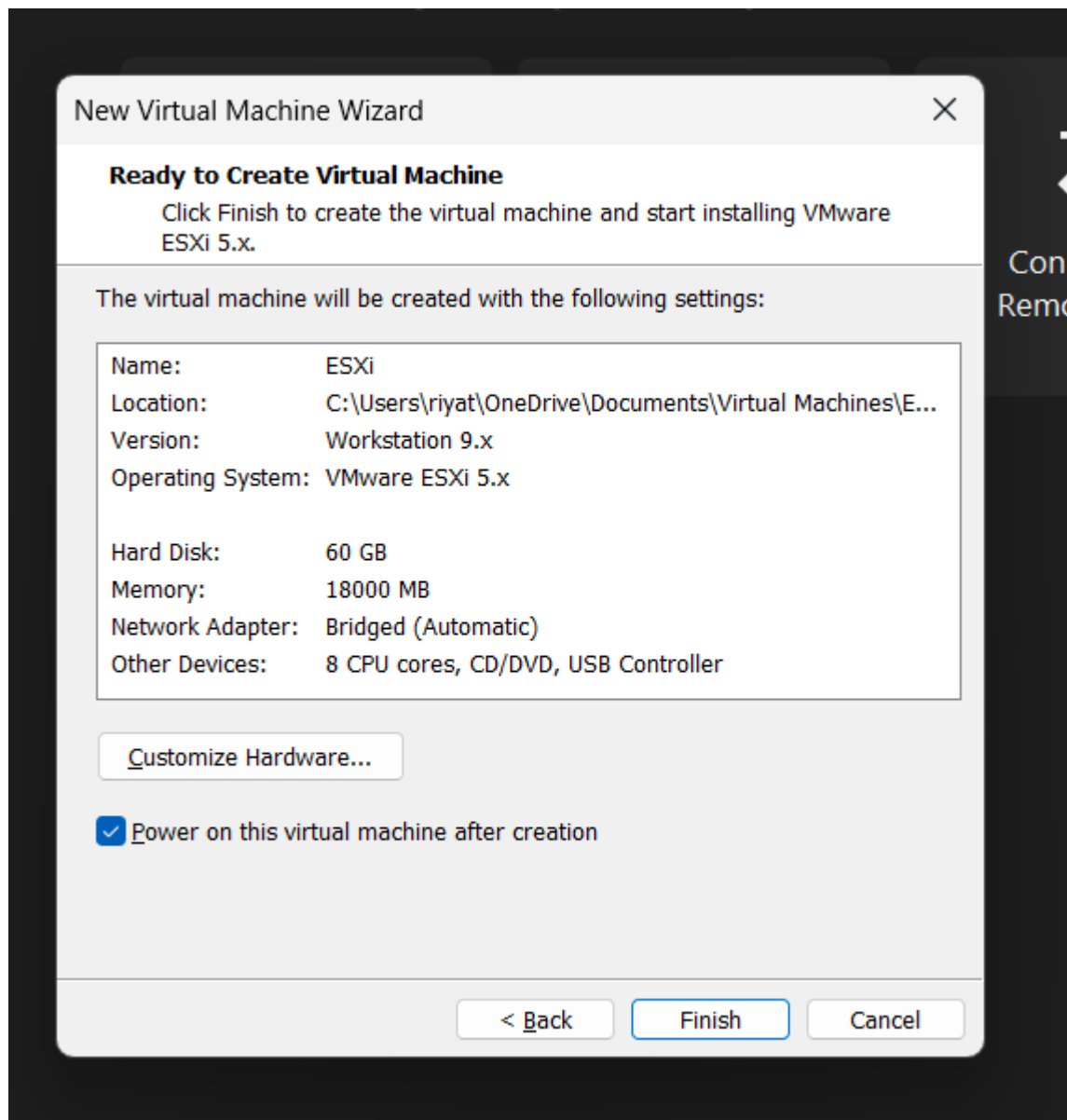
Step 12: Set maximum disk size and select store virtual disk as single file.



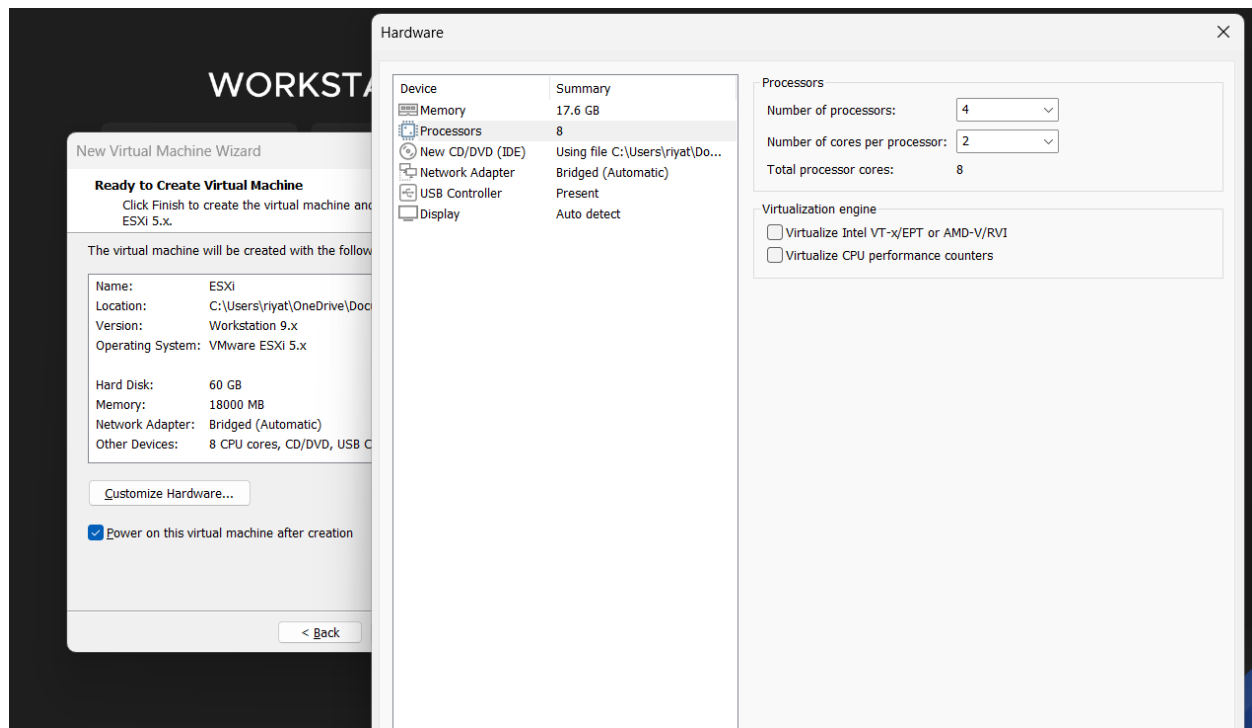
Step 13: Check the file name and click next.



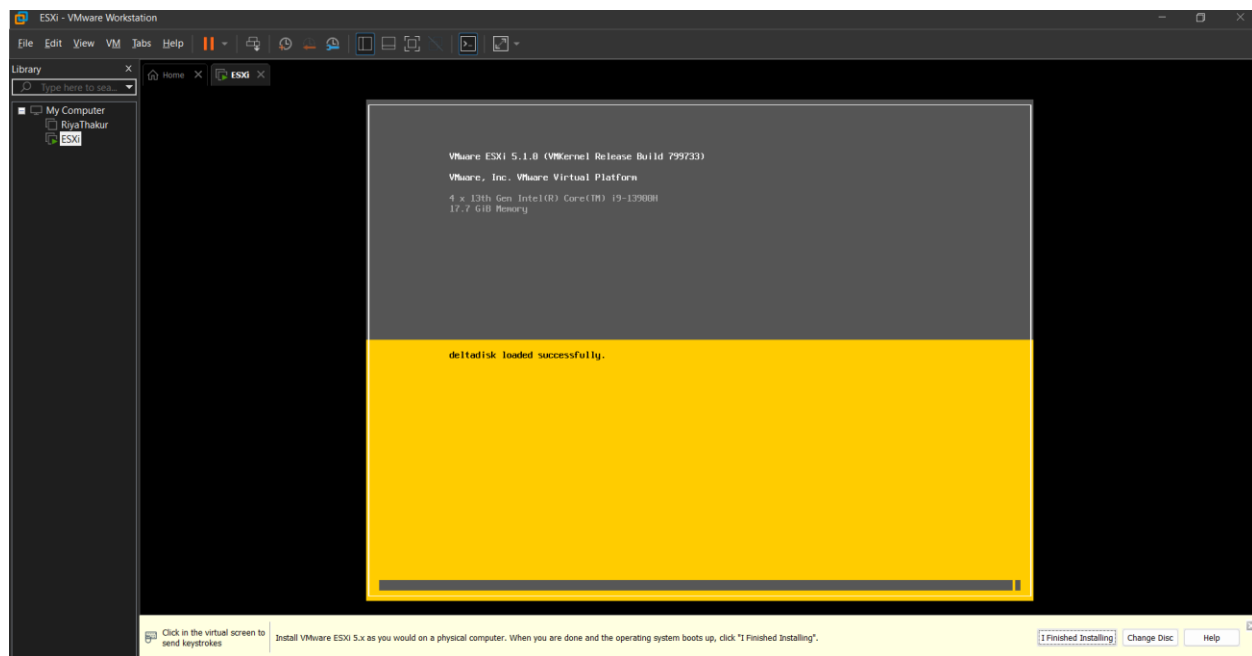
Step 14: Click customize hardware



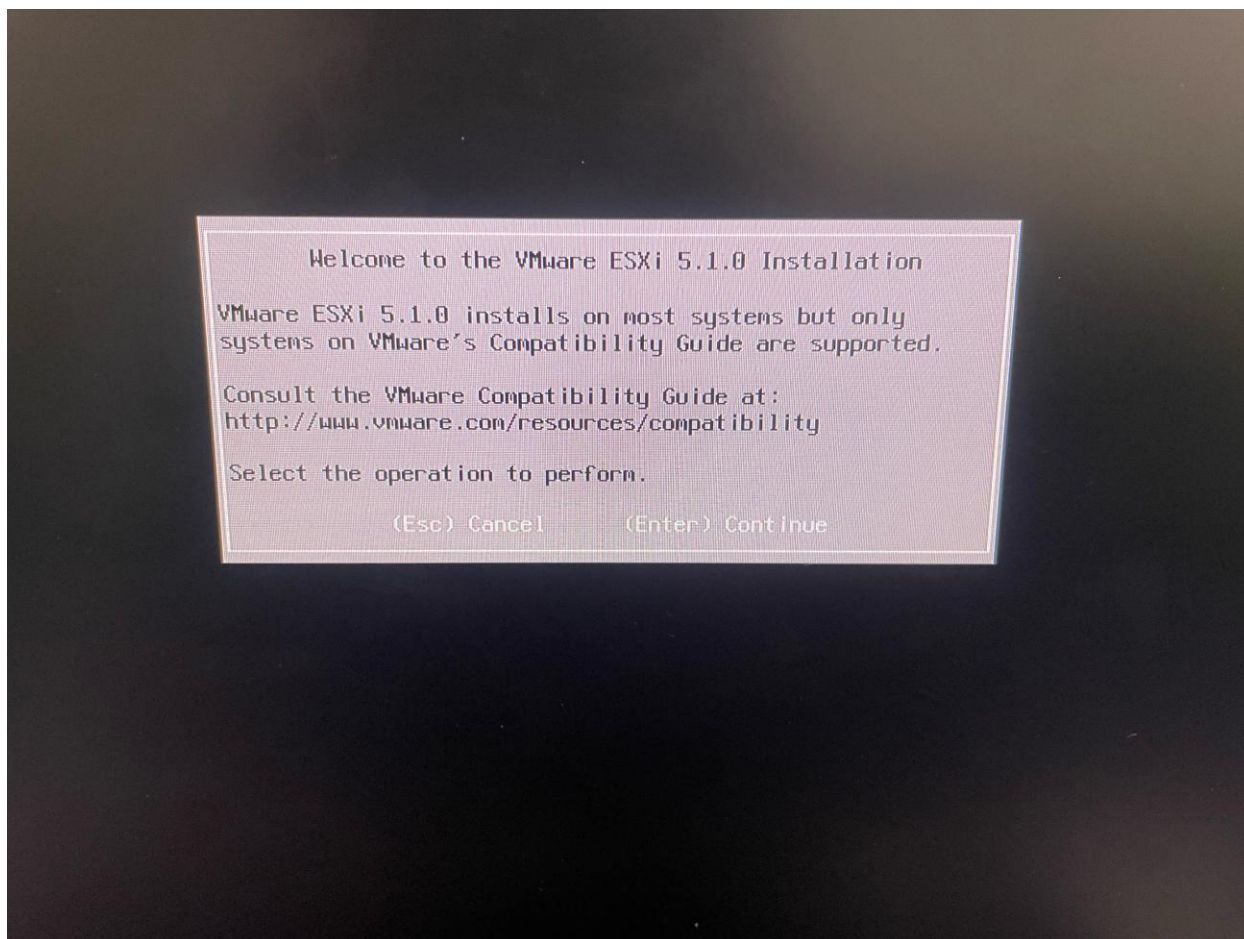
Step 15: Uncheck the Virtual Intel box

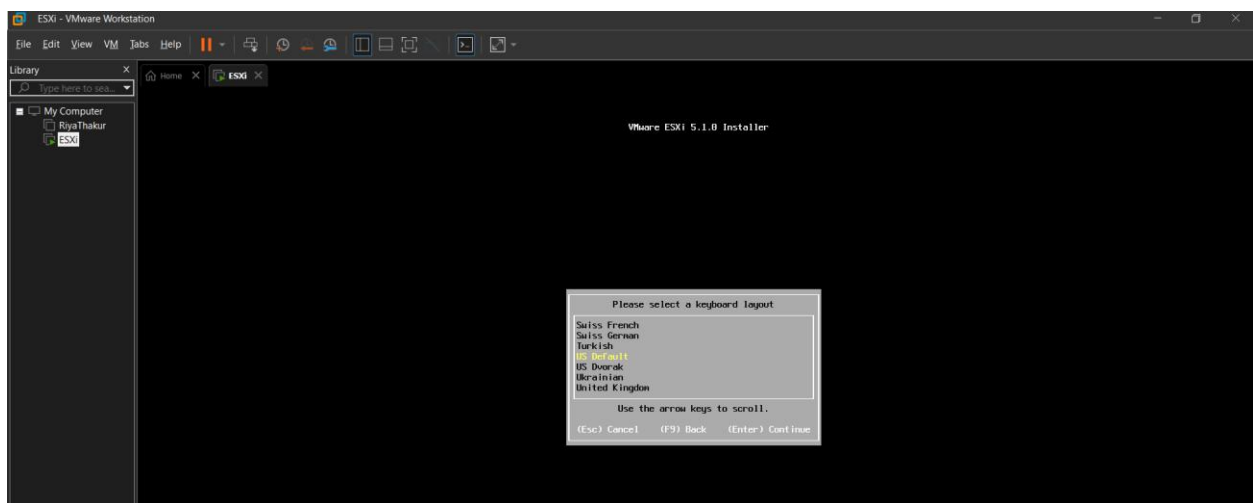
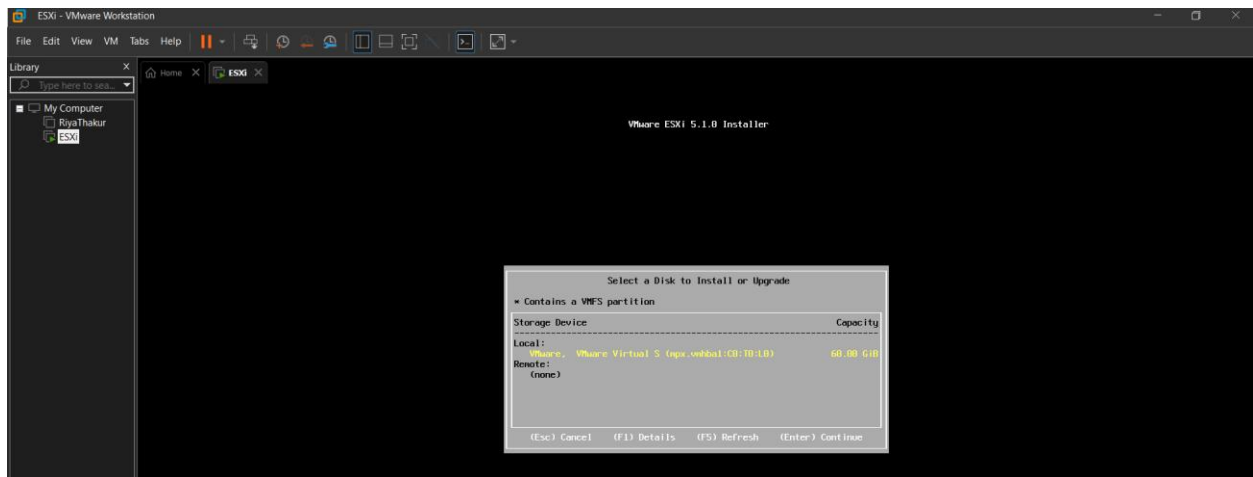
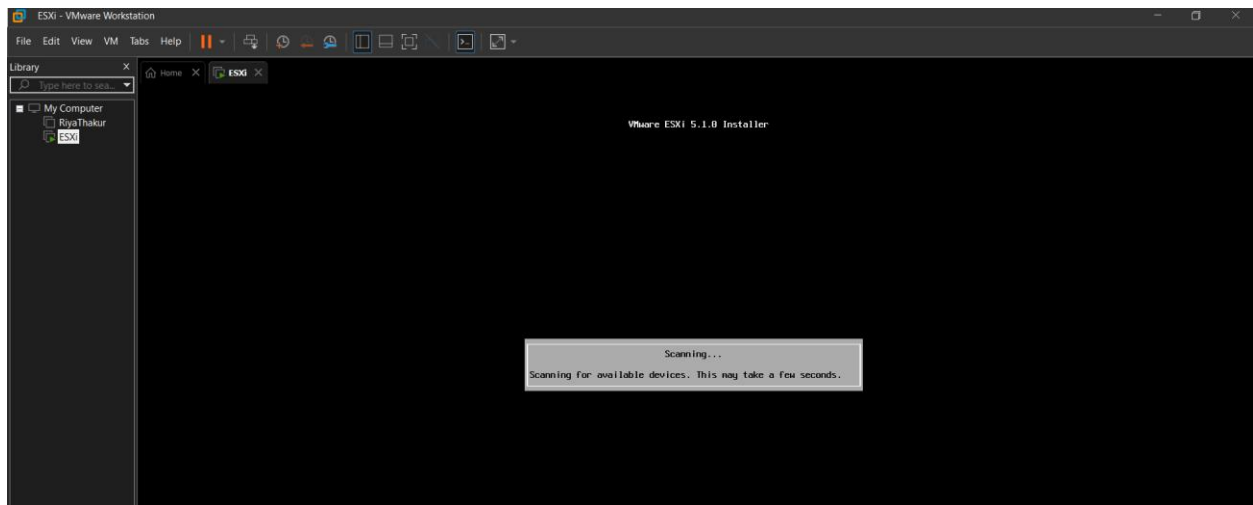


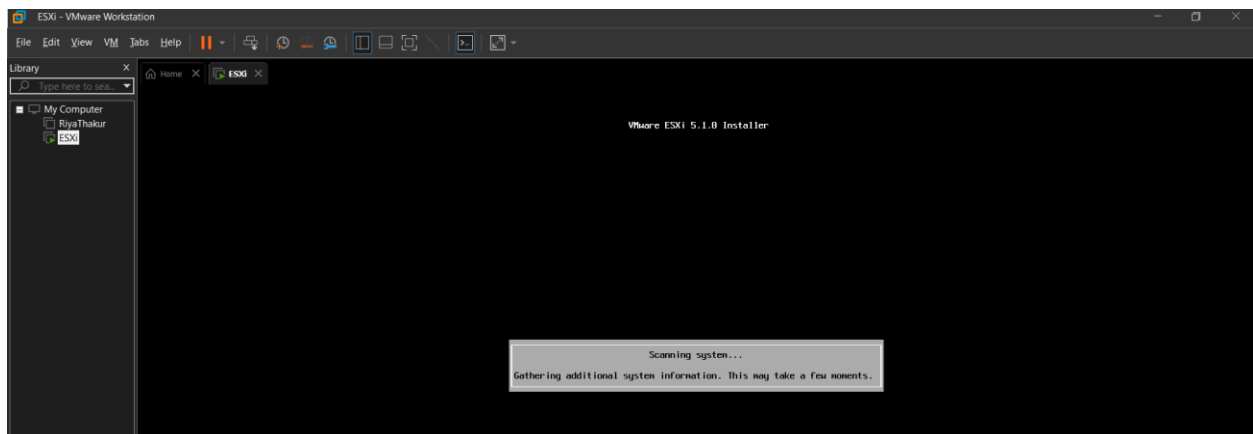
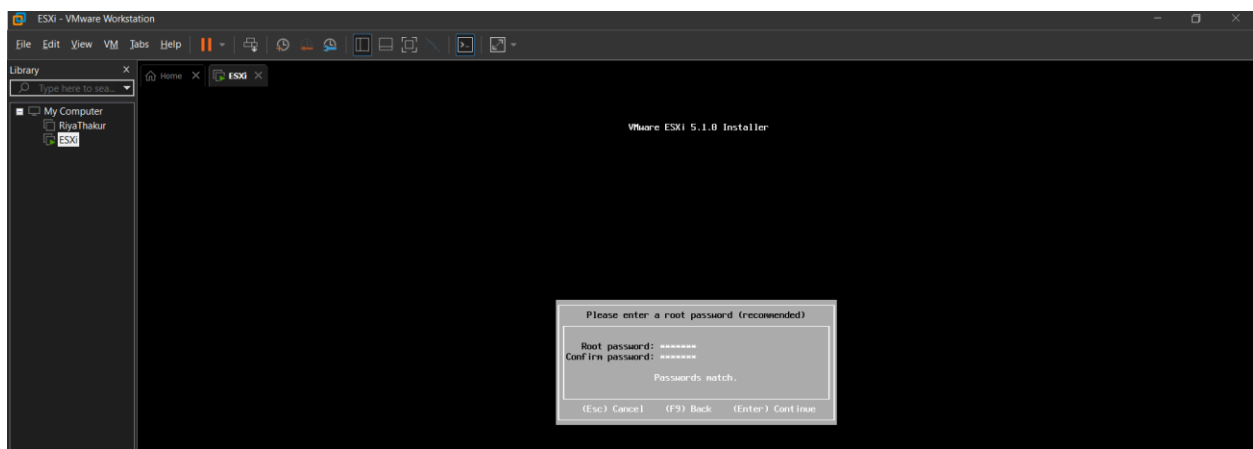
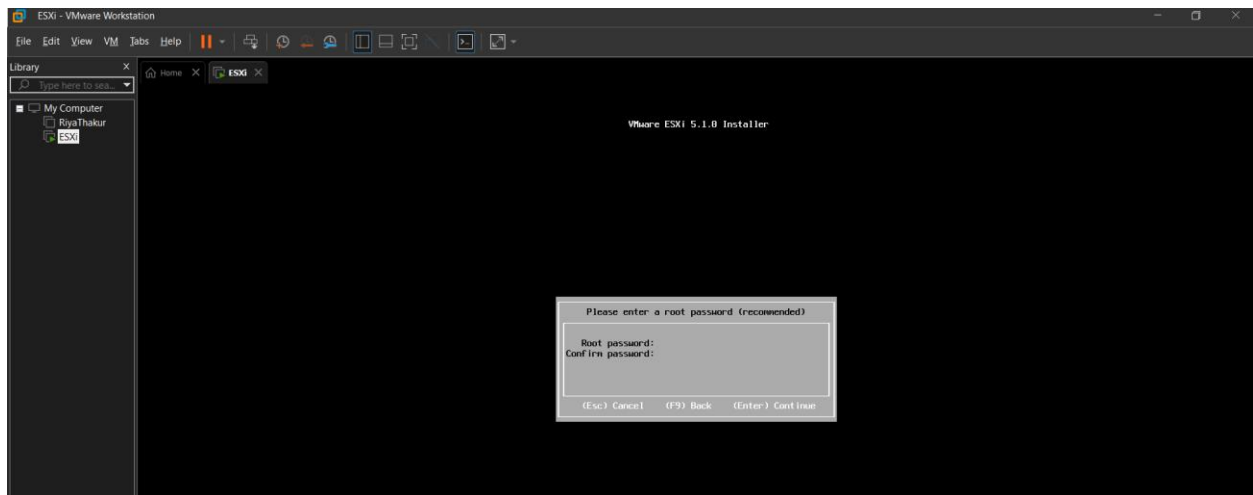
Step 16 : Start the virtual machine

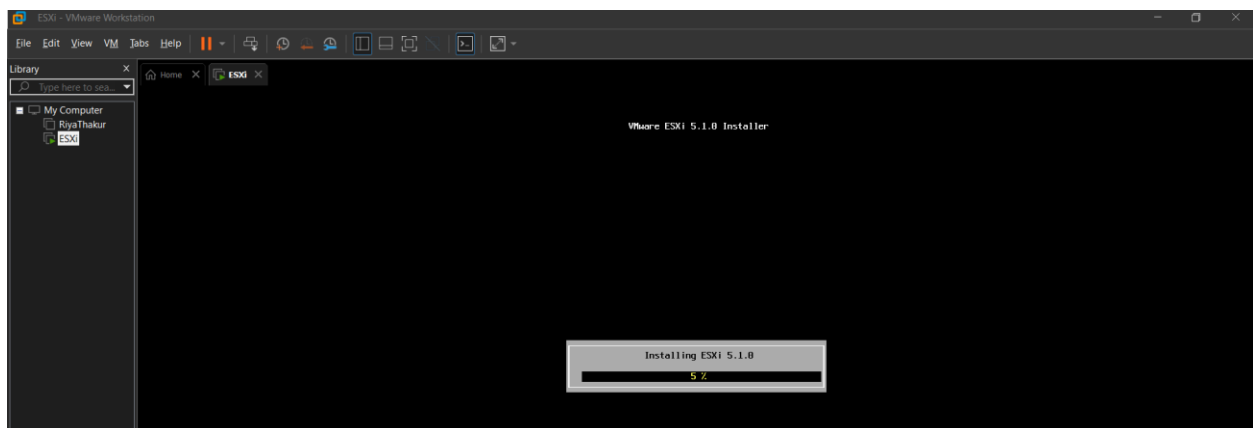
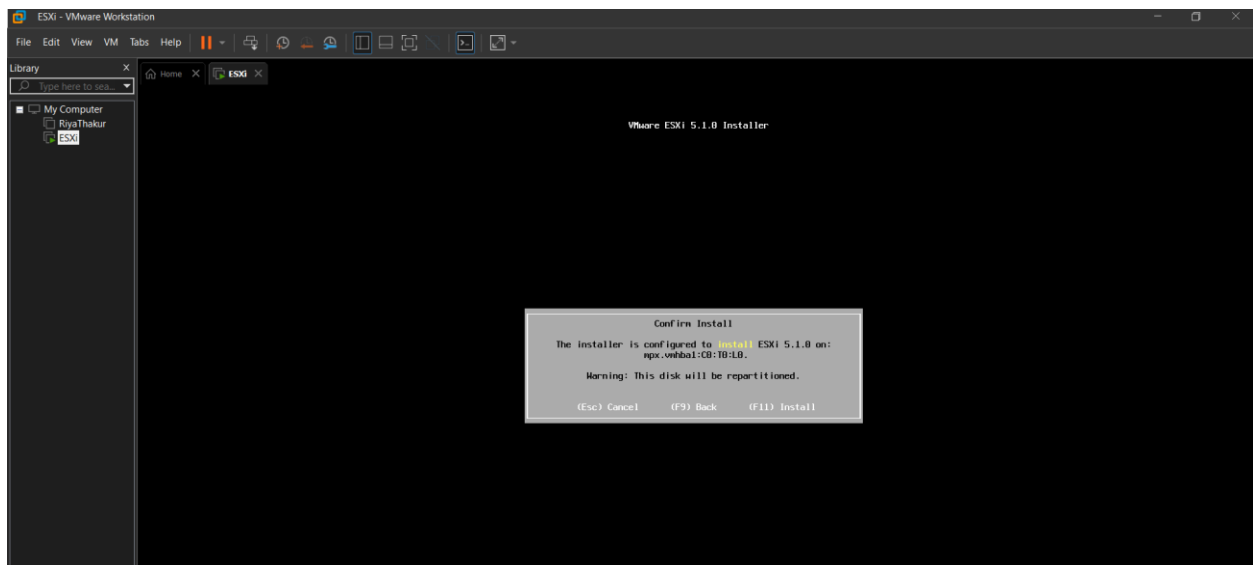
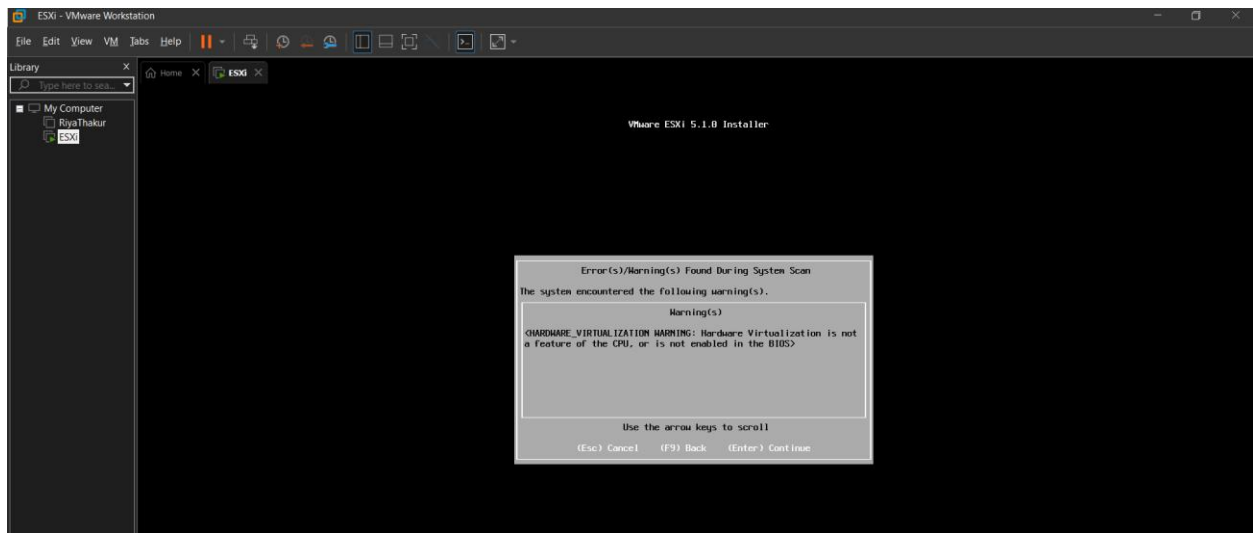


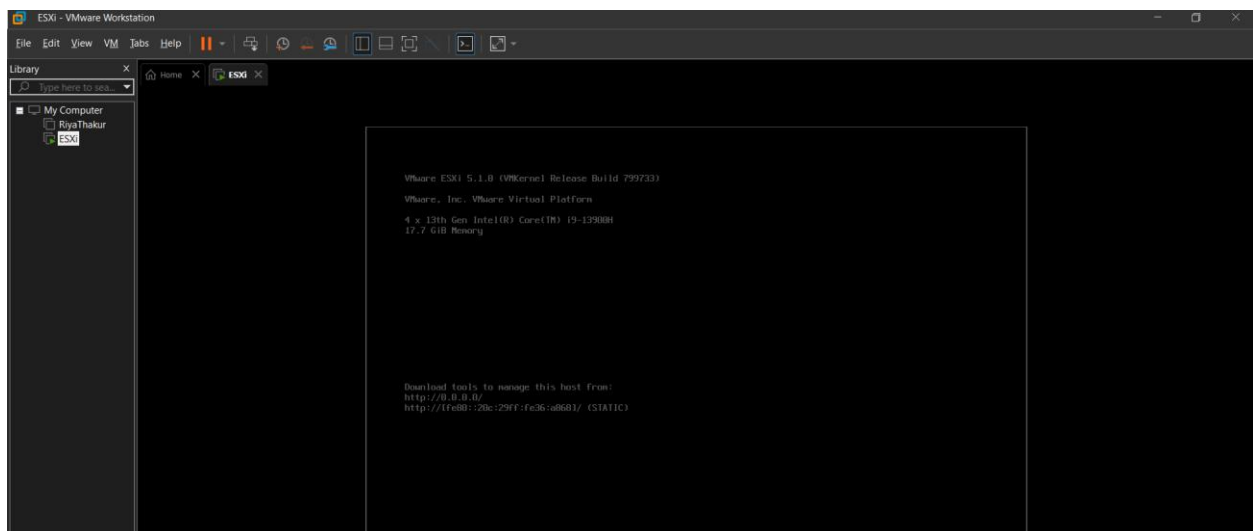
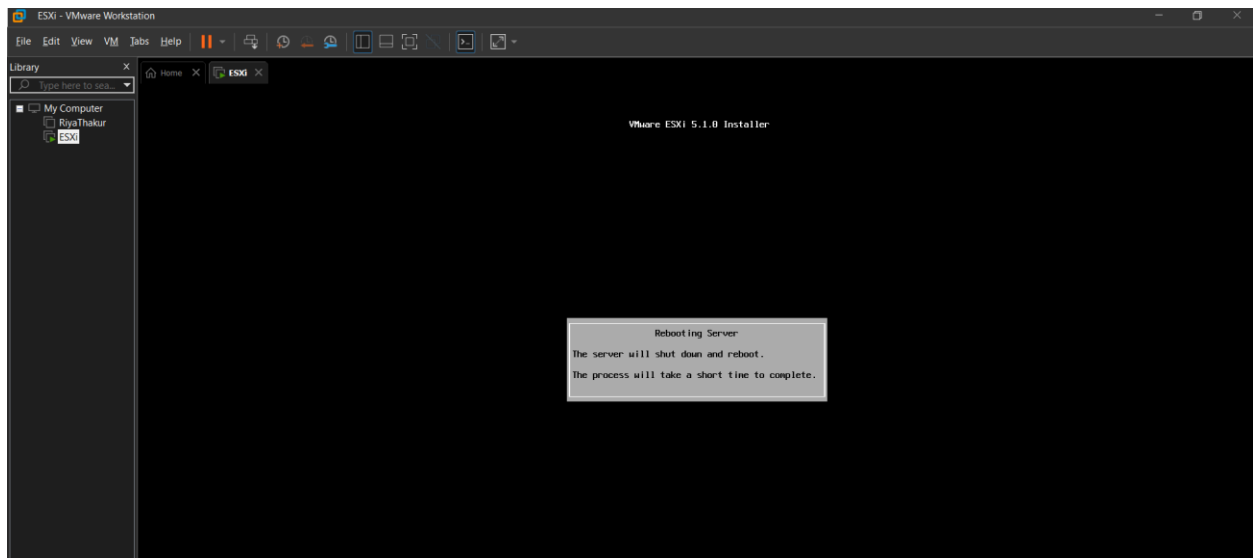
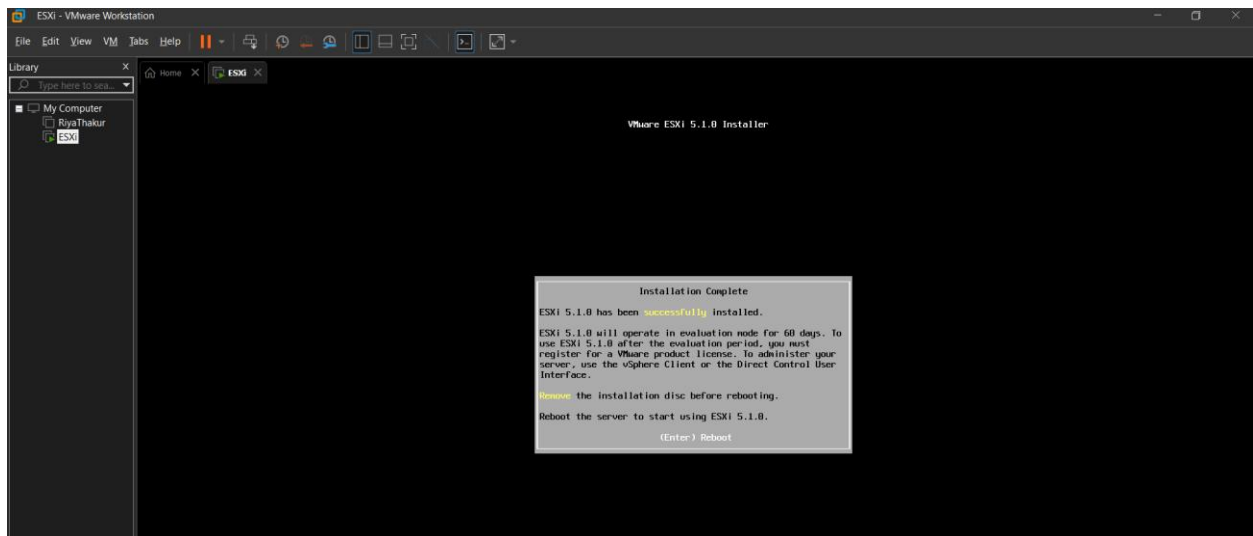
Step 17 : Now continue and accept.



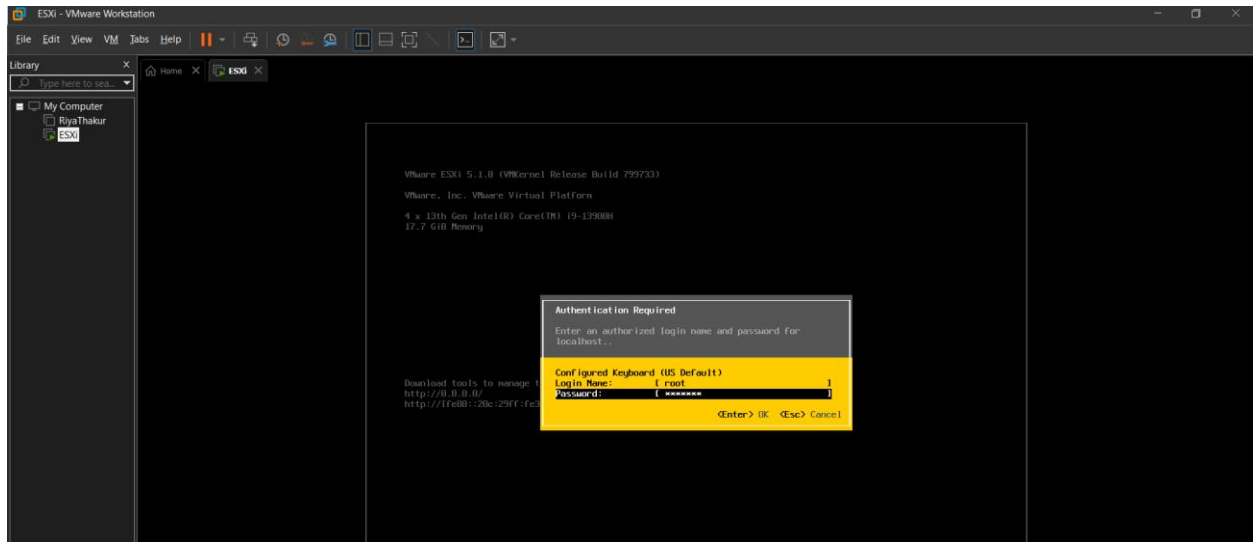




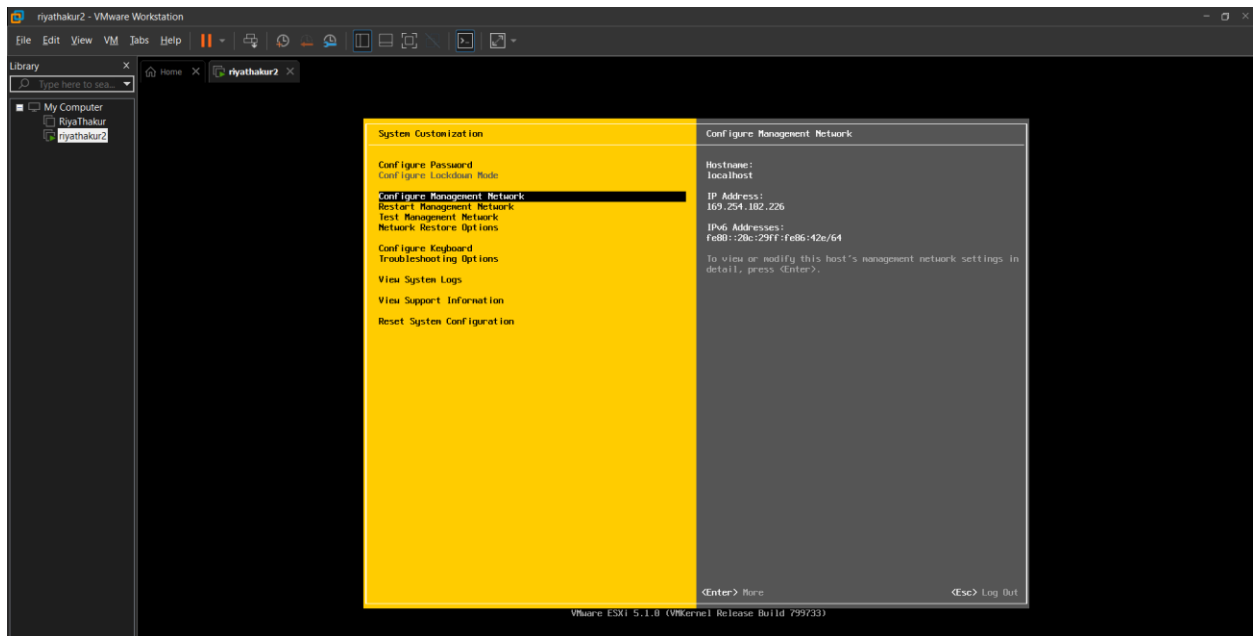




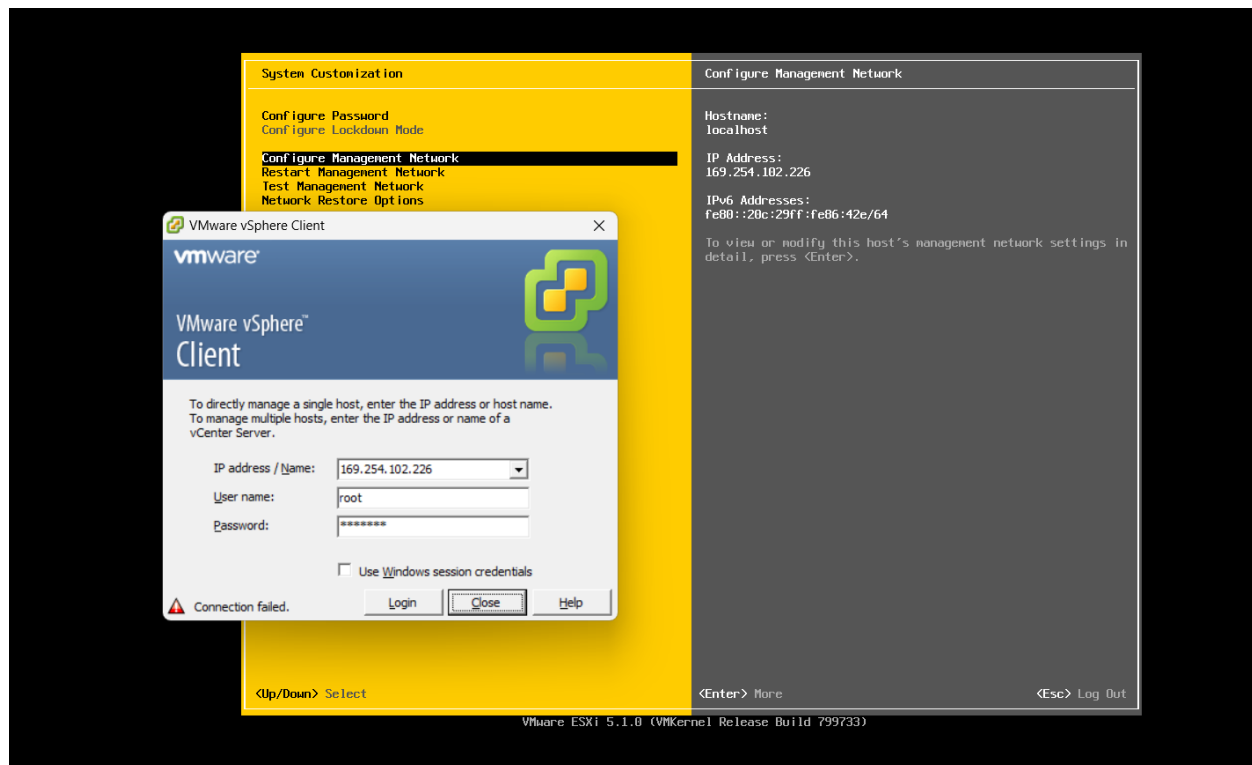
Step 17 : Set a password of 7 characters(1234567)



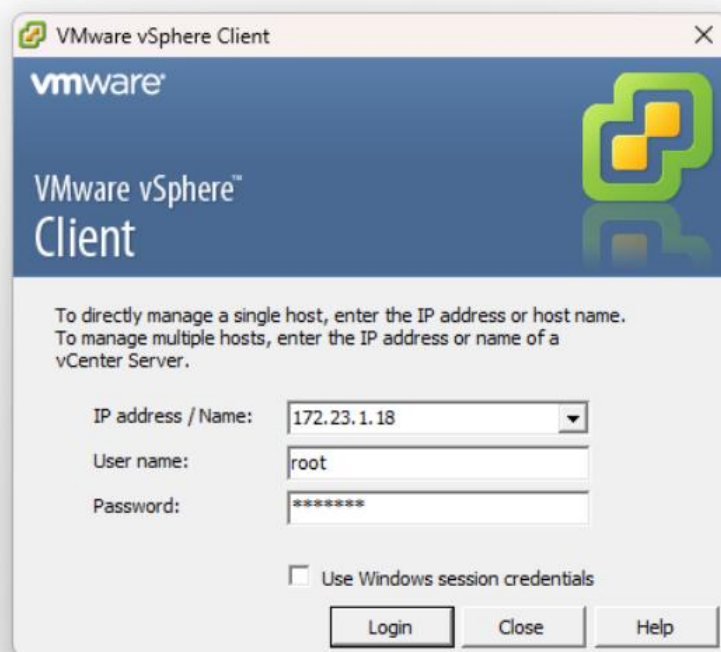
Step 18 : Check the IP address



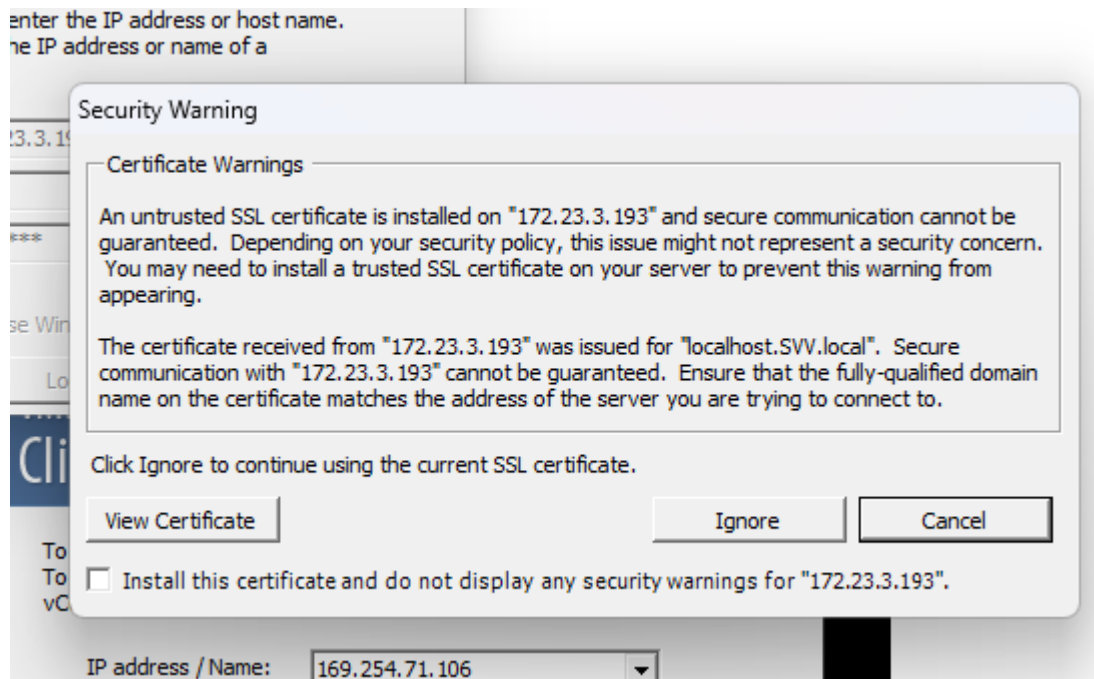
Step 19 : Open the VMware vSphere Client and try the IP address received along with the username and password previously set.



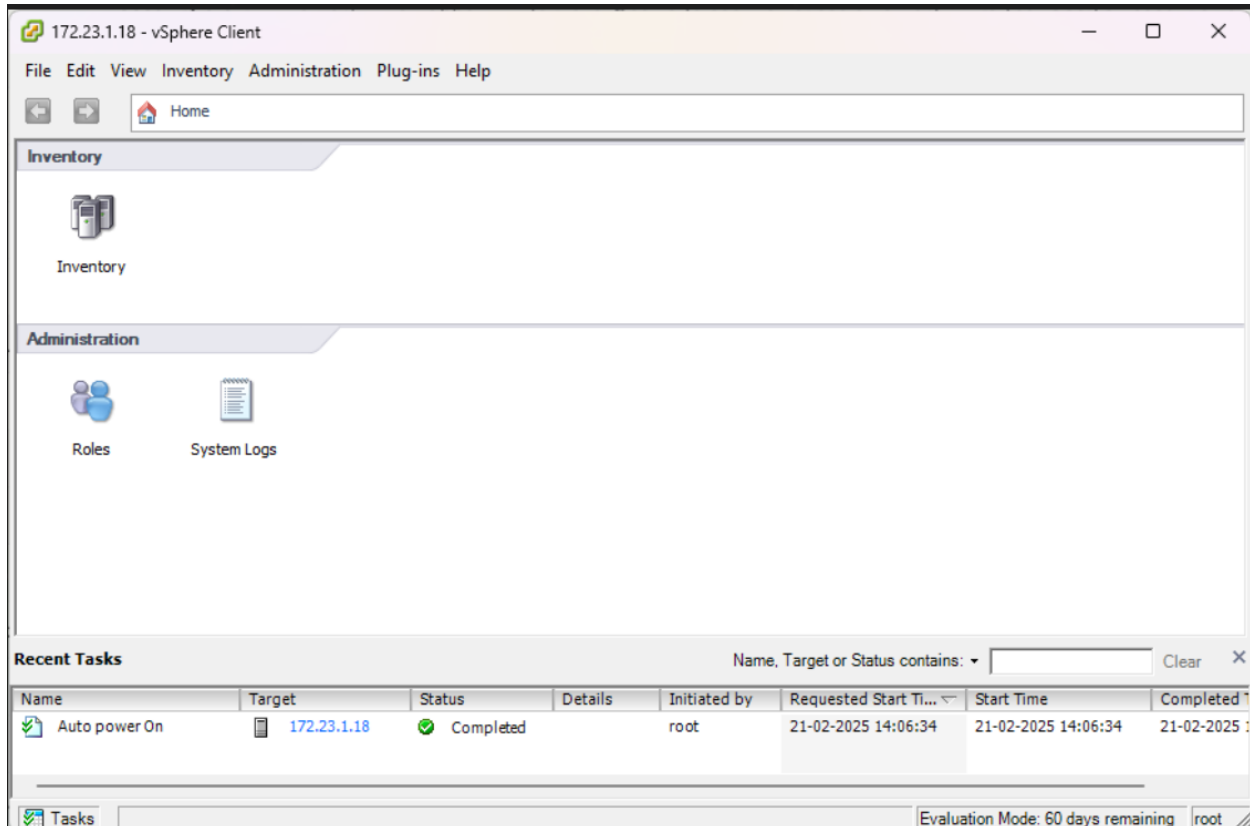
Step 20: Click Login

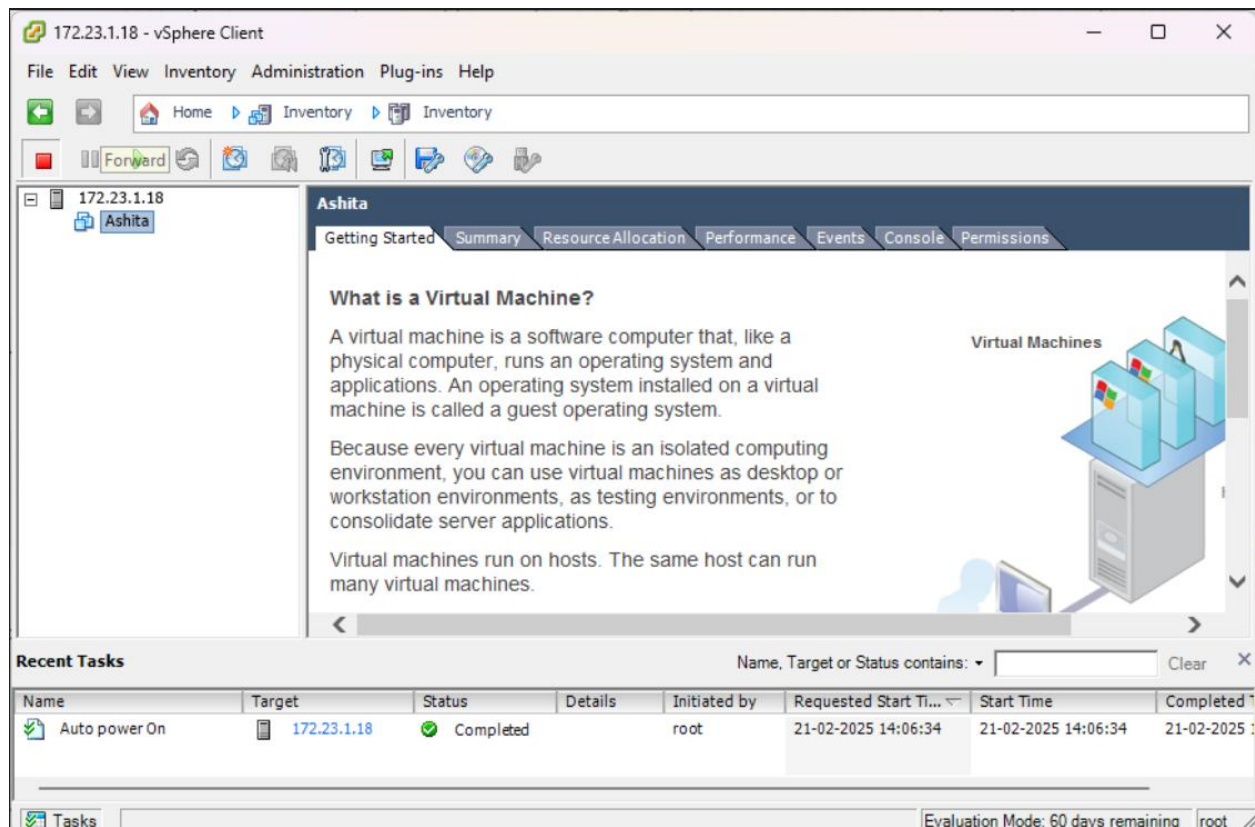


Step 21: Click ignore

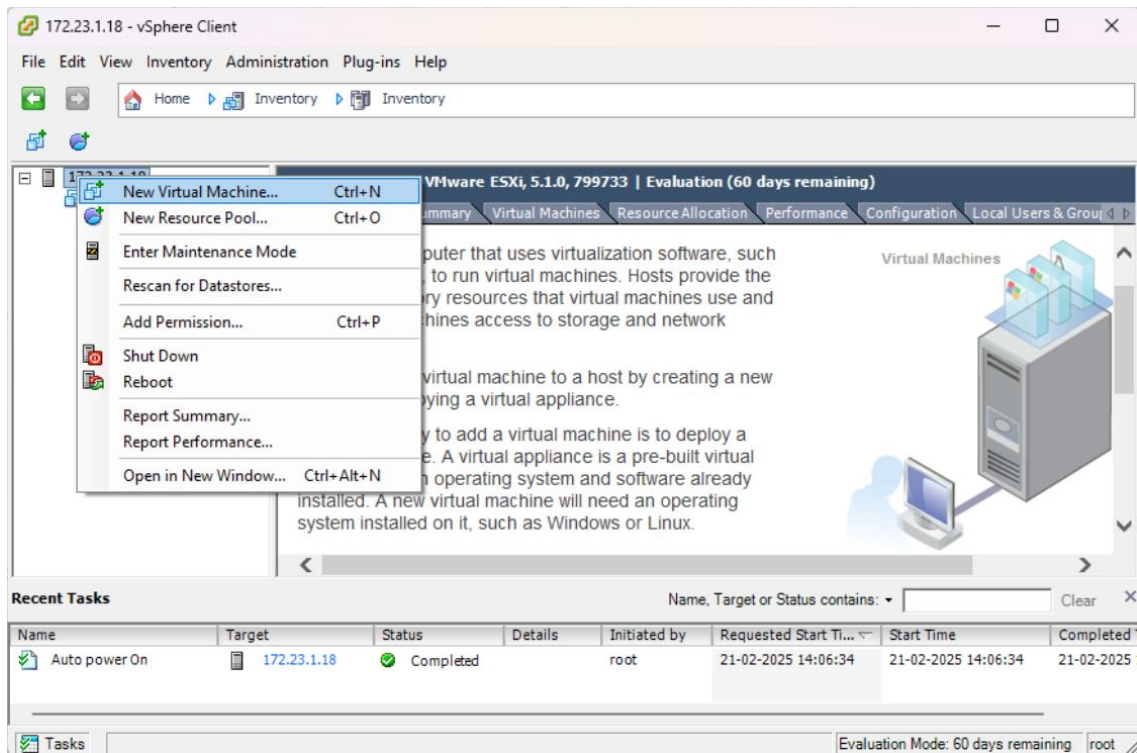


Step 22: Go to inventory

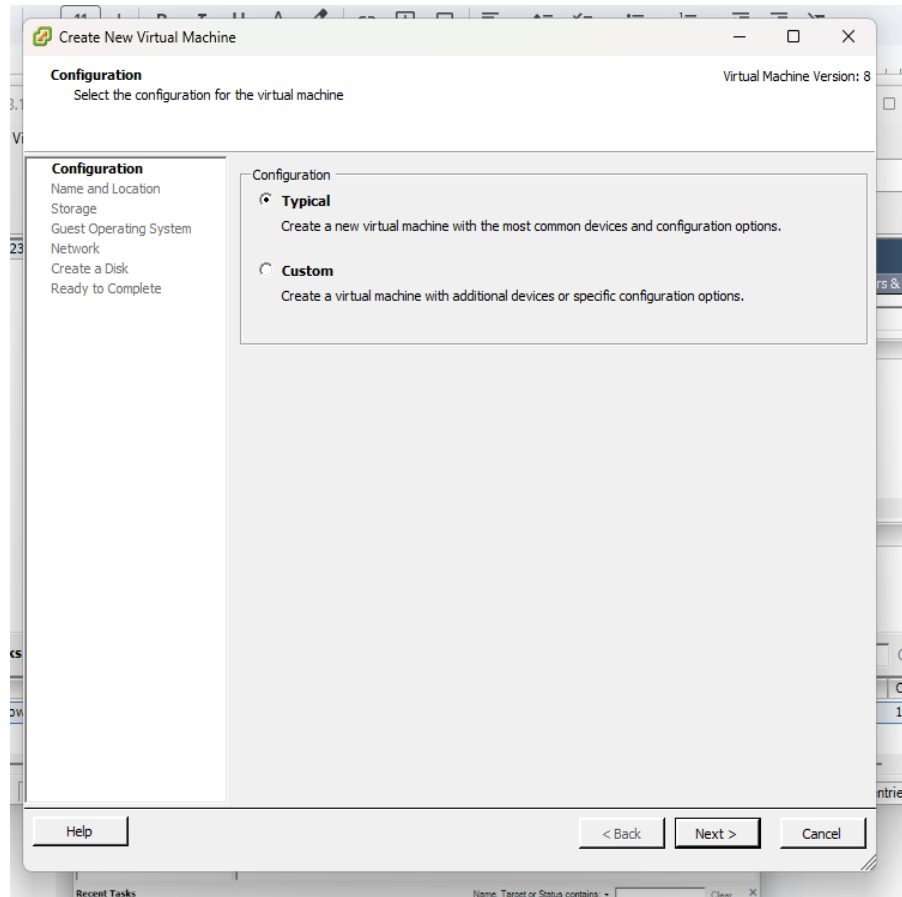




Step 23: Right click on the IP and select New Virtual Machine



Step 24: Select the 'Typical' radio button



Step 25: Set a name for your Virtual machine.

Create New Virtual Machine

Name and Location Virtual Machine Version: 8
Specify a name and location for this virtual machine

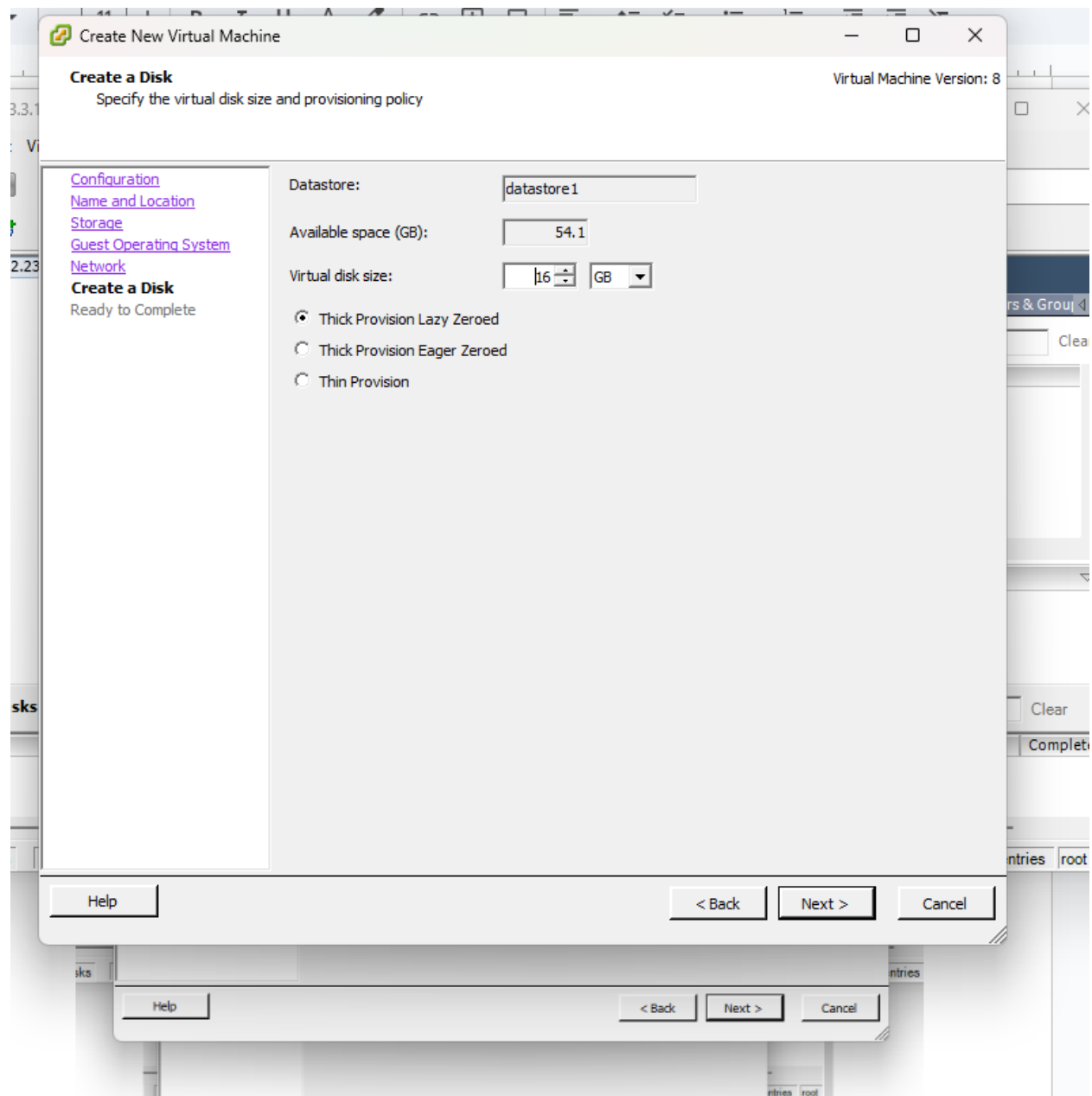
[Configuration](#)
Name and Location
Storage
Guest Operating System
Network
Create a Disk
Ready to Complete

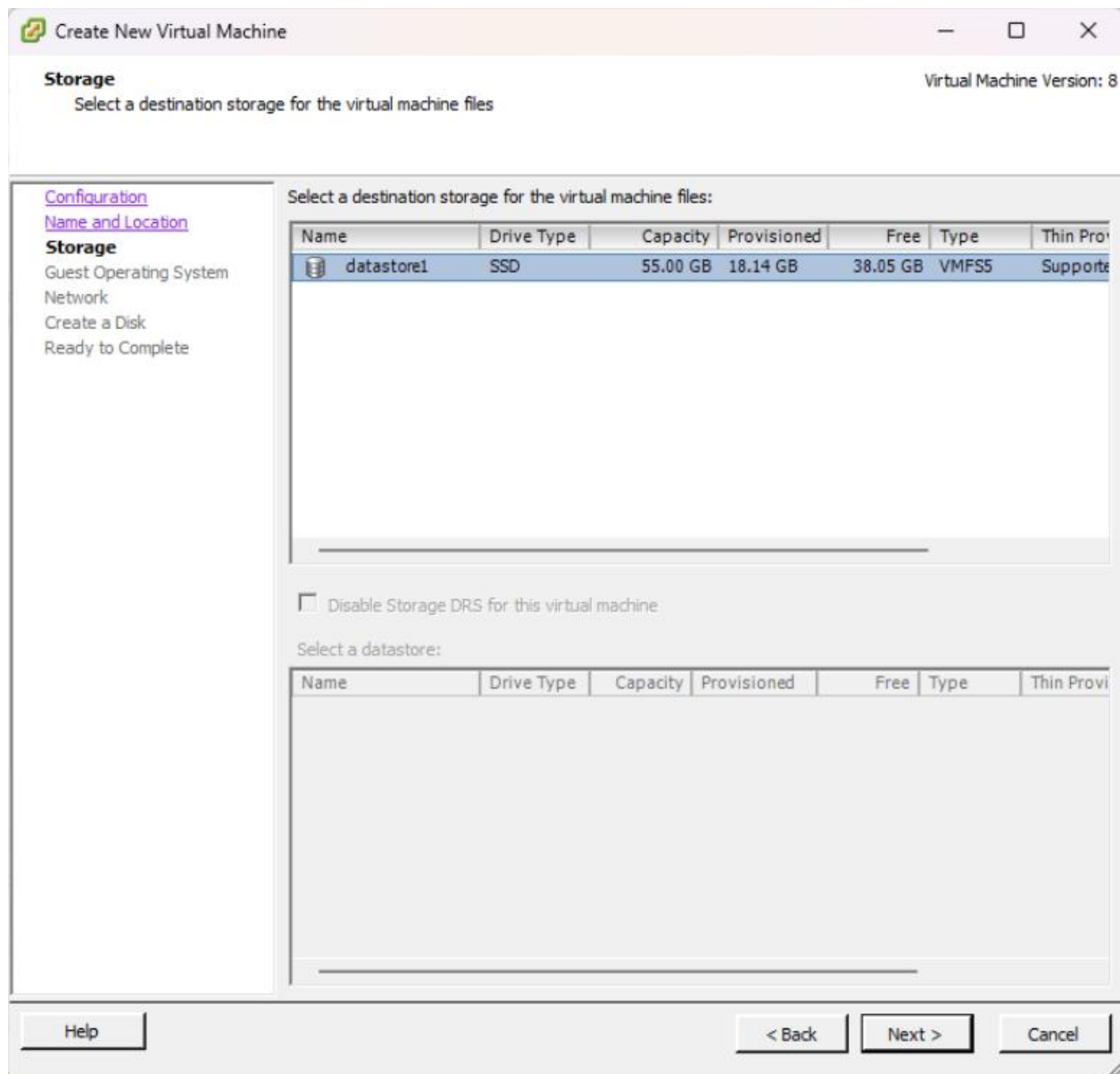
Name:
RiyaThakur

Virtual machine (VM) names may contain up to 80 characters and they must be unique within each vCenter Server VM folder.
VM folders are not viewable when connected directly to a host. To view VM folders and specify a location for this VM, connect to the vCenter Server.

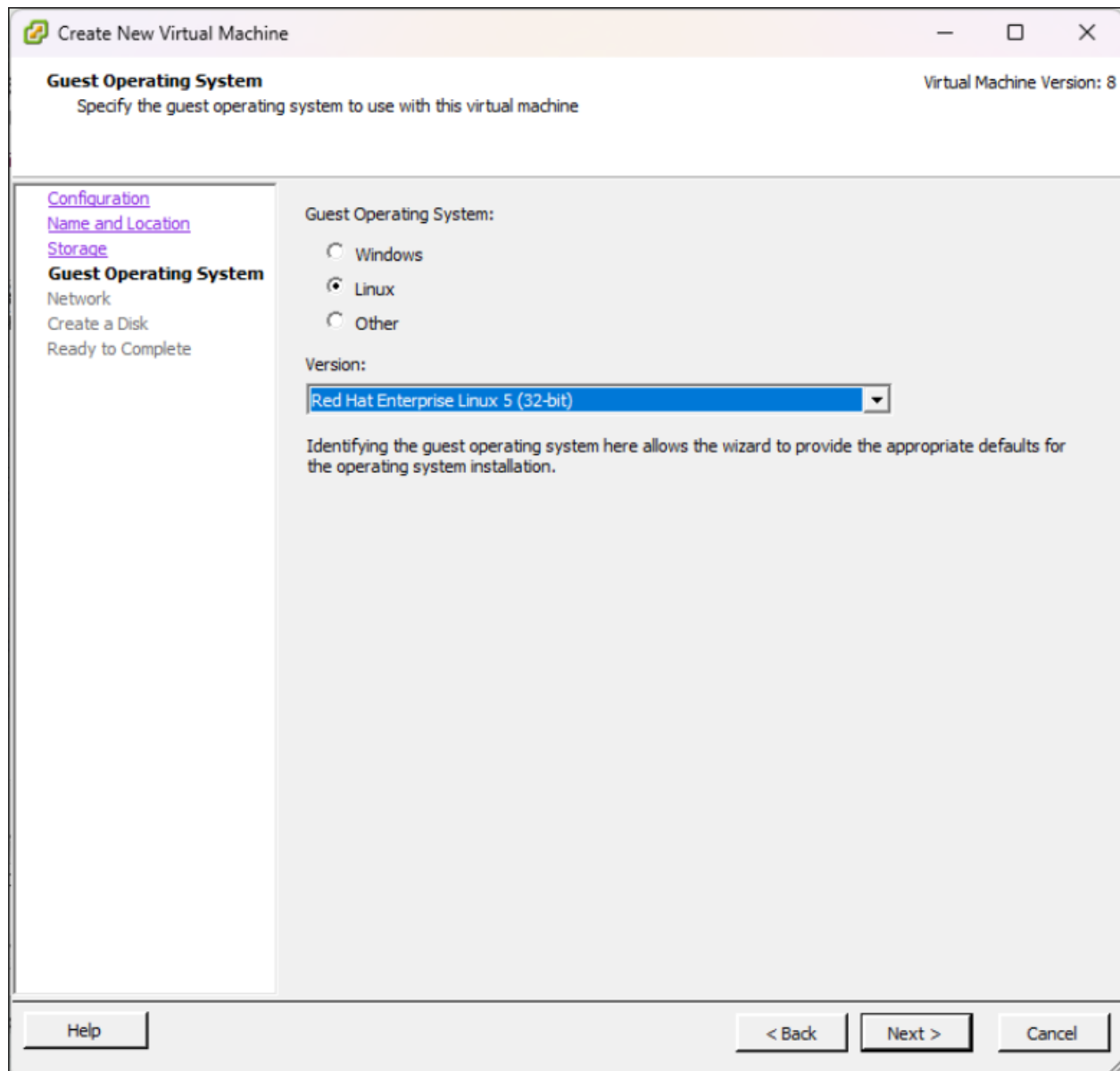
Help < Back Next > Cancel

Step 26 : Set virtual disk size and select thick provision lazy zeroed

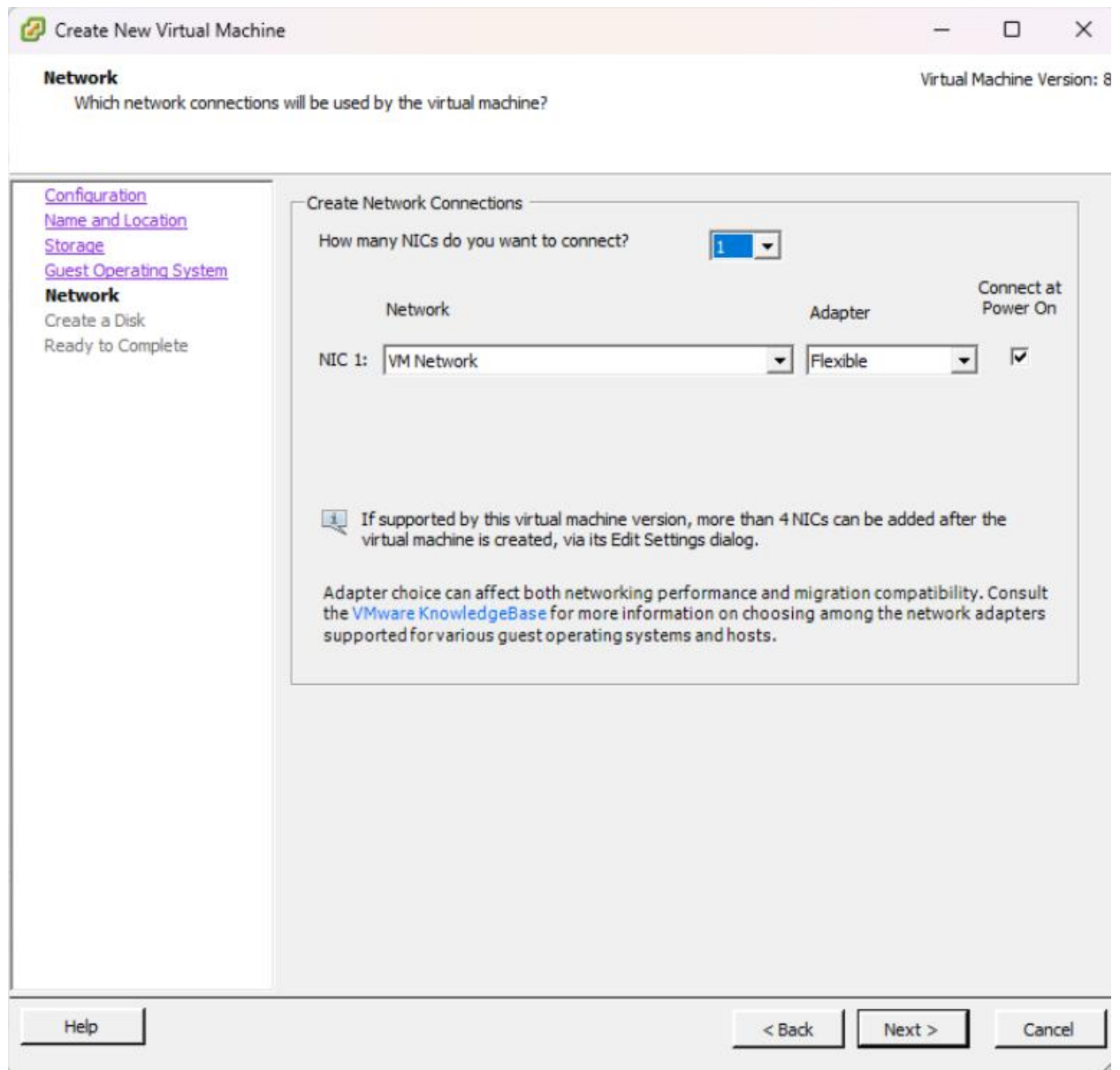




Step 27 : Select Linux and choose Red Hat Enterprise Linux 5(32-bit)



Step 28: Set the number of NIC connections.



Step 29 : Check the details and click Finish

Create New Virtual Machine

Ready to Complete Virtual Machine Version: 8
Click Finish to start a task that will create the new virtual machine

[Configuration](#)
[Name and Location](#)
[Storage](#)
[Guest Operating System](#)
[Network](#)
[Create a Disk](#)
Ready to Complete

Settings for the new virtual machine:

Name:	RiyaThakur
Host/Cluster:	localhost.SVVJocal
Datastore:	datastore1
Guest OS:	Red Hat Enterprise Linux 5 (32-bit)
NICs:	1
NIC 1 Network:	VM Network
NIC 1 Type:	Flexible
Disk provisioning:	Thick Provision Lazy Zeroed
Virtual Disk Size:	5 GB

☐ Edit the virtual machine settings before completion

 Creation of the virtual machine (VM) does not include automatic installation of the guest operating system. Install a guest OS on the VM after creating the VM.

[Help](#) [< Back](#) [Finish](#) [Cancel](#)

Step 30 : Check if you new virtual machine is created

172.23.1.18 - vSphere Client

File Edit View Inventory Administration Plug-ins Help

Home Inventory Inventory

172.23.1.18

- Ashita
- RiyaThakur

localhost.SVV.local VMware ESXi, 5.1.0, 799733 | Evaluation (60 days remaining)

- Getting Started
- Summary
- Virtual Machines
- Resource Allocation
- Performance
- Configuration
- Local Users & Groups

A host is a computer that uses virtualization software, such as ESX or ESXi, to run virtual machines. Hosts provide the CPU and memory resources that virtual machines use and give virtual machines access to storage and network connectivity.

You can add a virtual machine to a host by creating a new one or by deploying a virtual appliance.

The easiest way to add a virtual machine is to deploy a virtual appliance. A virtual appliance is a pre-built virtual machine with an operating system and software already installed. A new virtual machine will need an operating system installed on it, such as Windows or Linux.

Virtual Machines

Recent Tasks

Name, Target or Status contains: Clear

Name	Target	Status	Details	Initiated by	Requested Start Time	Start Time	Completed Time
Create virtual machine	172.23.1.18	Completed		root	21-02-2025 14:12:56	21-02-2025 14:12:56	21-02-2025 14:12:56
Auto power On	172.23.1.18	Completed		root	21-02-2025 14:06:34	21-02-2025 14:06:34	21-02-2025 14:06:34

Tasks

Evaluation Mode: 60 days remaining root

172.23.1.18 - vSphere Client

File Edit View Inventory Administration Plug-ins Help

Home Inventory Inventory

172.23.1.18

- Ashita
- RiyaThakur

RiyaThakur

- Getting Started
- Summary
- Resource Allocation
- Performance
- Events
- Console
- Permissions

Basic Tasks

▶ Power on the virtual machine

▶ Edit virtual machine settings

vSphere Client

Recent Tasks

Name, Target or Status contains: Clear

Name	Target	Status	Details	Initiated by	Requested Start Time	Start Time	Completed Time
Create virtual machine	172.23.1.18	Completed		root	21-02-2025 14:12:56	21-02-2025 14:12:56	21-02-2025 14:12:56
Auto power On	172.23.1.18	Completed		root	21-02-2025 14:06:34	21-02-2025 14:06:34	21-02-2025 14:06:34

Tasks

Evaluation Mode: 60 days remaining root

172.23.1.18 - vSphere Client

File Edit View Inventory Administration Plug-ins Help

Home Inventory Inventory

172.23.1.18

Ashita

RiyaThakur

RiyaThakur

Getting Started Summary Resource Allocation Performance Events Console Permissions

close tab X

What is a Virtual Machine?

A virtual machine is a software computer that, like a physical computer, runs an operating system and applications. An operating system installed on a virtual machine is called a guest operating system.

Because every virtual machine is an isolated computing environment, you can use virtual machines as desktop or workstation environments, as testing environments, or to consolidate server applications.

Virtual machines run on hosts. The same host can run many virtual machines.

Virtual Machines

Host

vSphere Client

Basic Tasks

Power Off the virtual machine

Suspend the virtual machine

Edit virtual machine settings

Recent Tasks

Name	Target	Status	Details	Initiated by	Requested Start Time	Start Time	Completed Time
Power On virtual mach...	RiyaThakur	Completed		root	21-02-2025 14:13:22	21-02-2025 14:13:22	21-02-2025 14:13:23
Create virtual machine	172.23.1.18	Completed		root	21-02-2025 14:12:56	21-02-2025 14:12:56	21-02-2025 14:12:57
Auto power On	172.23.1.18	Completed		root	21-02-2025 14:06:34	21-02-2025 14:06:34	21-02-2025 14:06:34

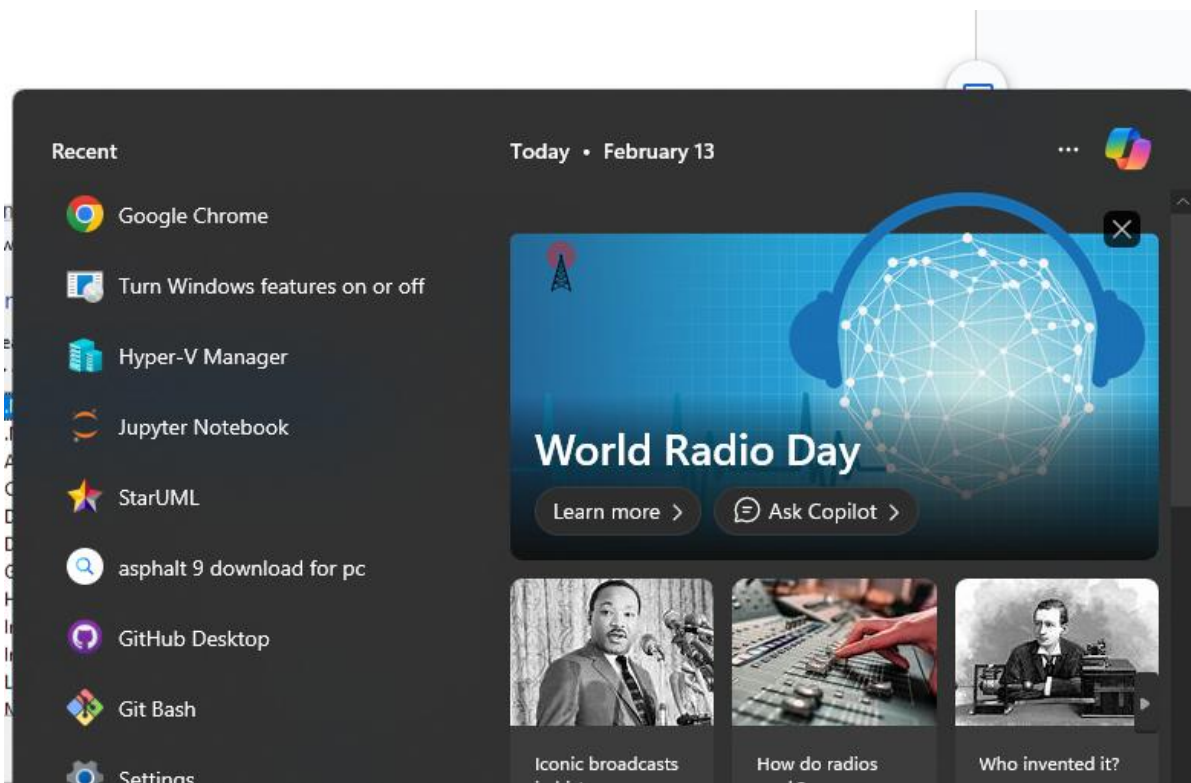
Tasks

Practical 08

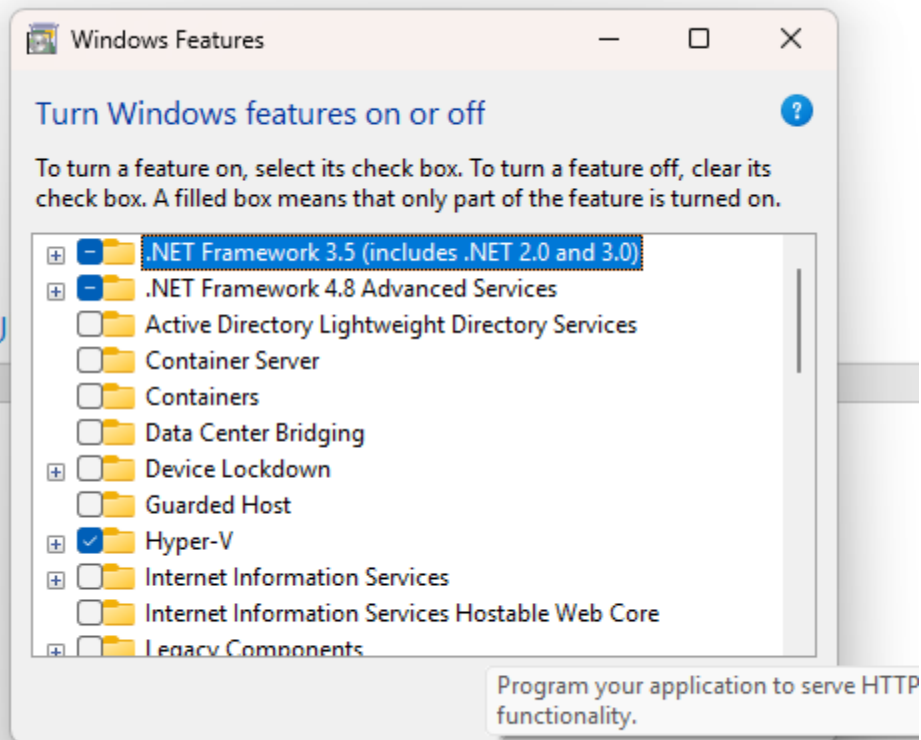
Aim: Implement Windows Hyper-V virtualization.

Steps:

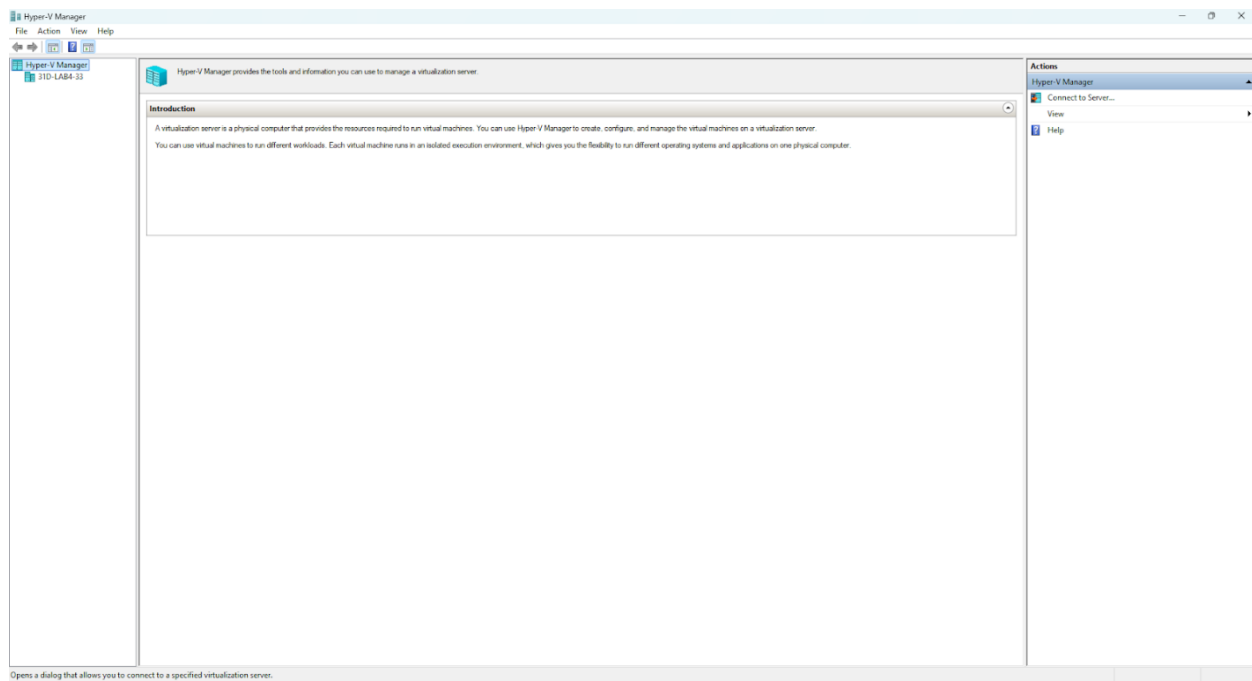
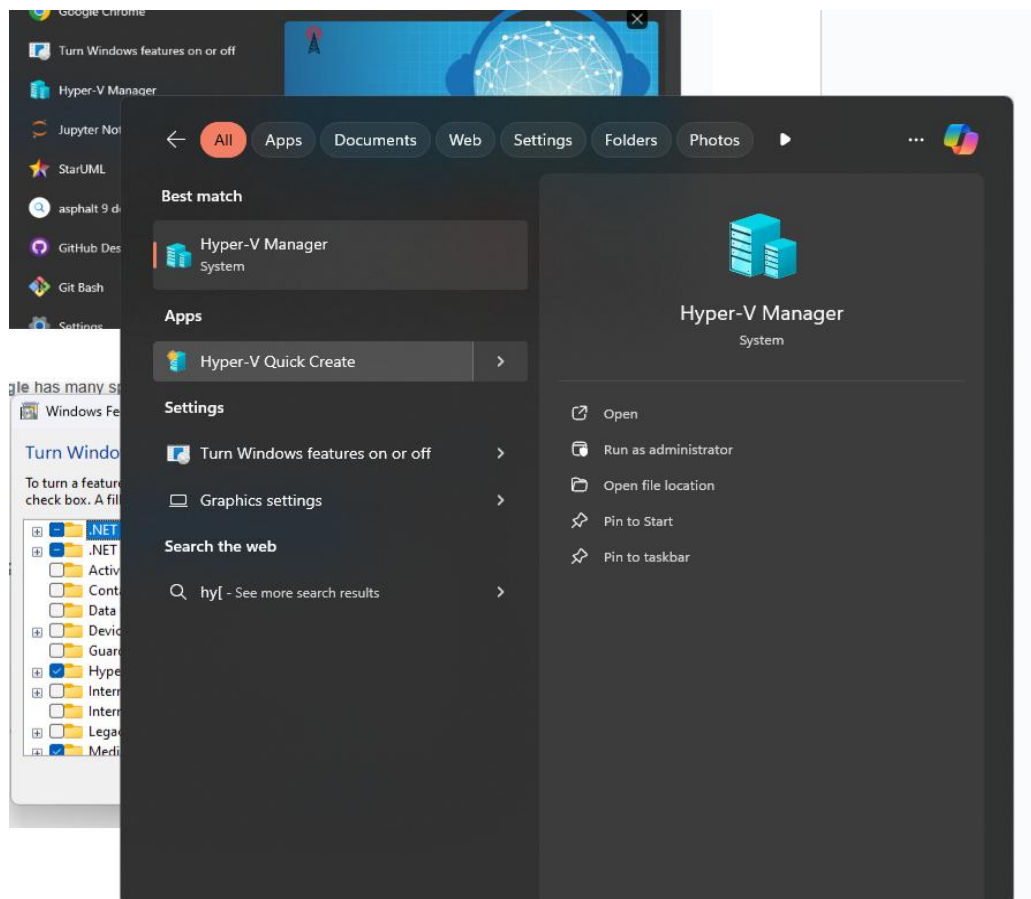
Step 1: Go to windows and go to click features and select hyper v and click ok



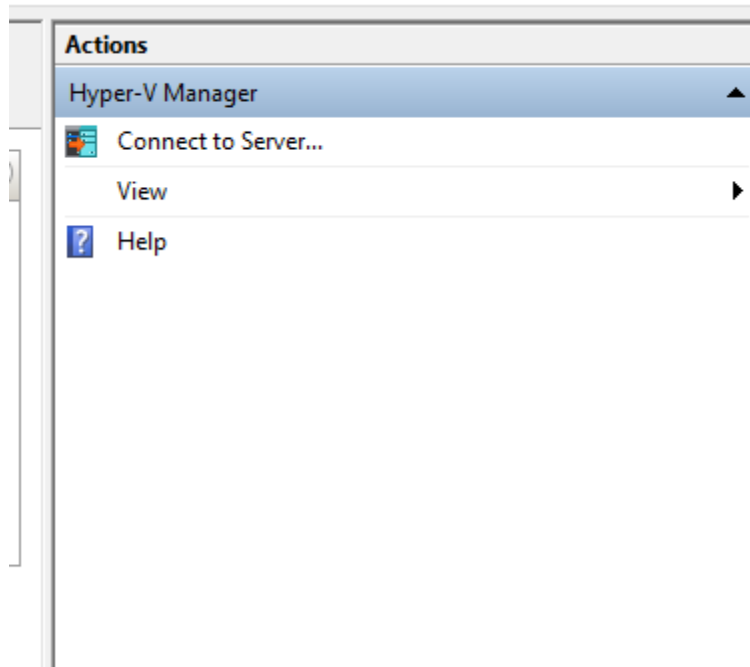
je 'U



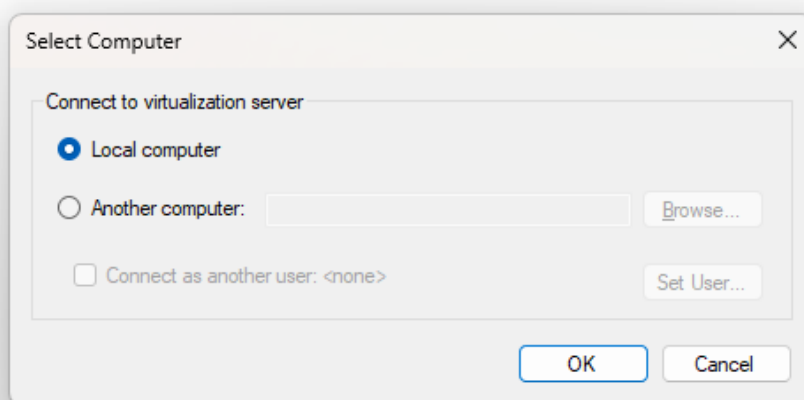
Step 2 : Go to hyper-v- manager



Step 3: Go to connect to server



Step 4: Select local computer and ok



Step 5 : Select quick create

Actions

31D-LAB4-33 ▲



Quick Create...

New ▶



Import Virtual Machine...



Hyper-V Settings...



Virtual Switch Manager...



Virtual SAN Manager...



Edit Disk...



Inspect Disk...



Stop Service



Remove Server

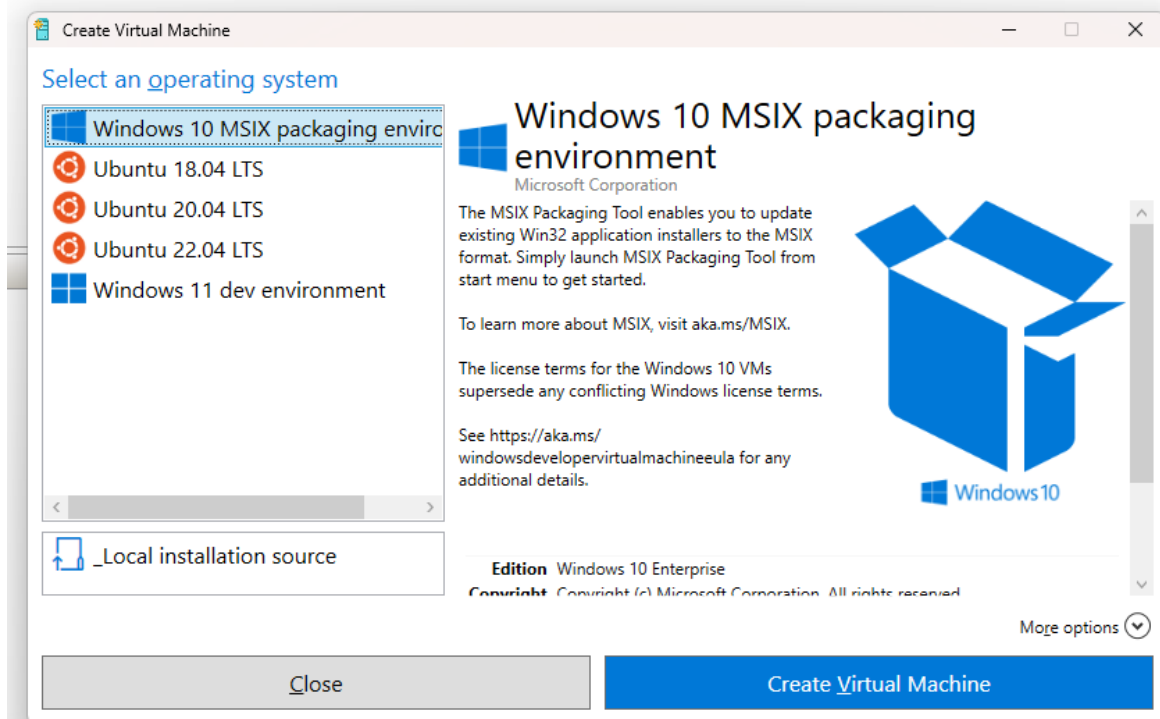


Refresh

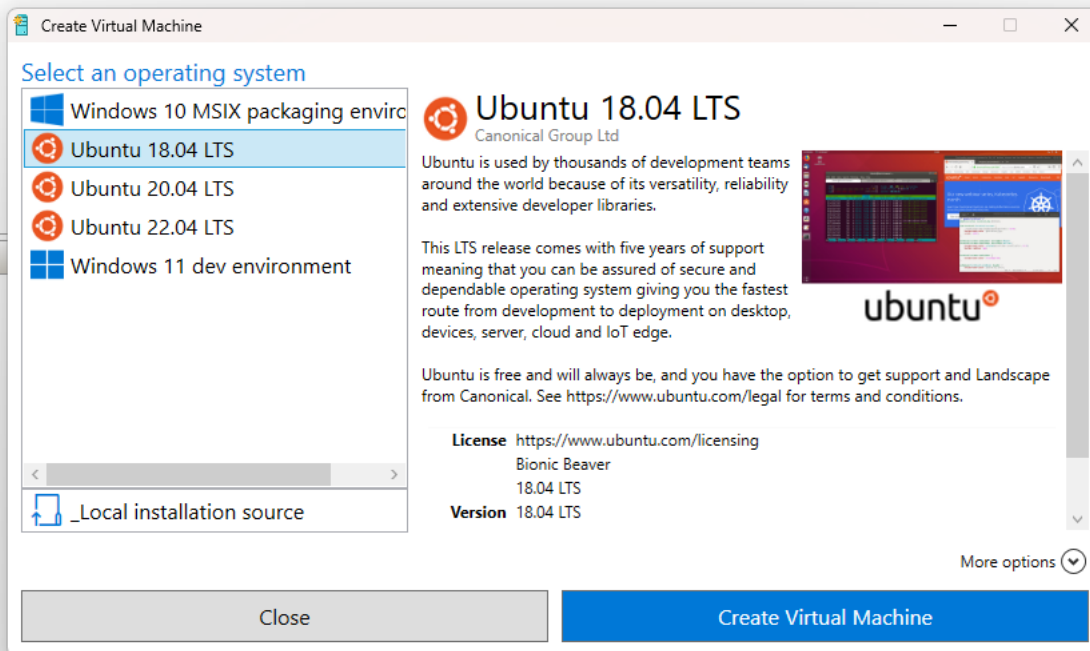
View ▶



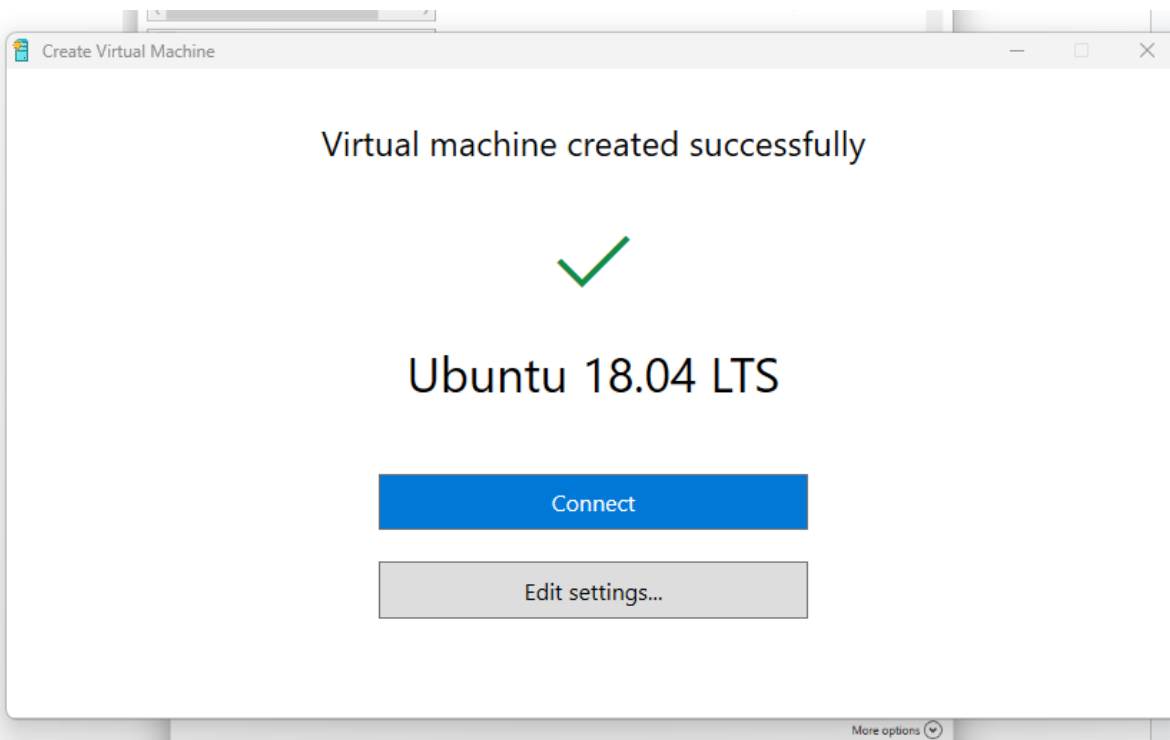
Help

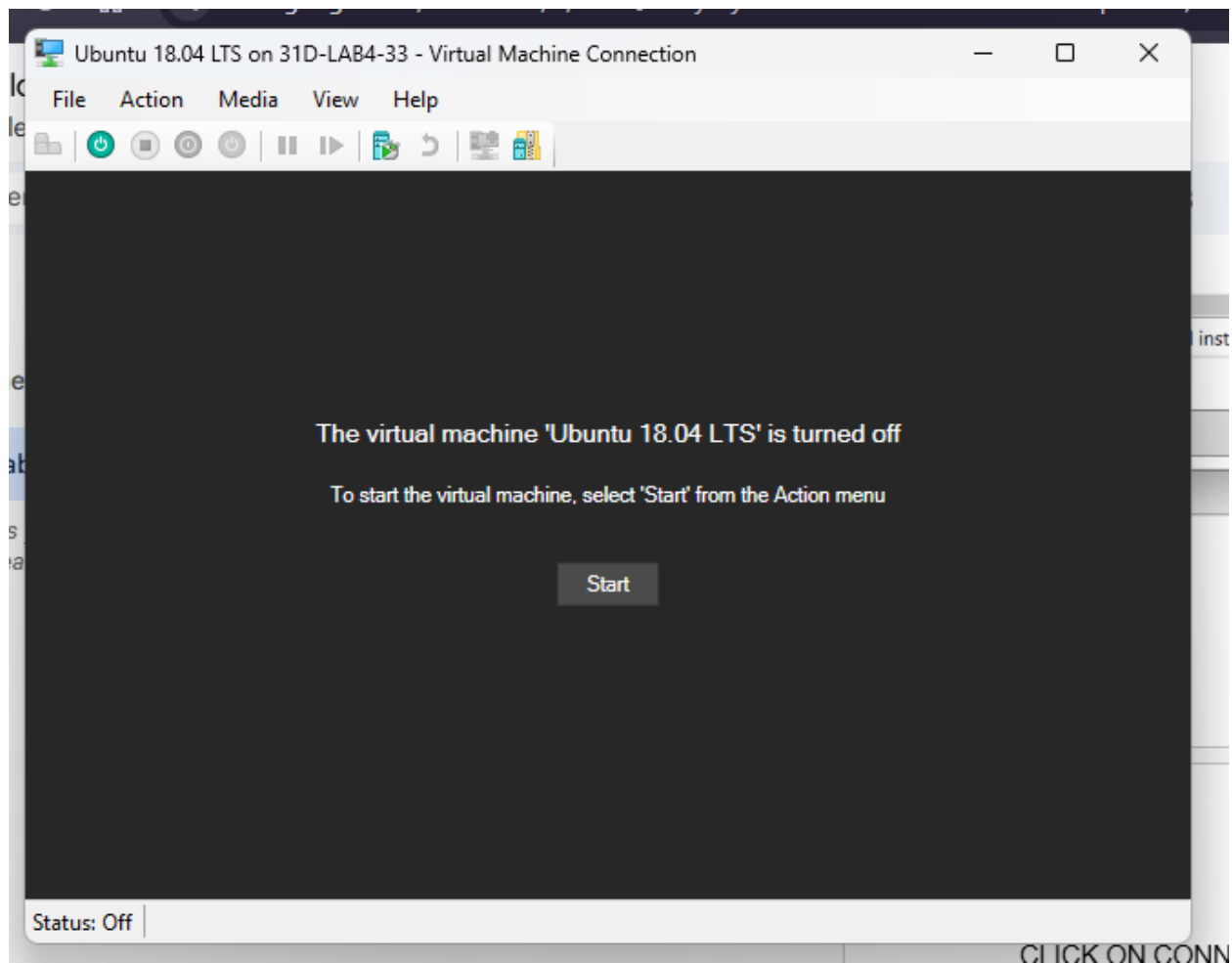


Step 6 :Select ubuntu 18.04 LTS and click on create virtual machine

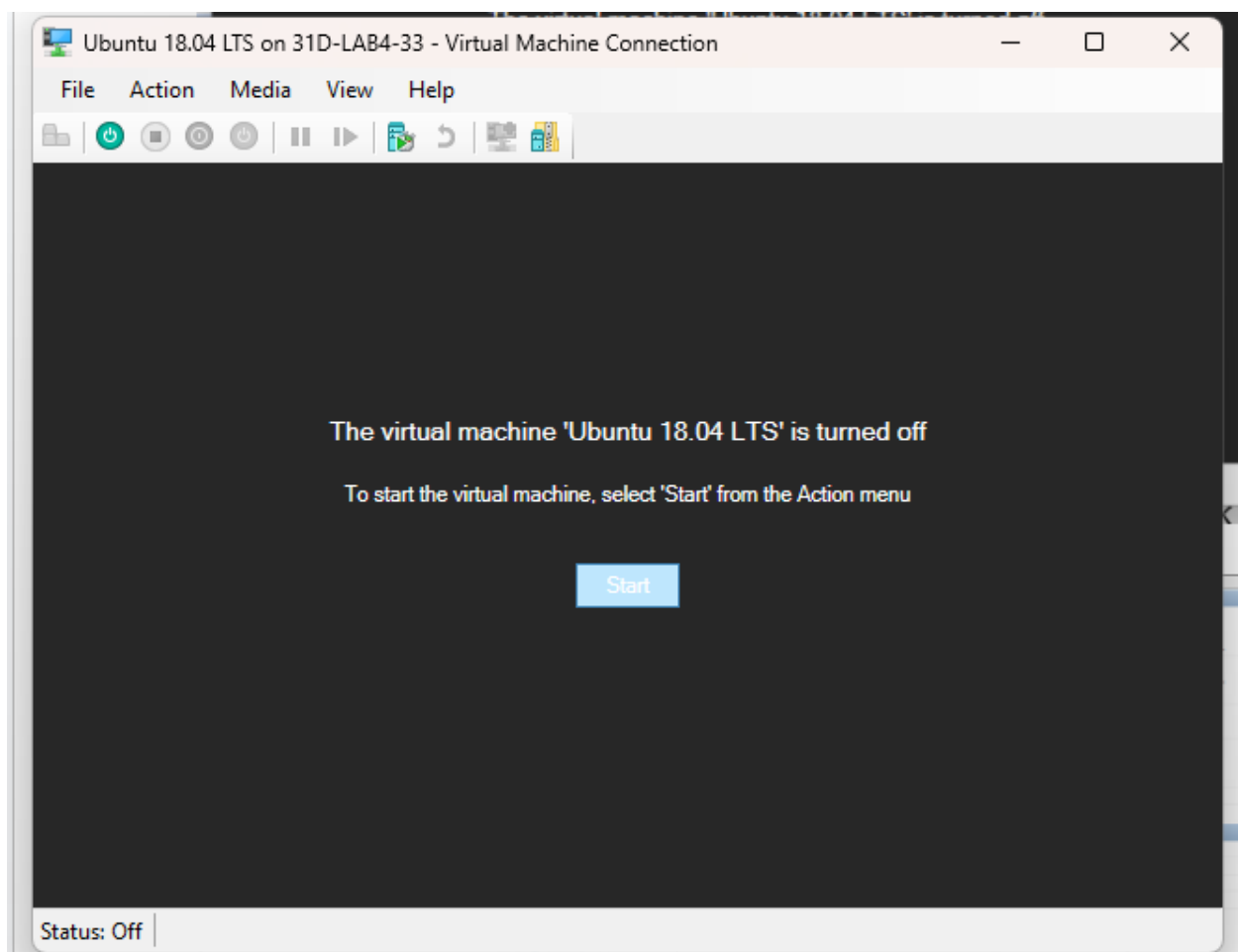
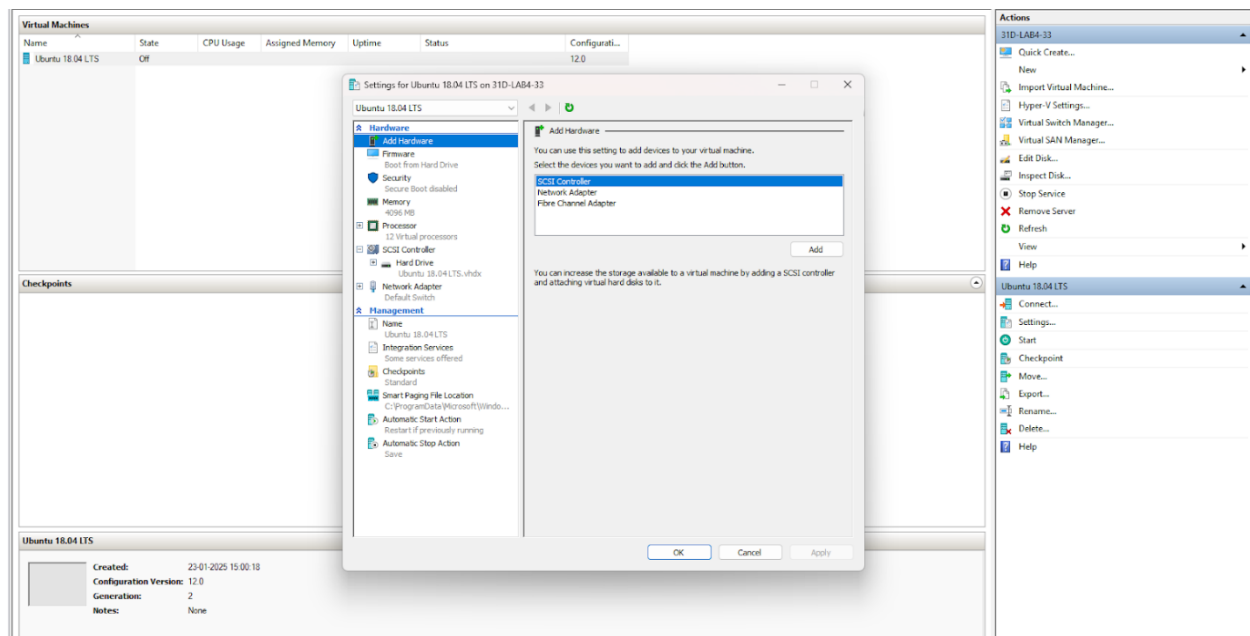


Step 7 :Click on Connect and CancelS



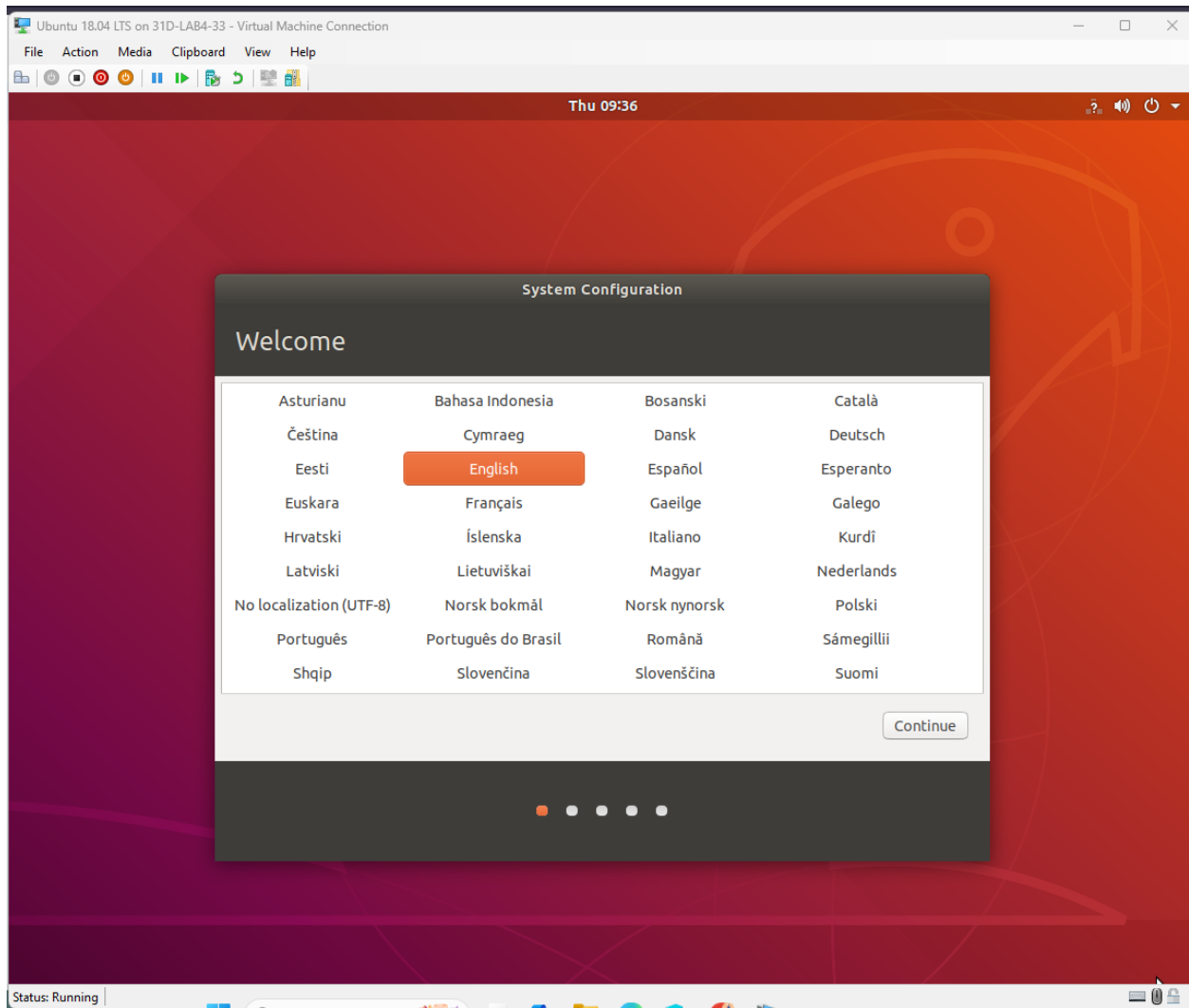


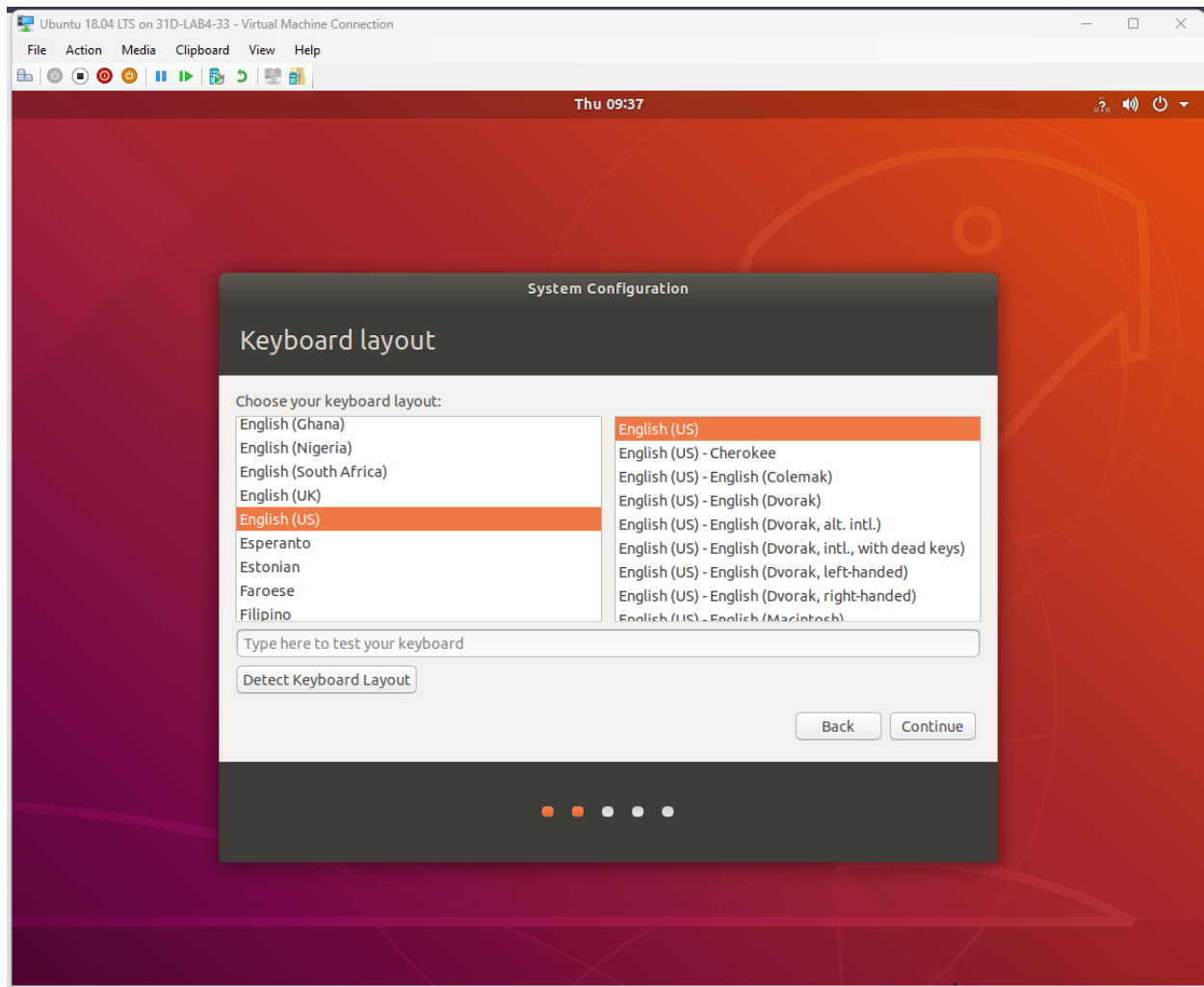
Step 8 : Click on Settings- Look at the memory and processor



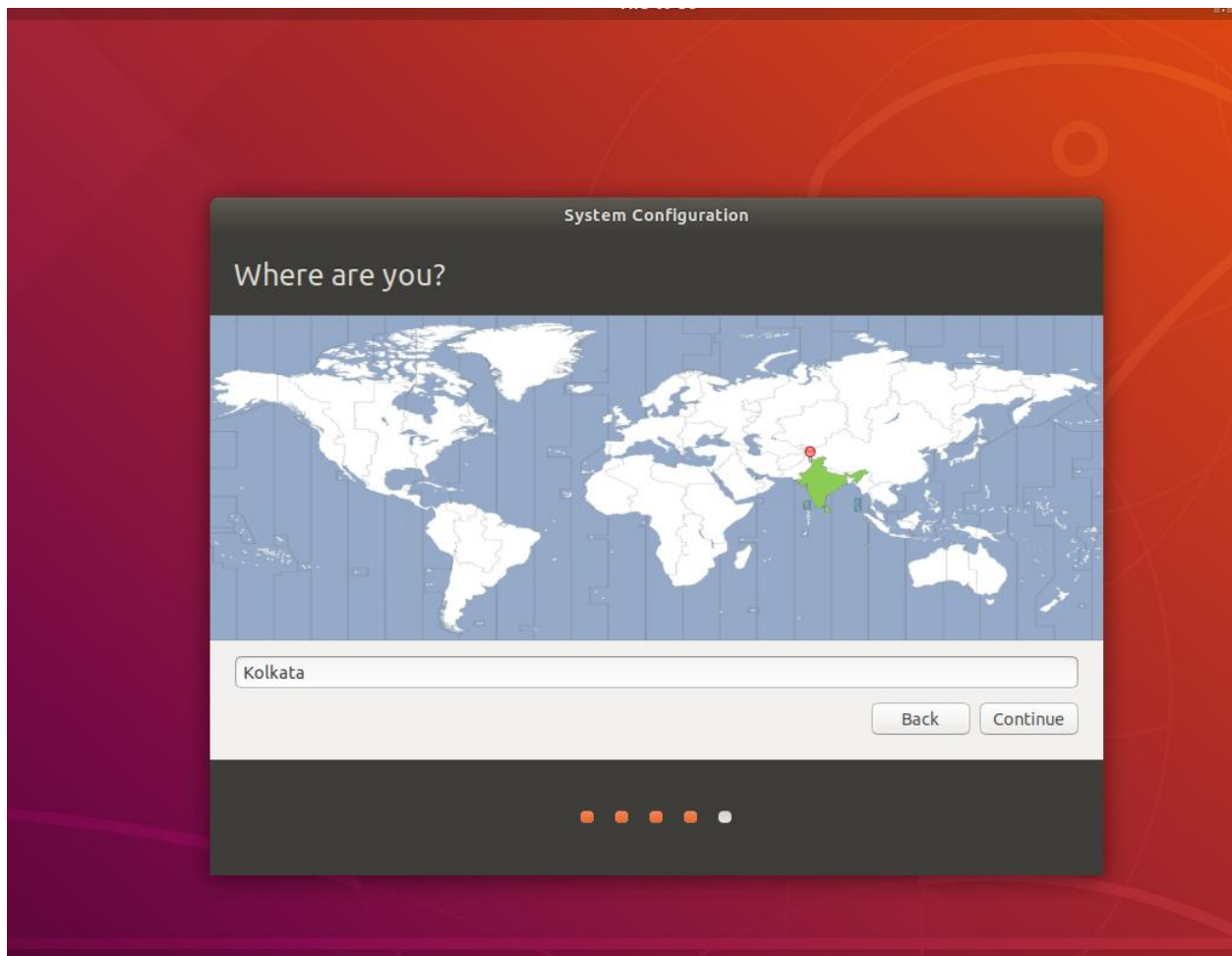
Step 9: Click on Start

Step 10 : Select the language

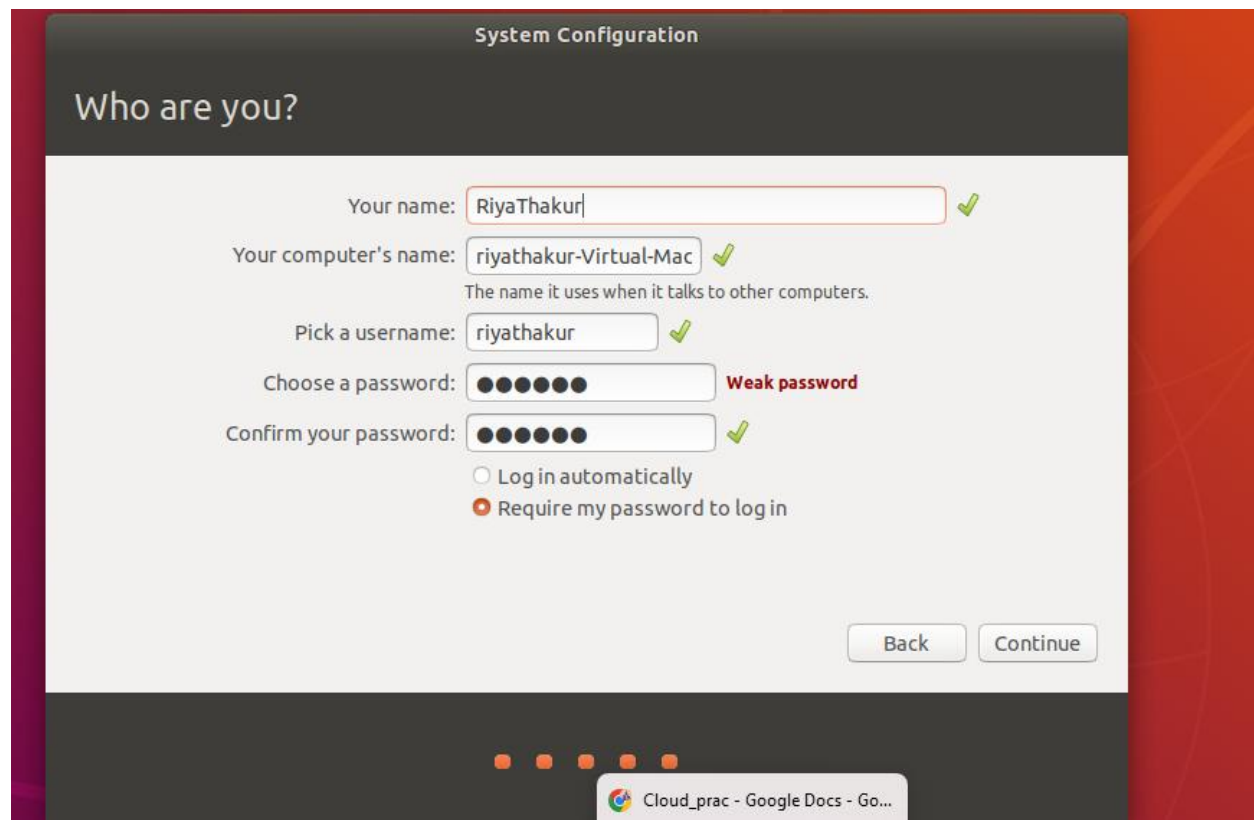


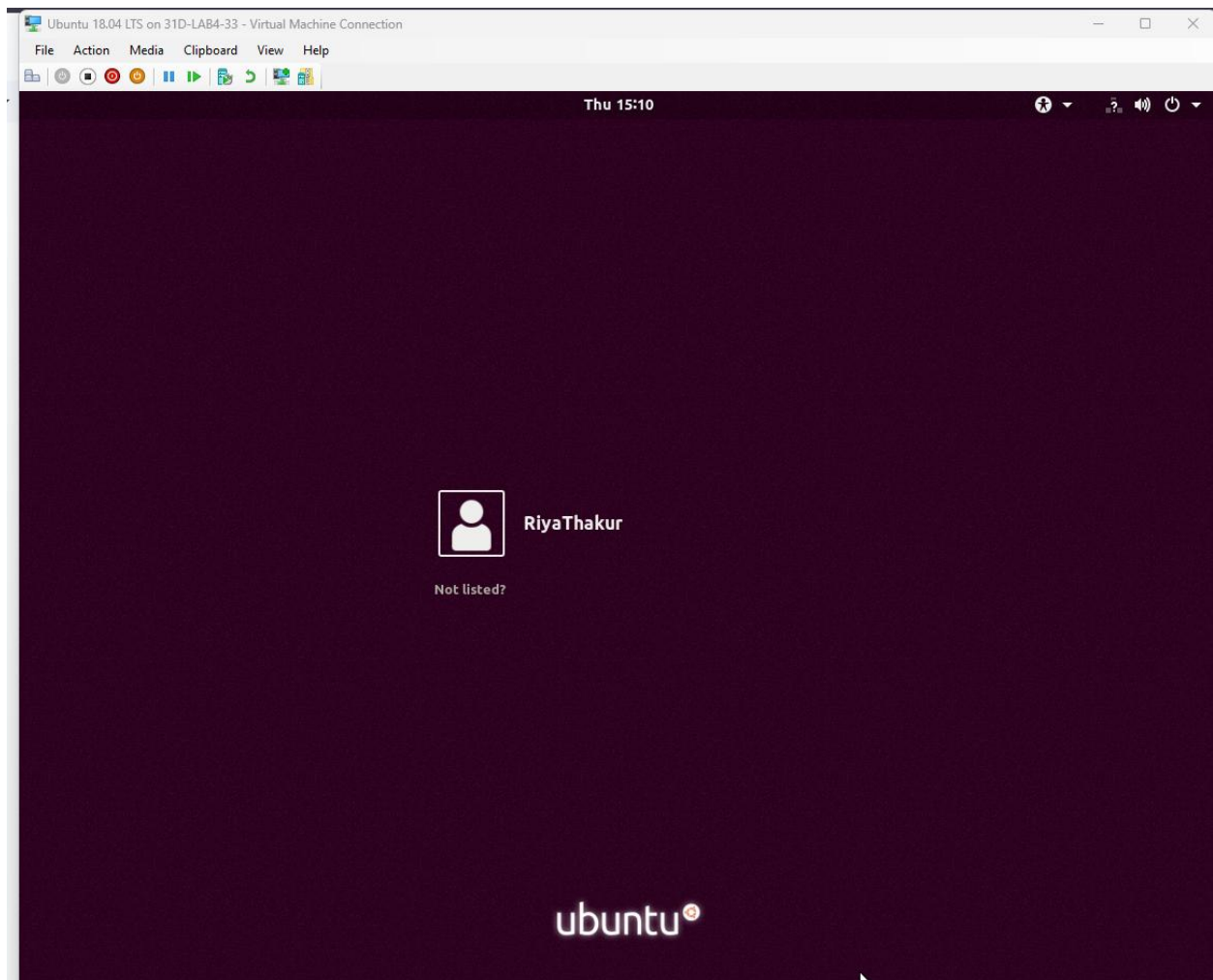


Step 11: Pin Kolkata and click Continue

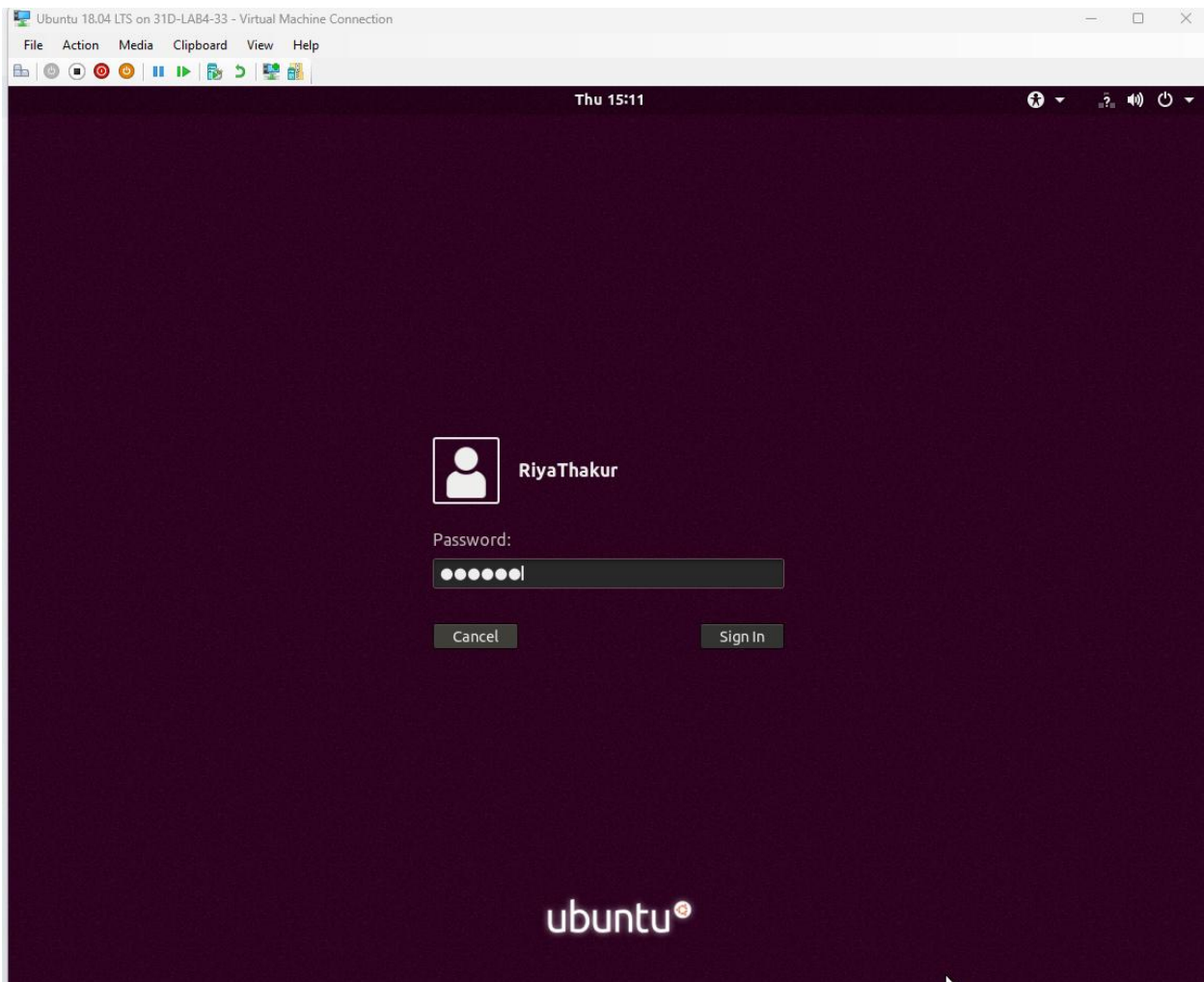


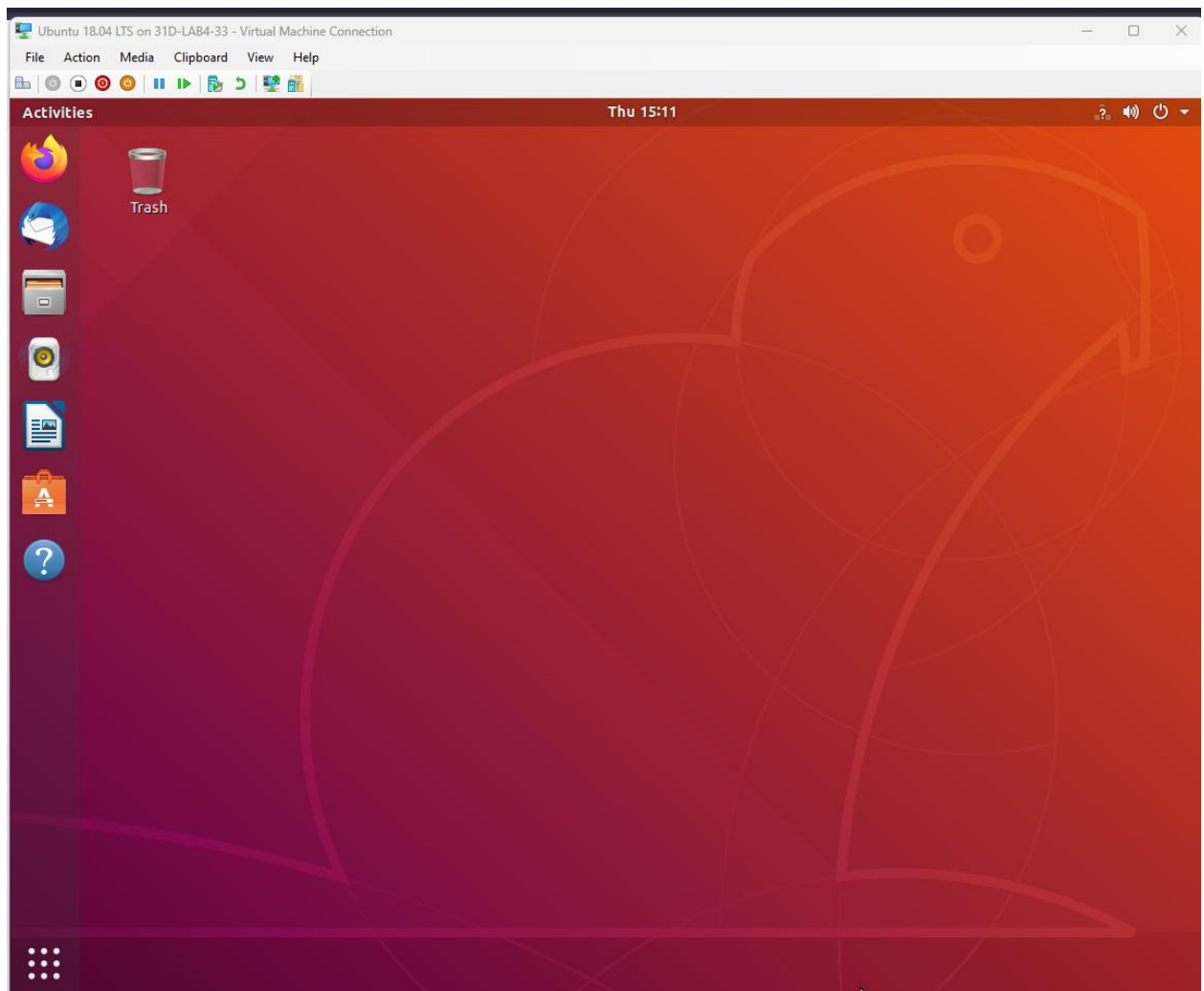
Step 12: Fill the required information.

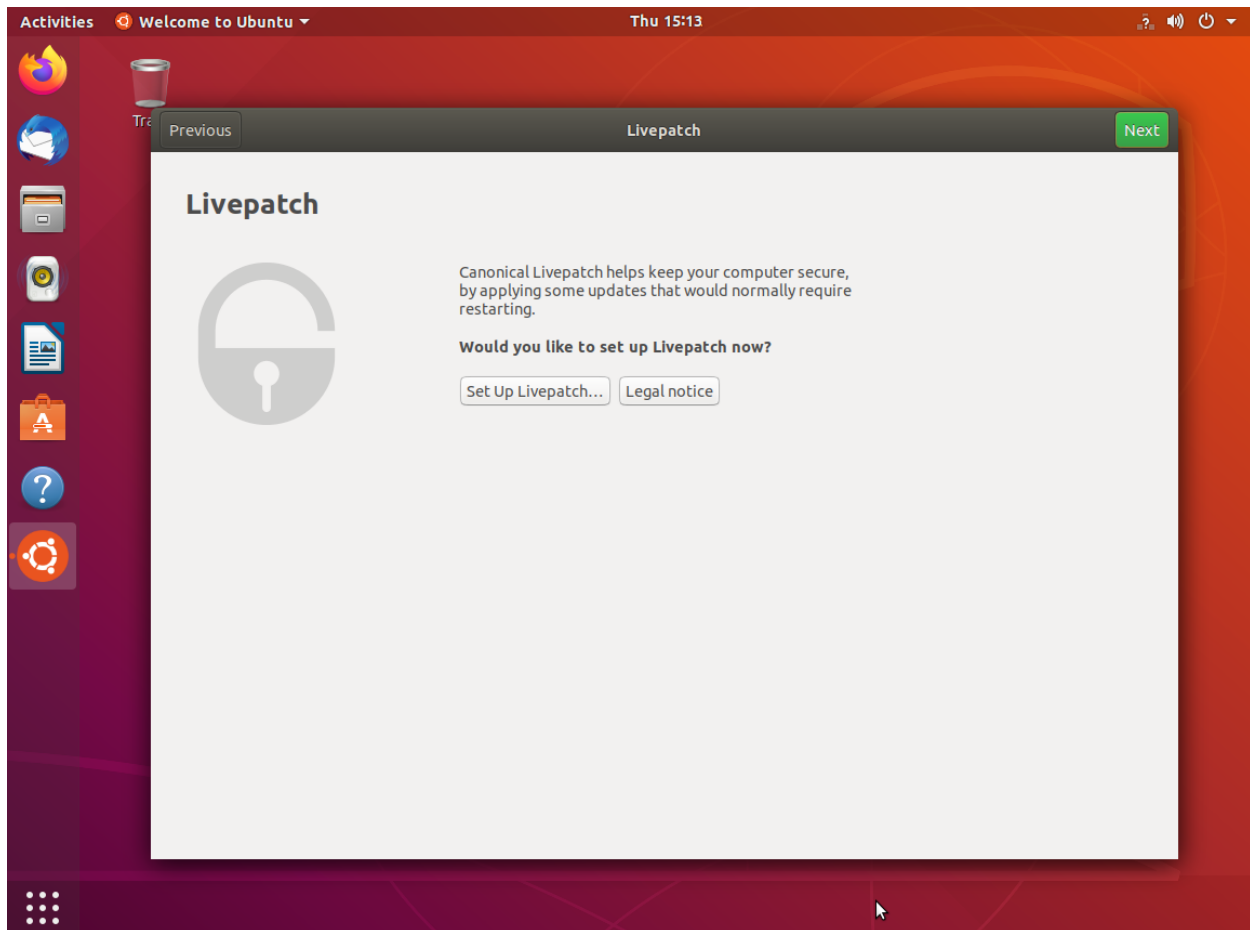




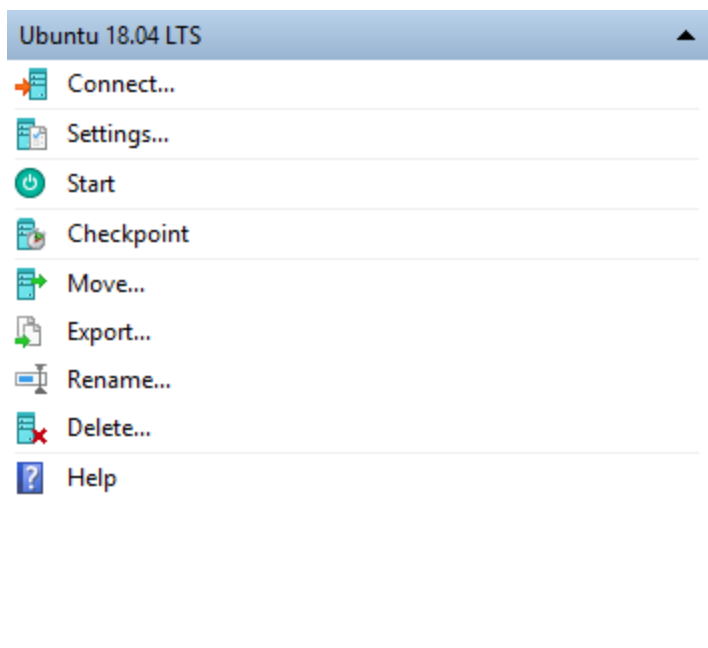
Step 13 : Now login using the appropriate password



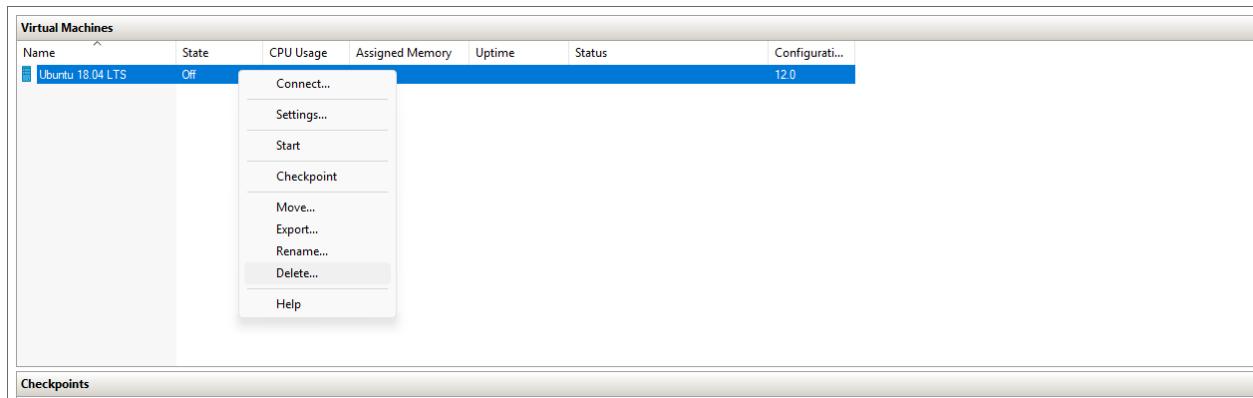




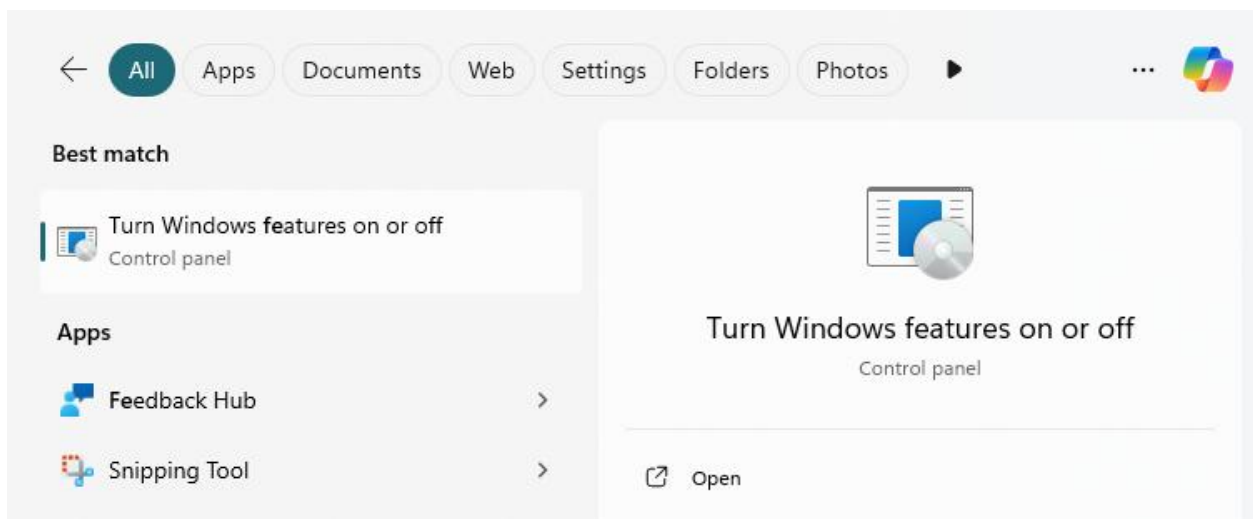
Step 14: Cancel and turnoff



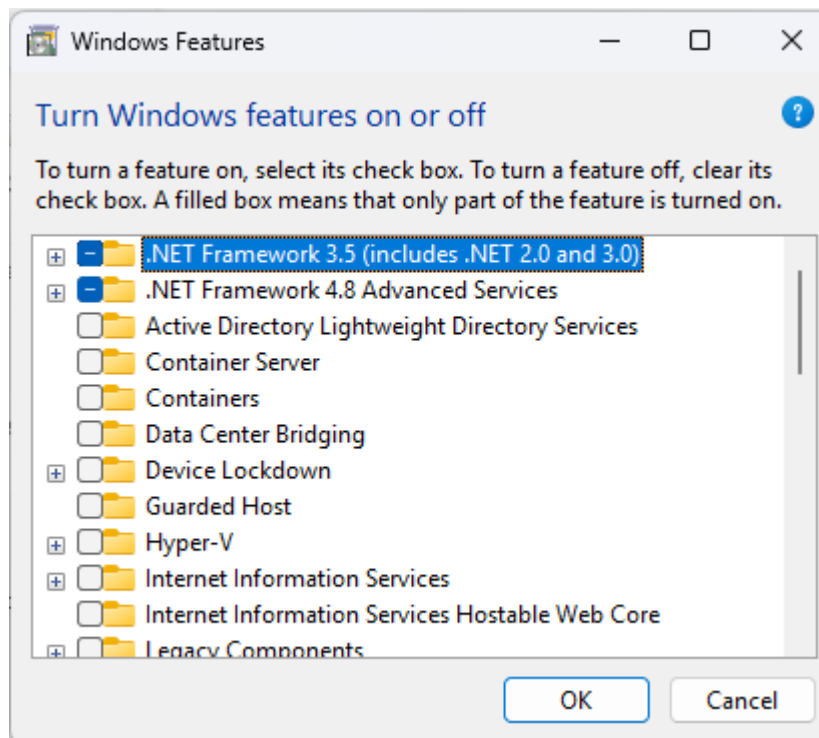
Step 15: Select Ubuntu and delete



Step 16 : Go to features



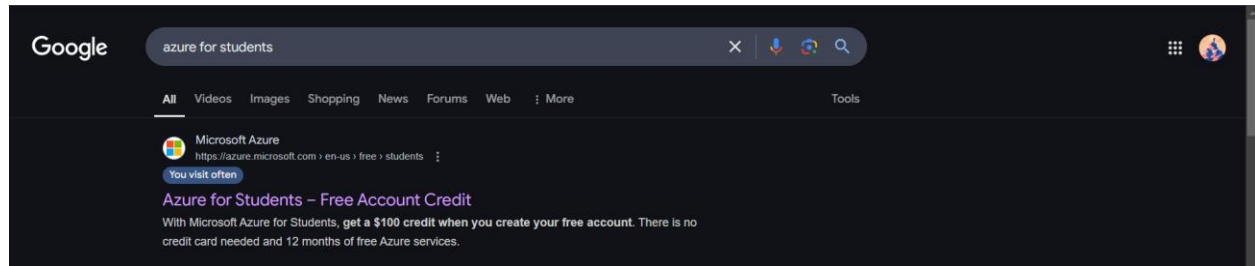
Step 17: Click on hyper v and deselect and click ok



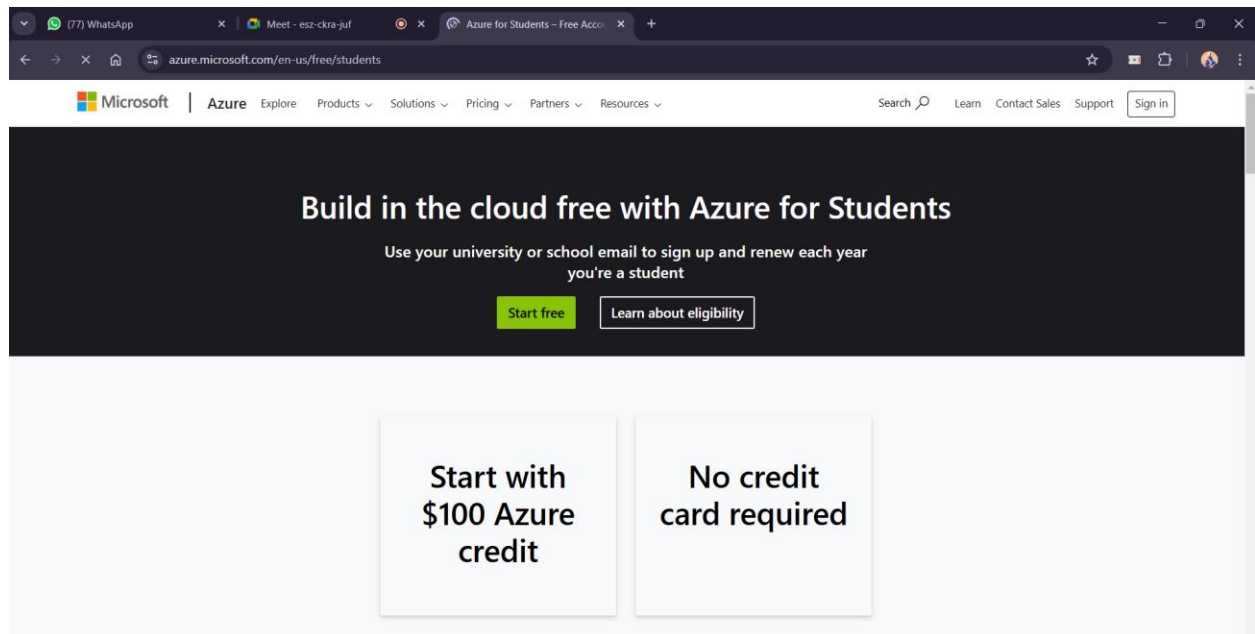
Practical 09

Aim: Develop Virtual machine using Microsoft Azure

Step1 : Go to your browser and search “azure for students” and click on the first link.



Step2: Now click on “Start Free”



Step 3: Fill in the details and verify yourself

Academic Verification

Start by entering your name as per the school records. Select your school's country and enter your school's name. Enter your date of birth as per the school records. The email address may be used to reach you if we have trouble verifying your application, so please enter your school provided email address.

First name

Last name

Country

If your country is not listed, the offer is not available in your region. [Learn More](#)

School name

School name will help provide Microsoft with additional information for verification. If available, please enter it here.

Date of birth

School email address
If you have received a verification code, please enter that here.

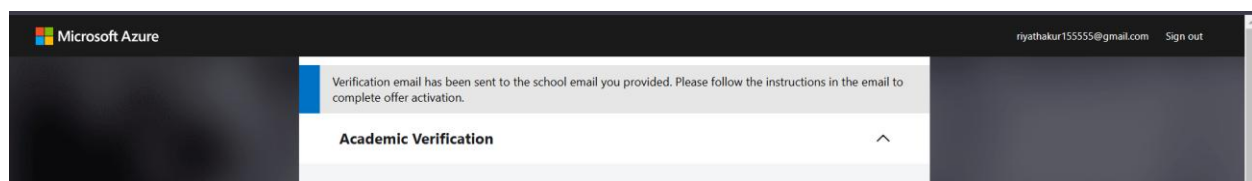


Verification complete!

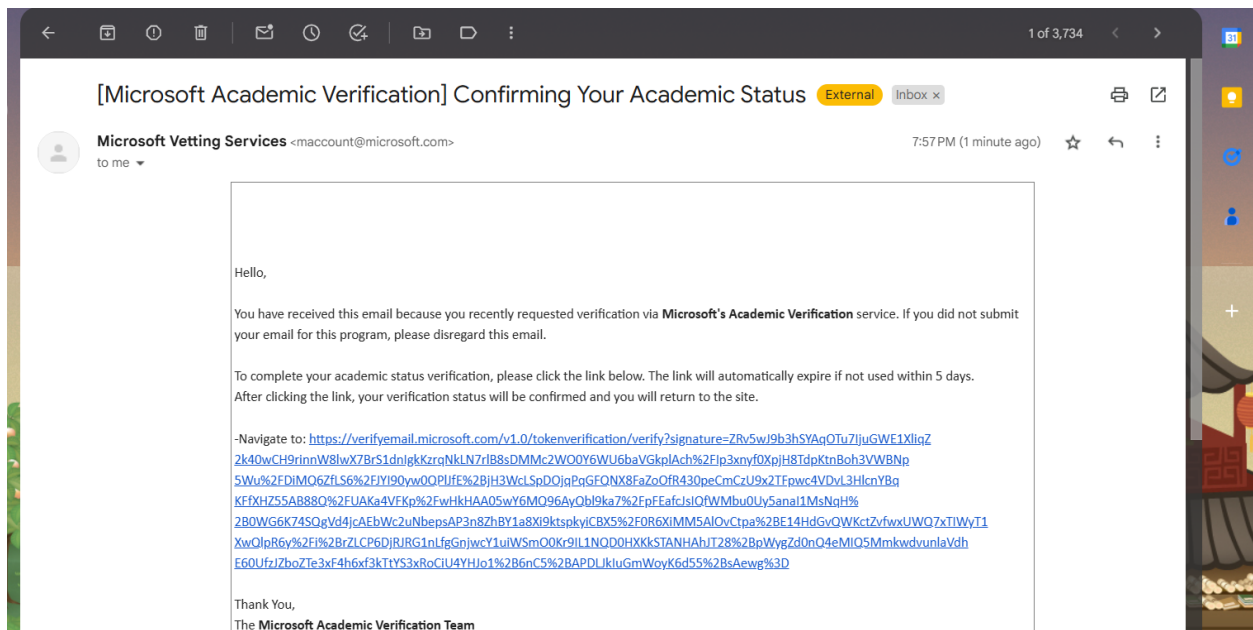
You've proven you're a human.
Continue your action.

[Verify academic status](#)

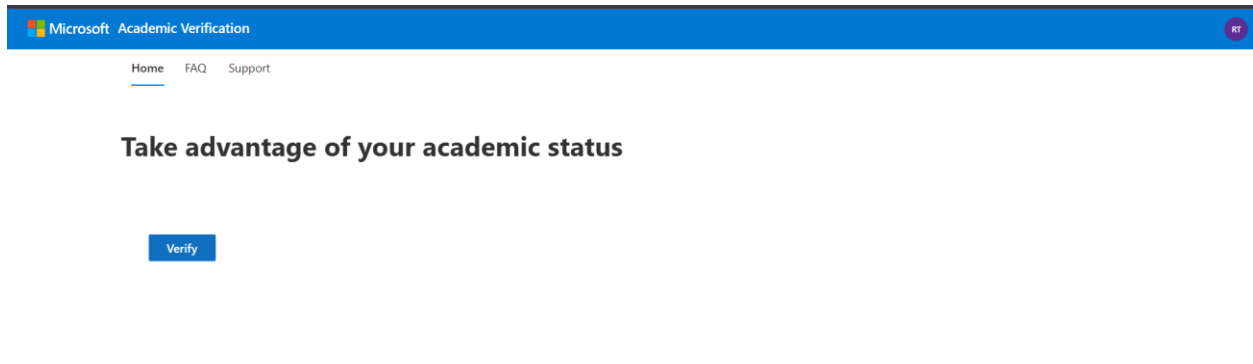
[Start Chat](#)



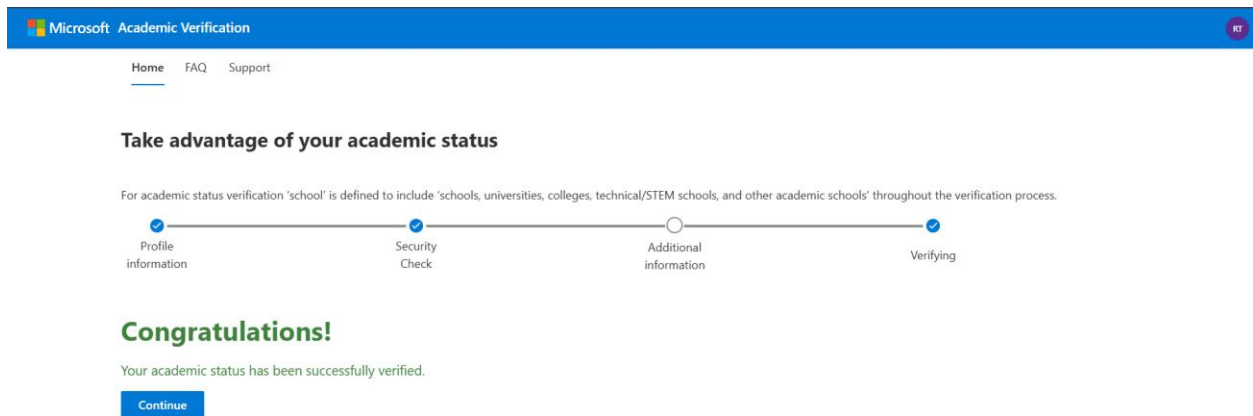
Step 4: Click on the verification link that is sent on your respective academic email address.



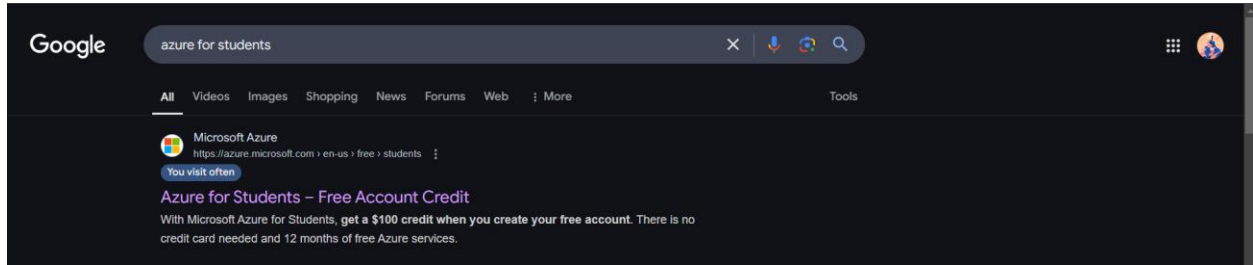
Step 5: Click on Verify.



Step 6: Click on Continue



Step 7: Open new browser and again search “azure for students”



Step 8: Fill in the required details and click sign-in up

Your profile

Country/Region

India

Choose the location that matches your billing address. **You cannot change this selection later.** If your country is not listed, the offer is not available in your region. [Learn More](#)

First name

Riya

Middle name (Optional)

Vinod

Last name

Thakur

Email address for important notifications

riyathakur155555@gmail.com

Phone

98332-32424

Azure for Students

Get \$100 in Azure credits and free access to popular cloud services plus developer tools like Visual Studio Code

State

Maharashtra

Postal Code

400088

I understand that Microsoft may contact me about my free account.

☒ I agree to the [subscription agreement](#), [offer details](#).

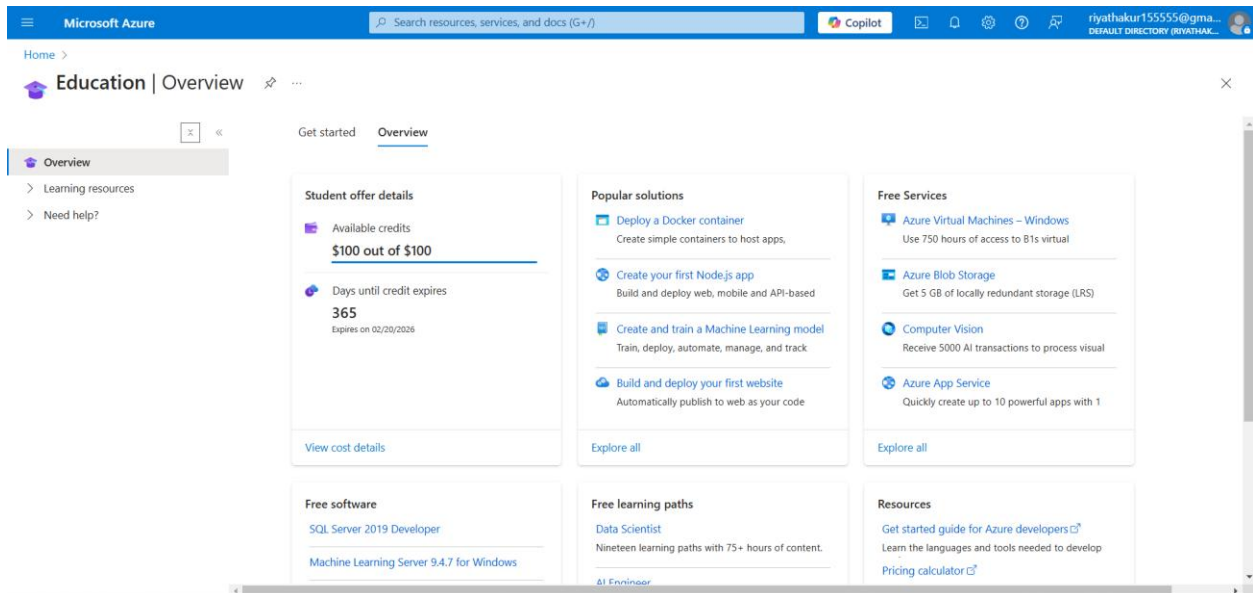
I will receive information, tips, and offers about Azure, including Azure Newsletter, Pricing updates, and other Microsoft products and services.

☒ I would like Microsoft to share my information with select partners so I can receive relevant information about their products and services.

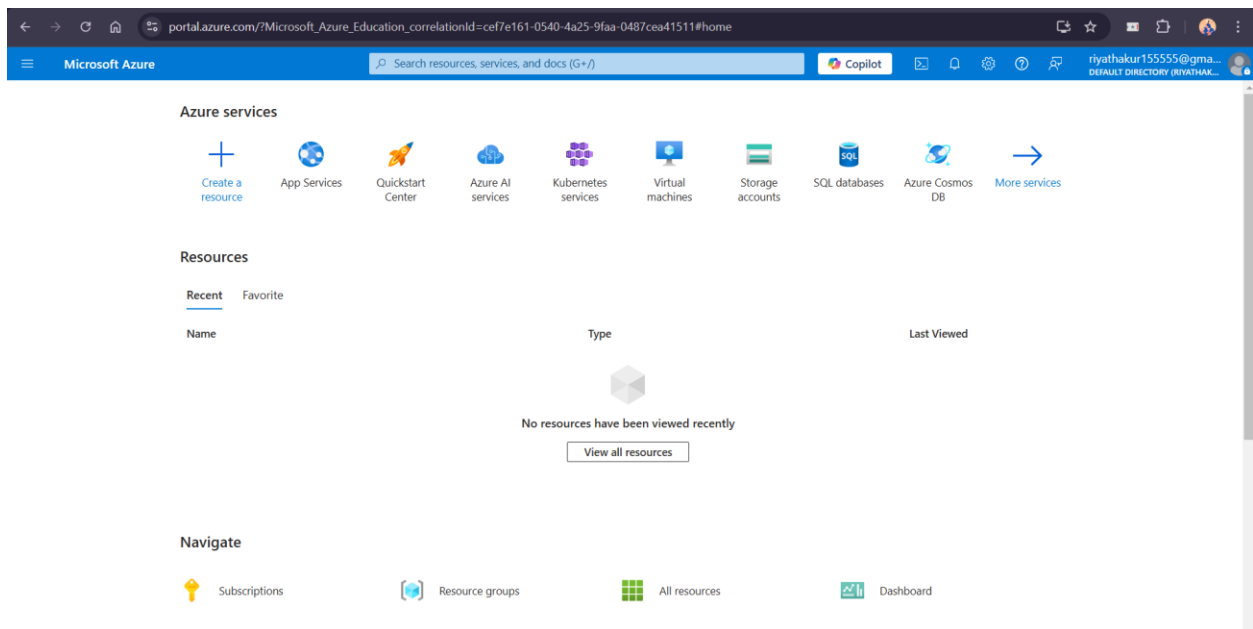
Read our [privacy statement](#) for information on how your data is handled.

Sign up

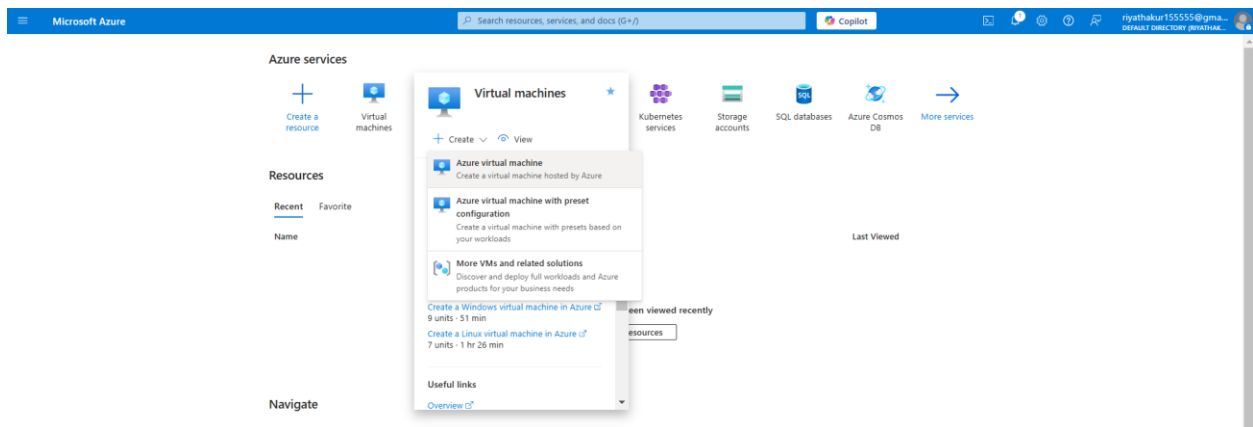
Step 9 : Check to see if \$100 credits are available to you or not.



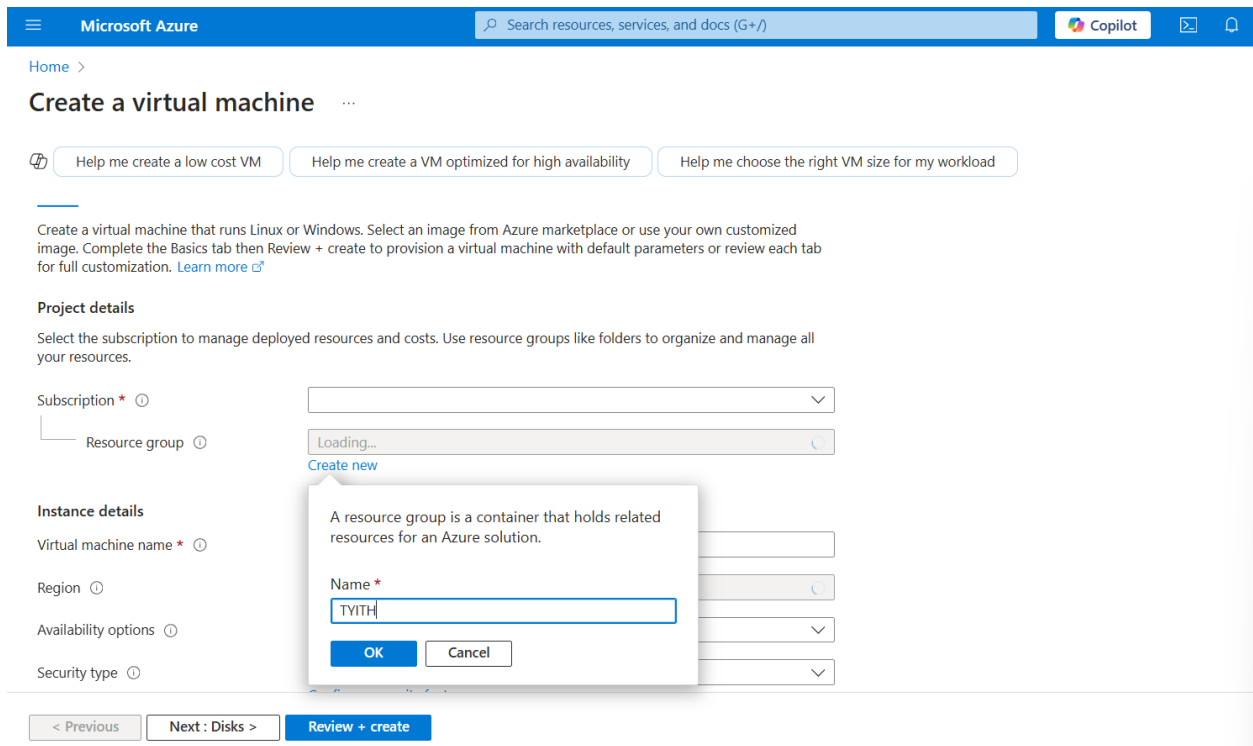
Step 10 : Go to home and hover over virtual machine



Step 11 : Hover over create and select azure virtual machine



Step 12: In resource group – go to create new and give a name and click ok



Step 13: Fill in the required details and click next until last page.

Microsoft Azure

Search resources, services, and docs (G+J)

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Home >

Create a virtual machine ...

Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Subscription * ⓘ

Azure for Students

Resource group * ⓘ

(New) TYITH

Create new

Instance details

Virtual machine name * ⓘ

RiyaThakur

Region * ⓘ

(Asia Pacific) Central India

Availability options ⓘ

Availability zone

Zone options ⓘ

☒ Self-selected zone

Choose up to 3 availability zones, one VM per zone

☐ Azure-selected zone (Preview)

Let Azure assign the best zone for your needs

☐ Using an Azure-selected zone is not supported in region 'Central India'.

Availability zone * ⓘ

Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

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Create a virtual machine ...

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Help me choose the right VM size for my workload

SSH Key Type

☒ RSA SSH Format

☐ Ed25519 SSH Format

☐ Ed25519 provides a fixed security level of no more than 128 bits for 256-bit key, while RSA could offer better security with keys longer than 3072 bits.

Key pair name *

RiyaThakur_key

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☒ None

☐ Allow selected ports

Select inbound ports

Select one or more ports

All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

< Previous

Next : Disks >

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Create a virtual machine

Help me create a low cost VMHelp me create a VM optimized for high availabilityHelp me choose the right VM size for my workload

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk sizeImage default (30 GiB)

OS disk typePremium SSD (locally-redundant storage)

Delete with VM

Key managementPlatform-managed key

Enable Ultra Disk compatibility

< PreviousNext : Networking >Review + create

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Create a virtual machine

Help me create a low cost VMHelp me create a VM optimized for high availabilityHelp me choose the right VM size for my workload

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network(new) RiyaThakur-vnet
[Create new](#)

Subnet(new) default (10.0.0.0/24)

Public IP(new) RiyaThakur-ip
[Create new](#)

NIC network security groupNoneBasicAdvanced

Public inbound portsNoneAllow selected ports

Select inbound portsSelect one or more ports

All traffic from the internet will be blocked by default. You will be able to

< PreviousNext : Management >Review + create

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Create a virtual machine

Help me create a low cost VMHelp me create a VM optimized for high availabilityHelp me choose the right VM size for my workload

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Configure management options for your VM.

Microsoft Defender for Cloud

Microsoft Defender for Cloud provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

Enable basic plan for free ☒

This will apply to every VM in the selected subscription.

Identity

Enable system assigned managed identity ☐

Microsoft Entra ID

Login with Microsoft Entra ID ☐

RBAC role assignment of Virtual Machine Administrator Login or Virtual Machine User Login is required when using Microsoft Entra ID login. [Learn more](#)

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Create a virtual machine

Help me create a low cost VMHelp me create a VM optimized for high availabilityHelp me choose the right VM size for my workload

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Configure monitoring options for your VM.

Alerts

Enable recommended alert rules ☐

Diagnostics

Boot diagnostics ☒ Enable with managed storage account (recommended)
☐ Enable with custom storage account
☐ Disable

Enable OS guest diagnostics ☐

Health

Enable application health monitoring ☐

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Create a virtual machine ...

Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Extensions provide post-deployment configuration and automation.

Extensions

⊖

Select an extension to install

VM applications

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. [Learn more](#)

?

Select a VM application to install

Custom data and cloud init

Pass a cloud-init script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. [Learn more about custom data for VMs](#)

?

Custom data

Custom data on the selected image will be processed by cloud-init.
[Learn more about custom data for VMs](#)

?

User data

Pass a script, configuration file, or other data that will be accessible to your applications throughout the lifetime of the virtual machine. Don't use user data for storing your secrets or passwords. [Learn more about user data for VMs](#)

?

Enable user data ☐

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Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

?

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name ⊖	Value ⊖	Resource
RiyaThakur	123456789	Virtual machine ⌵
		14 selected ⌵

< Previous

Next : Review + create >

Review + create

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Create a virtual machine ...

Validation passed

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BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Price

1 X Standard D2s v3
by Microsoft
[Terms of use](#) | [Privacy policy](#)

Subscription credits apply ⓘ
8.7354 INR/hr
[Pricing for other VM sizes](#)

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

Riya Thakur

Preferred e-mail address

riyathakur15555@gmail.com

Preferred phone number

Basics

Subscription

Azure for Students

Resource group

(new) TYITH

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Create a virtual machine ...

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BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview + create

Basics

Subscription

Azure for Students

Resource group

(new) TYITH

Virtual machine name

RiyaThakur

Region

Central India

Availability options

Availability zone

Self-selected zone

Zone options

1

Security type

Trusted launch virtual machines

Enable secure boot

Yes

Enable VTPM

Yes

Integrity monitoring

No

Image

Ubuntu Server 24.04 LTS - Gen2

VM architecture

x64

Size

Standard D2s v3 (2 vcpus, 8 GiB memory)

Enable Hibernation

No

Authentication type

SSH public key

Username

azureuser

SSH key format

RSA

Key pair name

RiyaThakur_key

Public inbound ports

None

Azure Spot

No

Disks

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